INDUSTRIAL INSTRUMENTS N5

BEREA TECHNICAL COLLEGE
539 SMITH STREET
DURBAN 4001

JS Smith

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Moncebiso Mair

1. FLOW

1.1 PROPERTIES OF FLUID FLOW

INTRODUCTION

Liquids and gases are called fluids unlike solid objects, fluids can flow. An important part of process control deals with the measurement of fluid flow, which is the continuous movement of a liquid, or a gas, from one place to another. In order to understand fluid flow, you need to know a few basic properties of fluids.

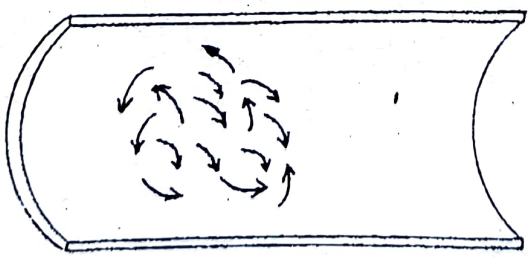
1.2 FLOW CHARACTERISTICS (VELOCITY PROFILES)

Fluid can flow in two different ways. The kinds of fluid flow are LAMINAR and TURBULENT.

LAMINAR flow, is the linear flow of liquids. The central line of the fluid moves the fastest and the outermost slowest against the pipe wall, due to friction.

TURBULENT flow is the irregular movement of a fluid. It is made up of many smaller currents, swirling and moving in all directions.

Turbulent flow causes a great deal of frictional resistance to fluid flow.



IMPORTANT: Unless turbulence is needed, any object you place into a stream of fluid should be as smooth as possible and thereby reducing turbulence to a minimum.

The type of fluid flow you have in a pipe depends on several factors:

a) The fluids velocity

b) The fluids viscosity THICK, STICKY STATE

c) The fluids specific gravity DENSITY IN RELATION TO ANOTHER SUBSTANCE.

d) The size and smoothness of the pipe

Up to a certain velocity range a fluid will flow in a laminar shape. Above this transition range, the flow becomes turbulent.

1.3 PRESSURE AND VELOCITY IN FLUID FLOW

Fluids flowing through a pipe of uniform cross-sectional area, maintain a constant velocity, but if the same fluid is forced to flow through a gradually converging (shrinking) pipe, it's velocity begins to increase and if it flows through a diverging (enlarging) section of the pipe, the fluid velocity drops.

When fluids flow through any pressure differential device there is a permanent loss in pressure downstream from the restriction, dependent on velocity and the magnitude of restriction. need example of flow meter

1.4 GETTING FLUIDS TO FLOW

There are two ways of setting up a pressure so that fluids can flow. One is by means of gravity, the other is with the aid of a pump.

1.5 INDICATION OF FLUID FLOW RATE

When working with process instruments, you may find the rate of fluid flow expressed in several ways:

VOLUME FLOW RATE

Volume flow rate tells you the amount of fluid moved in a given length of time. You can calculate volume flow rate by using a simple equation:

$$Qv = A \times v$$

and $$v = \sqrt{2GH}$$

$$\therefore \quad Qv = \sqrt{2GH} \cdot A$$

Where
- Qv = Total flow in volume units at flowing conditions; may be M etc.
- A = Cross-sectional area of the pipe - AREA OF CIRCLE
- v = Flow velocity in cubic meters per second
- G = Gravitational acceleration ± 9,80 centimetres per second squared.
- H = Head refers to the vertical distance between the free level of the source of supply and the free level on the discharge side in meters.

The actual flow rate will be less than the mathematical value of the equation. The reduction is caused by several factors, including frictional contraction of the stream within the pipe (these factors are expressed in a coefficient that can be accounted for in the calculation). The measurement unit for these flows can be in m^3/hour.

MASS FLOW RATE

The measurement unit for mass flow can be in kg/hour. You can calculate a mass flow rate if you know the volume flow rate and the fluid density, using the formula.

$$Qm = D \times Qv$$

Where
- Qm = Total flow in mass units
- D = Density in mass units
- Qv = Total flow (kg/hour)

1.6 FACTORS EFFECTING FLOW RATE

HEAD - HEIGHT PUMP CAN LIFT FLUID.

The rate at which a fluid flows through a channel or pipe depends upon three factors: Head, viscosity and frictional resistance. Head is a positive factor, while viscosity and friction are negative factors. The higher the head, the greater the volume flow rate, if all other factors remain constant.

The volume flow rate of a fluid in a pipe, is directly effected by the pipe size as well as the pressure head. If you double the diameter of a pipe (and keep the same pressure head), the flow rate will be four times as great. This is because doubling the pipe diameter increases the cross-sectional [redacted] of the pipe four times.
AREA

Pipe friction opposes fluid flow to a small degree, because of friction, the flow is actually slightly lower near the walls of the pipe and faster at it's centre. Corrosion inside pipes and the pipe walls also increases friction.

In some ways, viscosity resembles friction. It is an internal resistance to flow caused by the difficulty of the fluid's molecules in moving around one another. The higher the viscosity, the lower the flow rate.

1.7 TYPES OF FLOW MEASUREMENT

Instruments used in the measurement of flow can be divided into two main classes:

1. Rate-of-flow meters
2. Quantity meters (volumetric) / INFERENTIAL

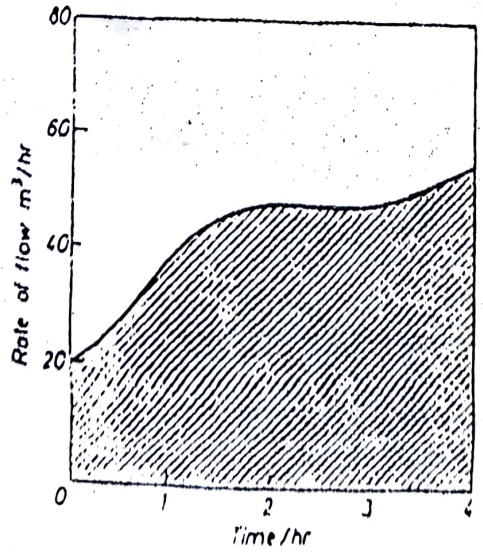
In meters of the first class the actual rate of flow is measured; if the total quantity which flows in a given time is required, the quantities flowing in each small interval of time must be summed or integrated. If the rate of flow is reasonably steady, the total quantity will equal the average rate of flow × time.

In the second class the total quantity which flows in a given time is measured; to obtain the average rate of flow this quantity must be divided by the time taken to flow.

If the flow is not steady, however, the total quantity will be represented by the area under the graph obtained by plotting the rate of flow against the time.

This type of graph is obtained from recording flowmeters having square root compensation, and the total quantity will be represented in m^3 by the shaded area as:

$$\textit{Quantity} = \textit{Rate of flow} \times \textit{time} = \frac{m^3}{\textit{hours}} \times \textit{hours} = m^3$$



Finding the area under the curve is called integrating the rate of flow, and is done mechanically in many instruments. The portion of the instrument which does this is called an *integrator*, and will be described later.

The instruments in these two sections are then further subdivided according to the nature of their primary elements. Each instrument may be considered as two parts, the primary element and the secondary element. The primary element may be defined the instrument which converts the quantity which is being measured into a variable which operates the secondary element. The secondary element then measures this new variable which is arranged to depend upon the quantity being measured.

A. RATE OF FLOW MEASURING INSTRUMENTS

1.7.1 PRIMARY FLOW ELEMENTS

With these instruments, you can measure the flow of fluids to determine their velocity or volume flow rates. The measurement of flow using the differential pressure (Δ P) method is probably the most widely used technique and encompasses a wide variety of meter types: Venturi tubes, nozzles and the orifice plate, pitot tube.

Purpose of a primary flow element

The purpose of primary measuring devices is to create a difference in pressure in a flowing fluid. A secondary device then measures this difference in pressure to determine the flow velocity or volume flow rate of the fluid.

Bernoulli's Theorem - Incompressible Fluids

For a stream of ideal fluid, in steady flow, with no frictional forces involved but acting under gravitational forces only, the sum of the pressure energy, kinetic energy and potential energy is a constant. Expressed mathematically:

Pressure energy + kinetic energy + potential energy = k

$$\therefore \quad pV + \frac{mv^2}{2} + mgZ = k$$

where p = the pressure
v = the velocity
g = gravitational acceleration
Z = height

k = a constant
V = volume
m = mass

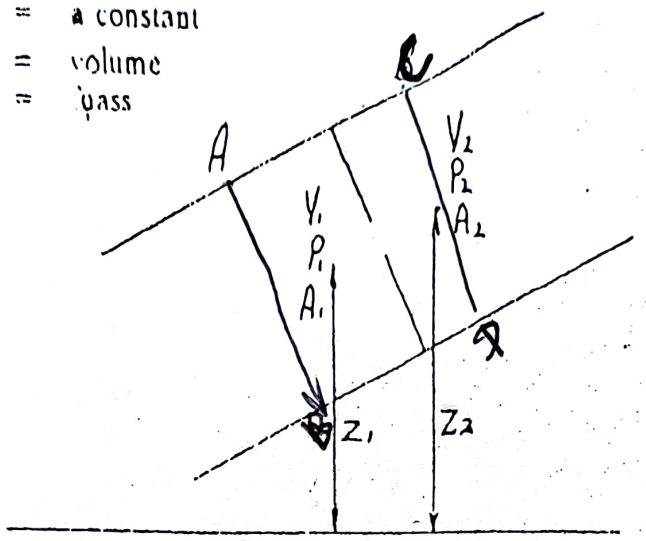


FIG A fluid stream constricted from area A_1 at AB to A_2 at CD

A fluid stream flows from a section area A_1 at AB to a smaller area A_2 at CD. The fluid, at first, will be taken to be incompressible, so that the density is unchanged from AB to CD. Continuity of flow demands that the quantity of fluid Q flowing through A_2 per second is the same as that through A_1. Since Q is the product of the area and velocity at each section.

$$Q = A_1v_1 = A_2v_2$$

where v_1 = the velocity at AB
v_2 = the velocity at CD

This means that the kinetic energies obtaining at AB and CD are:

$\frac{mv_1^2}{2}$ *and* $\frac{mv_2^2}{2}$ *respectively*

The potential energy at AB is mgZ_1 and that at CD mgZ_2.

A pressure change is involved from AB to CD. It can be deduced by considering the work done in transferring a small element of fluid from AB to CD. Let us designate the pressure energy at AB p_1V_1 and that at CD p_2V_2. The total energy at AB is the same as that at CD since no heat transfer, etc., is considered involved. Then the following relation holds:

$$p_1V_1 + \frac{mv_1^2}{2} + mgZ_1 = p_2V_2 + \frac{mv_2^2}{2} + mgZ_2 \qquad (1)$$

Fluids can not be compressed, therefore volume 1 must be equal to volume 2 and we know that

$$V = \frac{\text{mass } (m)}{\text{density } (\rho)}$$

We can rewrite equation (1) as follows:

$$\frac{p_1m}{\rho} + \frac{mv^2}{2} + mgZ_1 = \frac{p_2m}{\rho} + \frac{mv_2^2}{\rho} + mgZ_2$$

Divide above equation with mg:

$$\therefore \quad \frac{p_1}{\rho g} + \frac{v^2}{2g} + Z_1 = \frac{p_2}{p^2} + \frac{v_2^2}{2g} + Z_2 \qquad (2)$$

Manipulating equation (2) gives:

$$\frac{v_2^2}{2g} - \frac{v_1^2}{2g} = \frac{(P_1 + Z_1)}{\rho g} - \frac{(P_2 + Z_2)}{\rho g} \qquad (3)$$

Pressure at AB with the height taken into account, will be higher than the pressure at CD, therefore difference in pressure with the height included (H)

$$(h) = \text{Pressure at } AB - \text{pressure at } CD$$

$$\therefore h = \frac{(P_1 + Z_1)}{\rho g} - \frac{(P_2 + Z_2)}{\rho g}$$

Substitute h into equation (3)

$$\frac{v_2^2}{2g} - \frac{v_1^2}{2g} = h$$

$$\therefore \quad V_2^2 - V_1^2 = 2gh \qquad (4)$$

We have seen that

$$Q = A_1v_1 = A_2v_2$$

If $m = \frac{A_2}{A_1}$

$$v_1 = mv_2$$

Substitute v_1 into equation (4)

$$v_2^2 - (mv_2)^2 = 2gh$$

$$v_2^2 - m^2v_2^2 = 2gh$$

$$v_2^2 (1 - m^2) = 2gh$$

$$v_2 = \frac{1}{\sqrt{1 - m^2}} \times \sqrt{2gh}$$

but the velocity approach factor (E)

$$(E) = \frac{1}{\sqrt{1 - m^2}}$$

$$\therefore \quad V_2 = E\sqrt{2gh}$$

Since we want to calculate the quantity of flow (Q) we can rewrite the equation as follows:

$$v_2 = \frac{Q}{A_2} = E\sqrt{2gh}$$

$$\frac{Q}{A_2} = E\sqrt{2gh}$$

$$\therefore \quad = A_2 E \sqrt{2gh} \ (m^3/h)$$

We can deduce the basic flow rate equation if A_2, E and 2g is taken as a constant values:

$$Q = k\sqrt{h} \ (m^3/h)$$

Out of above equation we can see that the flow rate depends on the square root of the differential pressure.

1.7.2 TYPES OF PRIMARY FLOW ELEMENTS

a) Concentric orifice plate

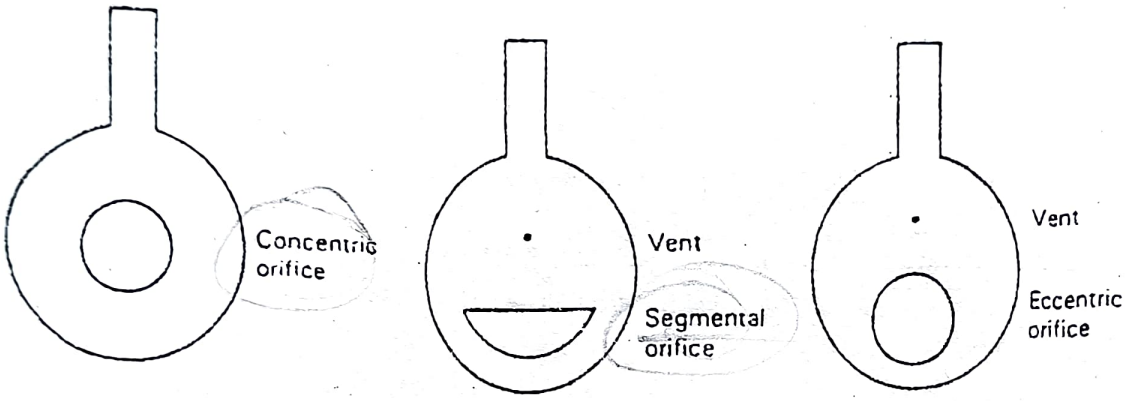
It is a circular metal plate with a central hole or orifice concentric with the circumference of the plate. It is fixed between the pipe flanges and is located by the flange bolts. The orifice is then concentric with the internal bore of the pipe.

b) The Segment Orifice Plate

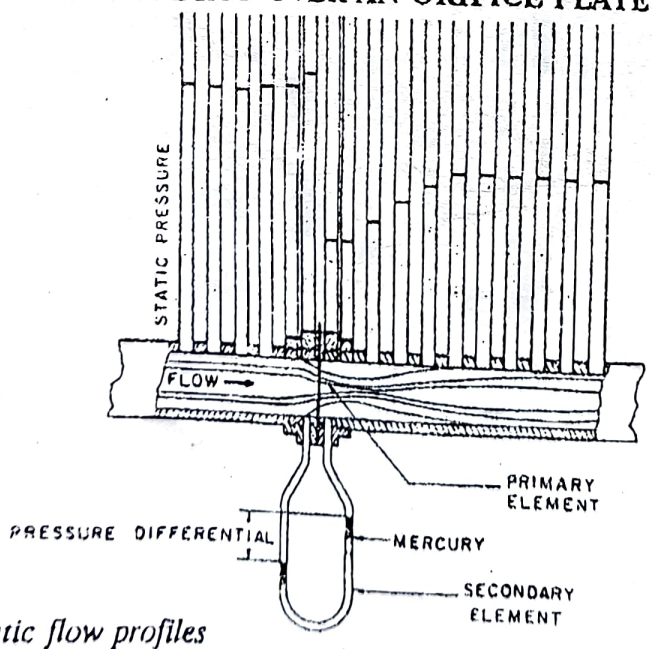
Used where dirty fluids are encountered. It consists of a plate with a segment cut out, the solid segment is at the top of the orifice plate with the open part at the bottom and flush with the pipe for this reason no solid material builds up on the upstream side of the orifice.

c) The Eccentric Orifice Plate

The eccentric plate has a hole that is bored off-centre with the lower part of it s orifice flush with the lower part of the pipe. Its shape is such that suspended material does not collect on the upstream side of the orifice.



1.7.3 PRESSURE DROP OVER AN ORIFICE PLATE



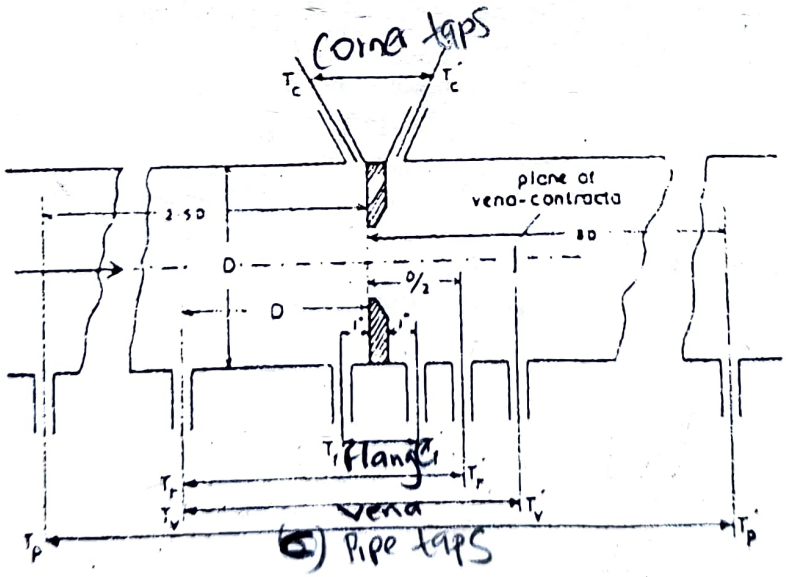
Dynamic static flow profiles

Suppose that tubes are inserted through the pipe wall at the position shown in the diagram. The pipe liquid will rise in these until the pressure due to the column of liquid

in each tube is equal to the static pressure at that position. The column heights are then a measure of the differential pressures, and from observing the different values we may trace the pattern of the pressure changes as we proceed along the pipe. At position A and B there is no pressure change worth specifying. At C and D, just before the orifice, we find a slight increase in pressure. The stream is then constrained to flow through the smaller area of the orifice, from which it issues as a jet. At positions E and F there are lower pressures than at the upstream positions due to the change in the stream sectional area. Since this is smaller the velocity has increased, and the pressure has fallen. The jet cross-section decreases in area after leaving the orifice until it reaches a point, indicated as G in the diagram, where it is a minimum and the velocity a maximum. This is mainly due to the liquid being directed inward as it approaches the orifice, and, through inertia effects, persisting in this direction for a distance after it leaves the orifice. The static pressure also reaches its minimum value at this position, which is known as the *vena contracta*. The distance from the orifice varies with the ratio of orifice diameter to pipe diameter, but an average value would be about one half the pipe diameter. From the vena contracta, the stream section expands until it reaches the pipe diameter at H.

1.7.4 THE POSITION OF PRESSURE TAPPINGS

The following are the main tapping positions.



A = Flange taps
B = Vena contracta taps
C = Pipe taps
D = Radius taps
E = Corner taps

A Flange Taps

These are situated 25,4 mm from the upstream and 25,4 mm from the downstream face of the orifice plate.

B Vena Contracta Taps

The upstream tapping is 1 pipe diameter from the upstream face, and the downstream tapping is determined from a curve relating the required dimension to the ratio of orifice to pipe diameter.

C Pipe Taps

These may be 2½ pipe diameter upstream and 78 diameters down stream from the upstream face of the orifice plate.

D Radius or Throat Taps (D and D/2 Taps)

The upstream pressure tapping is taken at one pipe diameter, D, upstream from the face of the orifice, and the downstream pressure tapping is taken at one half pipe diameter, D/2, downstream from the orifice face, approximately the vena contracta position.

E Corner Taps

Corner tappings are taken via holes cut obliquely through a flange or boss on the pipe, bringing the inside openings of the holes.

Advantages

- Low manufacturing costs
- Can be machined very accurately
- Easily installed
- Maintenance free

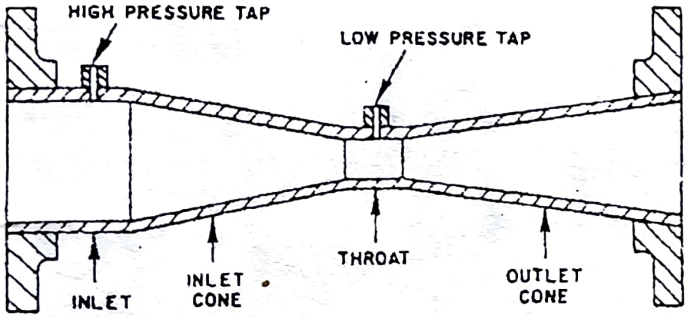
Disadvantages

- High pressure loss.
- In case the area ratio ($\frac{d2}{D2}$) is bigger than 0.55 the accurate measurement will be inaccurate.
- It cannot be used with liquids containing heavy materials which can form a layer of dirt.

1.7.5 THE VENTURI TUBE

A fairly high pressure loss is caused by the orifice plate, and, where pressure loss is important, it is necessary to turn to other methods of producing differential pressures. Let us consider devices with a gradual change in area.

There are virtually three sections to the tube: the inlet or upstream cone, the throat, and the outlet cone as indicated in the figure below.



The inlet cone tapers down from the pipe area to the throat section of the smaller area to produce the necessary velocity and pressure change. The outlet cone expands from the throat to the pipe area. Pressure tappings are taken at the inlet entrance to the cone, as well as at the throat.

Advantages

- The pressure loss is very small.
- It is useful for measuring the flow of slurries and suspensions of solids in liquids.

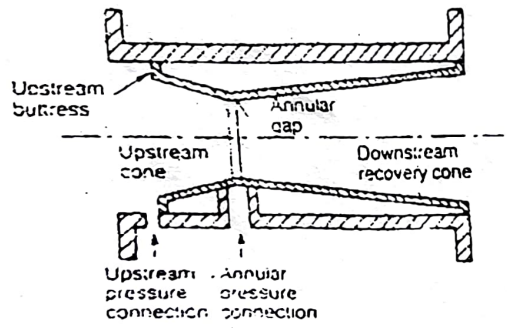
Disadvantages

- High cost.
- It is physically big.

1.7.6 THE DALL TUBE

The Dall tube is a modified venturi tube. The figure below shows a cross-section of a typical Dall flow tube.

Observe the diameter of the inlet cone is less than that of the pipe, resulting in a sharp step. This creates an impact pressure which is additional to the static pressure existing at the step. The high pressure connection is made just in front of the step.



The other connection is made at the throat where the relatively abrupt change in area results in a marked static pressure depression. The pressure loss across the Dall tube is 2 to 3 times the value of the Venturi tube.

Advantages

- Much shorter than the Venturi tube.
- Low cost.

- High pressure regaining factor.
- Turbulent flow has little effect on Dall tube.

Disadvantages

- Blockage in the throat area.
- Expensive with large types.

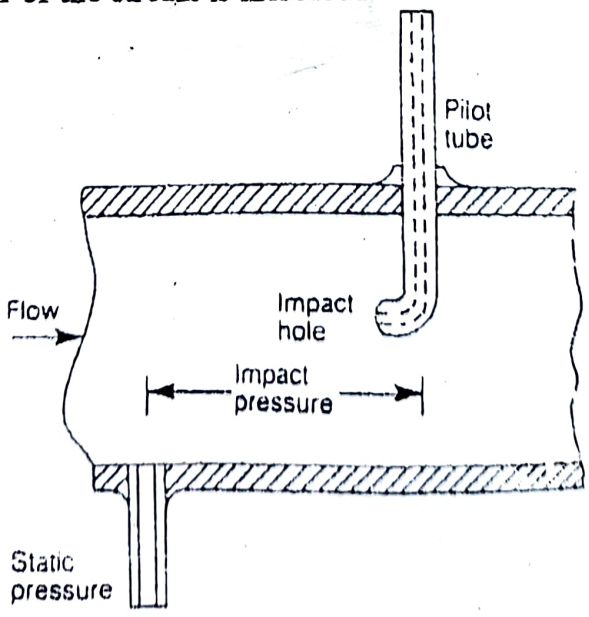
1.7.7 THE PITOT TUBE

As fluid approaches an obstruction in a flow stream, the velocity will decrease until it reaches zero at the point where it impinges on it. The kinetic energy is now converted into pressure energy. Thus, the pressure built up in the tube will be greater than that in the free stream by the impact pressure. The difference is measured between the pressure in the tube and the static pressure of the stream. The static pressure is measured by a tapping in the wall of the main. The difference between the pressure in the tube and the static pressure will be a measure of the impact pressure, and therefore of the velocity of the stream.

Types of pitot tubes

a) Single Hole Pitot Tube

This is a single hole pitot tube with a separate static tapping as shown in figure .... The tube must be strong enough to withstand the impact of the fluid stream and any vibration set up by the stream. This is achieved by giving the tube a streamlined section, so that the impact force and vibration is reduced, and the resistance to bending of the tube in the direction of the stream is increased.



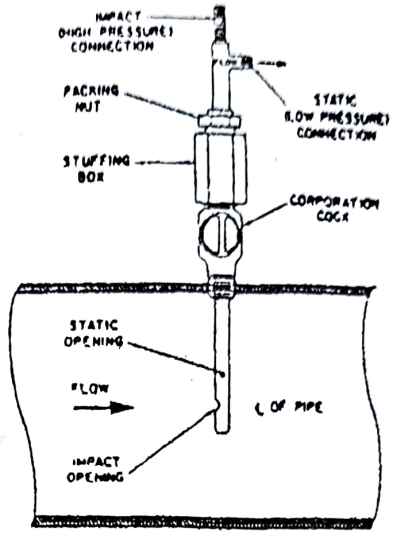
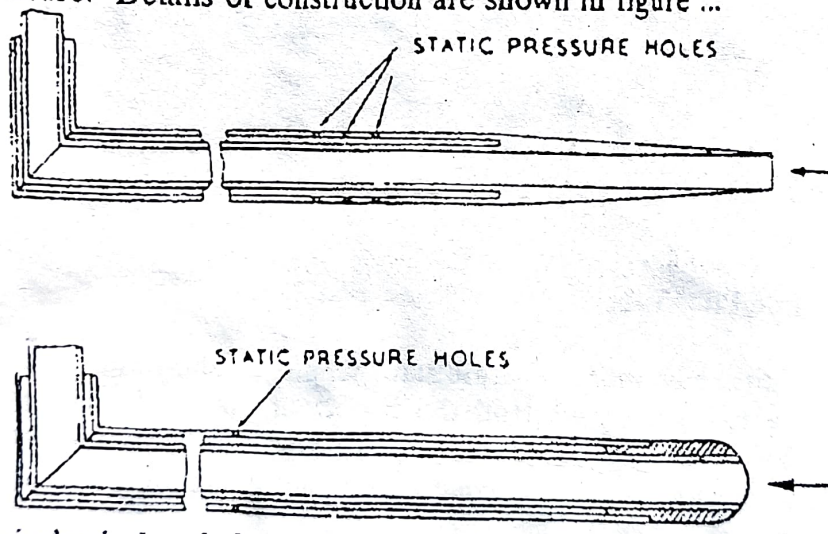


Fig. 2.13e Schematic of an industrial device for sensing static and dynamic pressures in a flowing fluid

b) Combined Type Pitot Tube

These combined types have been developed in two forms namely, the hemispherical head and sharp ended head. Both types consist of two concentric tubes. the inner one transmitting the impact pressure while the annular space between the tubes transmits the static pressure. Details of construction are shown in figure ...



The hemispherical-ended type is regarded as standard. It is less sensitive to variations of inclination of the tip to the direction of flow of the fluid.

Precautions in use

1. The axis of the pitot tube must be parallel to the axis of the pipe.

2. The fluid shall behave as a single-phase fluid and the velocity of flow must be between certain limits.

3. The length of unobstructed straight pipe upstream of the plane of measurement should be not less than 20 pipe diameters.

4. The diameter of the head of the pilot static tube must not exceed one twenty-fifth of the diameter of the duct.

Advantages

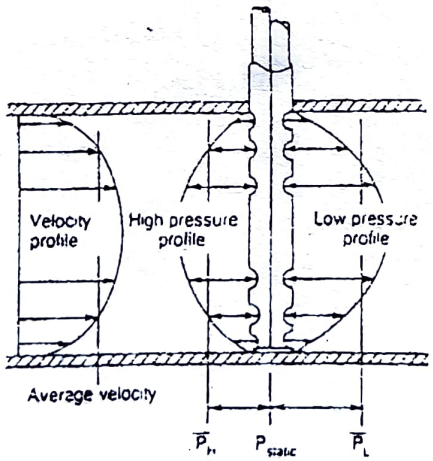
- The pressure loss caused by the pitot tube is very small.
- Low costs.

Disadvantages

- The fluid must be moving at a high velocity to produce a measurable differential pressure.
- The small openings may be blocked.

c) **The Annubar**

For the more permanent type of pitot-tube form of installation an Annubar may be used as shown in figure ... The pressure holes are located in such a way that they measure the representative dynamic pressure of equal annuli. The dynamic pressure obtained at the four holes facing into the stream is then averaged by means of the "intersertal" inner tube figure ... which is connected to the high pressure side of the manometer.



It may be inclined at an angle of up to 20° to the direction of flow without the inaccuracies brought about in the Velocity measurement exceeding 2 %. The tip, however, must be truly hemispherical, and the impact hole truly cylindrical, free from burrs and central with respect to the hemisphere.

The sharp ended tube must be maintained in good condition. The tip must not be allowed to become stump, distorted, or burred in any way, and the tube must be free from damage, kinks, or flattenings, upstream of the static pressure holes and for at least 5 tube diameters downstream.

Precautions in use

1. Laminar flow must be ensured at point where measurement is taken.

2. Obstruction of tube and static connections must be prevented.

3. Minimum slope of the tip to the direction of the flow of fluid must be ensured.

Advantages

1. Very low pressure loss.
2. Very useful for determining actual velocity profiles.
3. Low costs.

1.7.2 SECONDARY MEASURING DEVICES

The differential pressures obtained from the primary devices must be converted into a motion for indication or recording and this action is known as secondary measuring devices.

We have already noted in Bernoulli's theorem that the flow rate is not directly proportional to the differential pressure. This problem is known as the square root problem.

This square law effect must be incorporated into secondary measuring devices and some of these methods are:

a) Square root compensation.
b) Square root extraction

We will now take a closer look at these methods.

1.7.2.1 Square root compensation

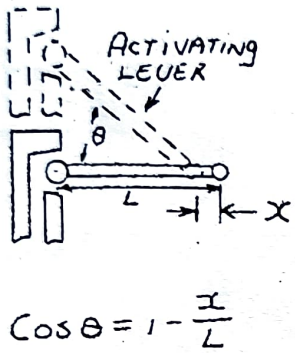
Different types of square root compensation:

1. Mercury monometer flowmeter (Inclined Tube).
2. Bell type flowmeter (Shaped Chamber).
3. Ring balance flowmeter.
4. Pen and paper method.

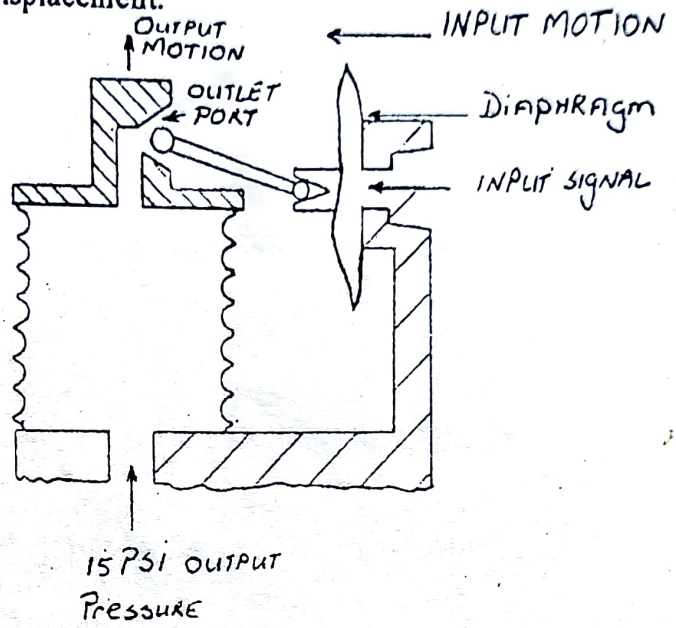
1.7.2.2 Square root extraction

The pneumatic square root angular displacement extractor works on the principle of balancing two forces by angular displacement.

Principle of operation



$$\cos\theta = 1 - \frac{X}{L}$$



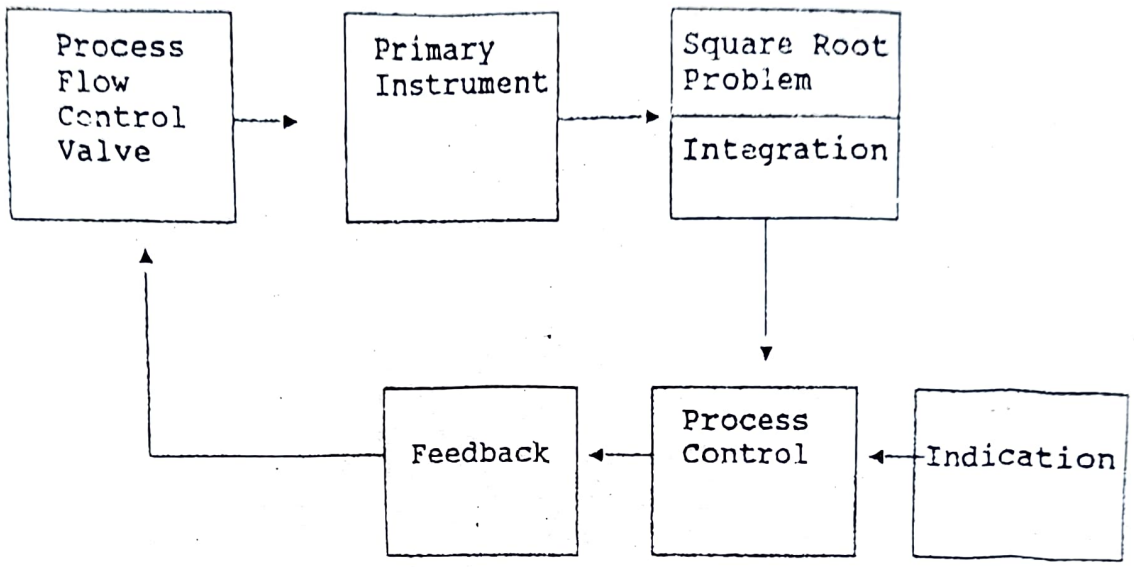
When the input pressure from the measured flow increases, the diaphragm will tend to move the activating lever to the right thus closing the outlet port. The output pressure will increase in the bellows resulting a motion to the top. This motion opens the outlet port until the ratio of input pressure and output pressure balances again. The angular displacement therefore compensate for the square root problem and will be directly proportional to the magnitude of the input signal. To calculate the angle of displacement we use the following equation:

$$\Theta = \cos^{-1}\left(1 - \frac{X}{L}\right)$$

where X = Displacement distance of activating lever
L = Length of activating lever

1.7.2.3 Integration

The purpose of an integrator is to measure the total volume of flow in a certain time. The function of the integrator in the process can be explained by the following block diagram.



Typical Flow Process

Process flow control valve : This valve controls the amount of flow of substance.

Primary instrument : This instrument generates a differential pressure of the flow.

Square root problem : The flow rate is the square root of the differential pressure.

Integration : This will measure the flow rate over a certain period.

Process control : This controls the process for an output for an instrument and if a error occurs to the feedback system.

Feedback : This system will generate a signal to the process flow control valve to rectify the fault.

Out of the above block diagram the integrator must be able to do the following:

- Divide units into time periods.
- Monitor the flowrate.
- Join time periods and flowrate.
- Send output to a recorder.

1.7.2.4 Basic components for an integrator

a) A meganism to connect the pen movement (flow rate output) to the integrator.
b) Time meganism.
c) A recording meganism to record the total flow in a certain time.

1.7.2.5 Types of Integrators

a) Mechanical Integrators

Two types of mechanical integrators are used and differs mainly in the counting part, namely, continuously and intermittently.

The counting part of the integrator is invariably a mechanism of the counter pattern.

The counter shaft may be driven at varying rates of revolution proportional to the flow rates, or it can be driven at constant speed but operated for varying periods according to the flow rate. For example, the shaft would be at rest for zero flow, continuously rotating for maximum flow, and operated and at rest for equal periods with 50 per cent low. Below 50 per cent the operated period would be less than the non-operated in proportion to the flow, and above 50 per cent the operated period would exceed the non-operated, again in proportion to the flow.

b) Continuously operated mechanical integrator

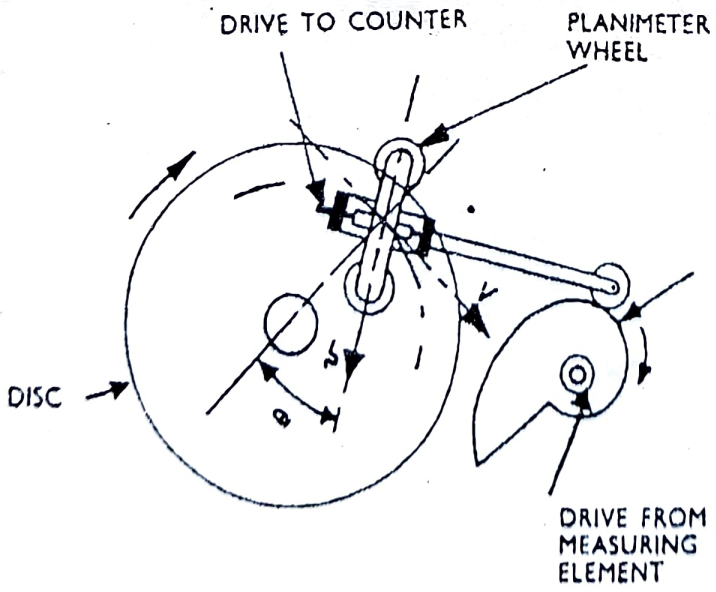


FIG ... Disc pattern integrator

If a disc is rotated at constant speed, the linear velocity of a point on a radius increases in direct proportion as one proceeds from the centre of the disc to the circumference. A small wheel placed in contact with the disc, its plane at right angles to the disc, will be rotated at rates of revolution proportional to its movement along a radius,assuming no slip. Now suppose the radial movement of the wheel is made directly proportional to the rate of flow, by transmitting its revolutions to a counter a form of integration is indicated.

The flowmeter mechanism rotates a cam, such that the angular displacement of this is proportional to the differential head. In contact with the edge of the cam is a roller mounted on a pivoted arm which, as it deflects, causes the planimeter wheel to rotate about its point of contact with the clock-driven disc.

Indicting the tangential velocity of the disc at its point of contact with the planimeter wheel by V and the angular displacement of the plane in which the planimeter wheel rotates from a plane passing through both the centre of rotation of the disc and the point of contact between it and the planimeter wheel by Θ, then $v = V \sin \Theta$, where v is the tangential velocity of the planimeter wheel. Since the value of V is constant, v is directly proportional to sin Θ. By cutting the cam to a profile which results in sin Θ being directly proportional to the square root of the differential head, the value of v is made directly proportional to the rate of flow.

Through a train of gears, the revolutions of the planimeter wheel are transmitted to the dials of the counter, and these dials register the total quantity passed, in any desired units.

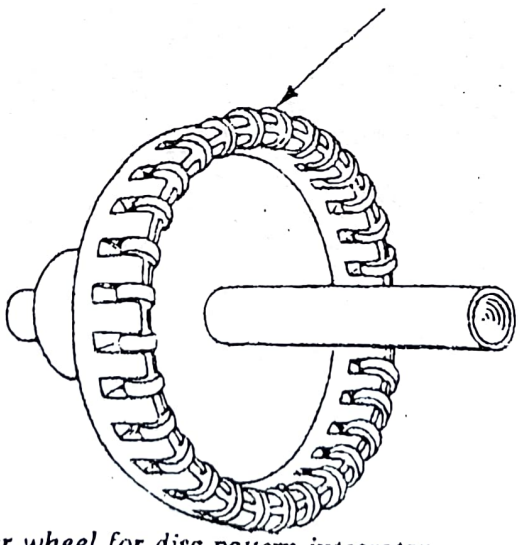


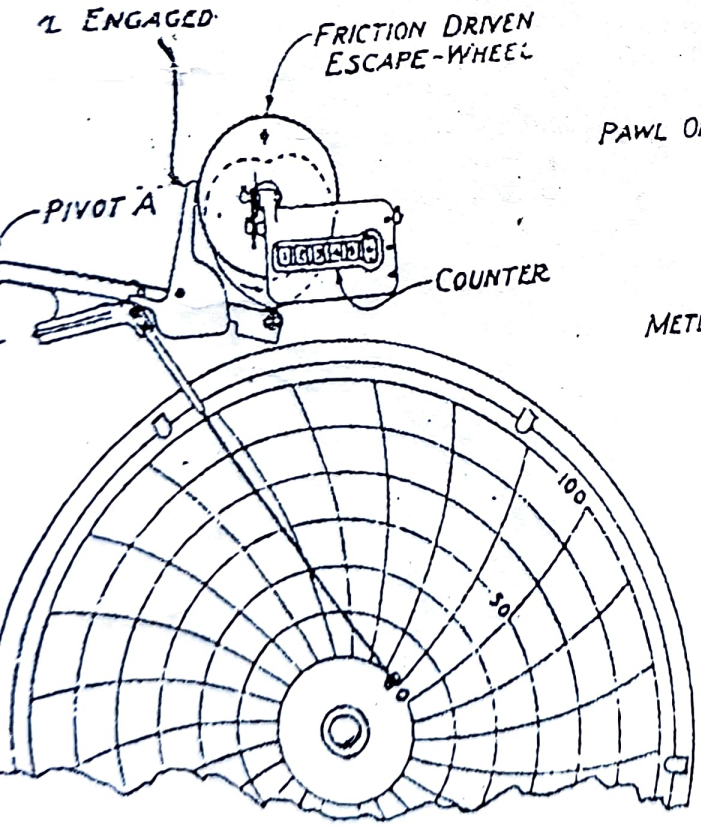
FIG ... *Planimeter wheel for disc pattern integrator*

It will be seen that this wheel is fitted with a number of serrated wheels round its circumference which are free to rotate in plane at right angles to the plane of rotation of the planimeter wheel.

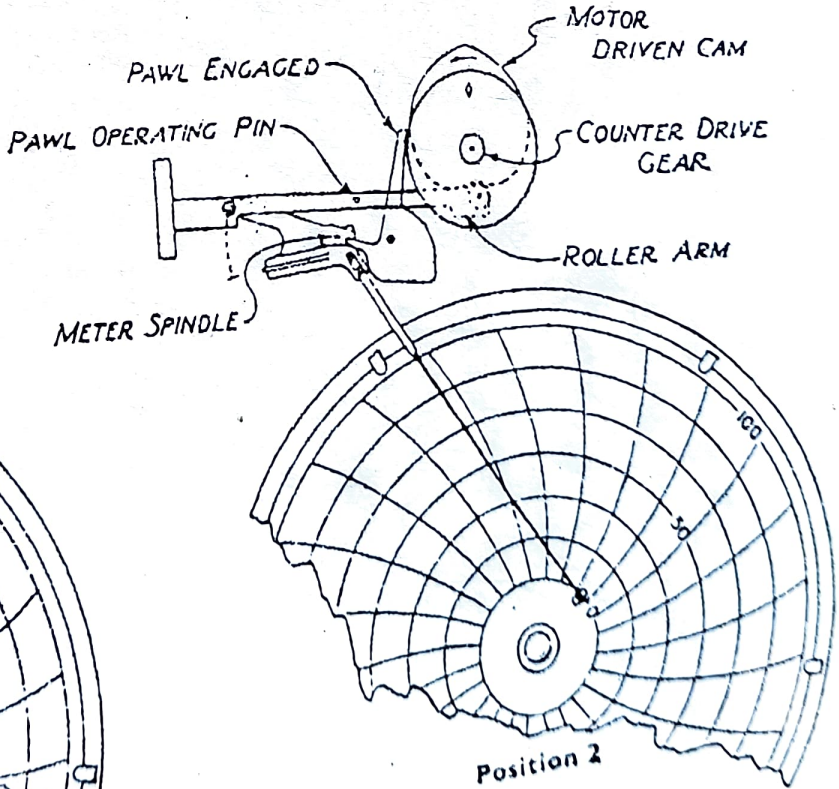
Intermittently operated mechanical integrators consist of a heart shaped cam. A roller arm follows the cam with a little roller mounted onto the end of the arm.

The cam is driven by a synchronous motor at two revolutions per minute, and the roller is kept in continuous contact by the weighted opposite end of the arm. The roller arm carries a pin which engages with a rod. The rod, in turn, combines with a toothed escape wheel driven by means of a friction clutch from the heart shaped cam. The escape wheel is geared to the indicator (counter). As long, therefore, as the wheel is rotating the counter is operated, and when the wheel is not rotating, the counter is at rest. At zero flow, the counter will not be operated, and at maximum flow it will be continuously operated. With intermediate flows, the periods of operation and non-operation must be adjusted proportionately.

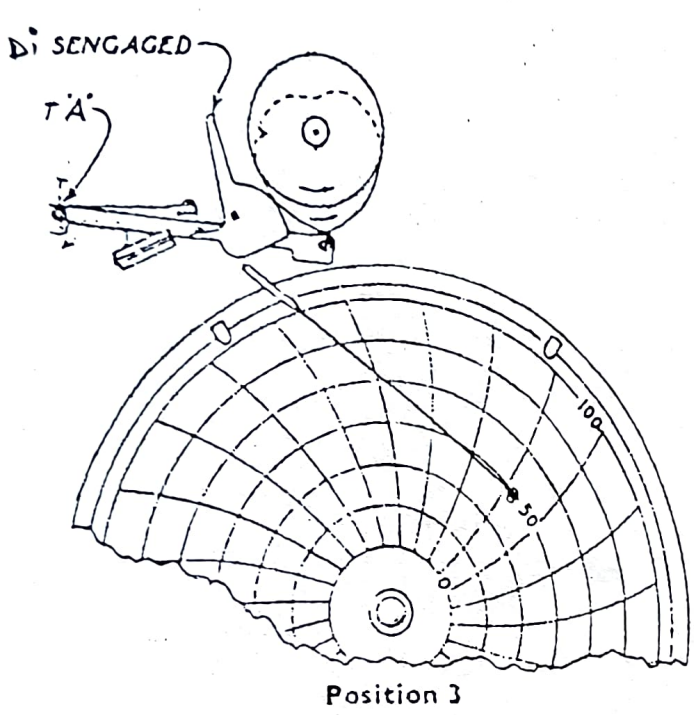
Integration is carried out at the rate of four times per minutes.



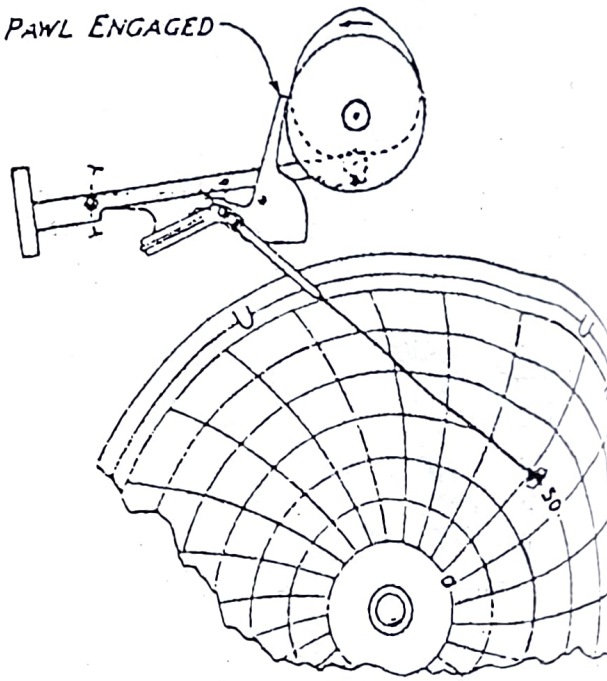
Position 1



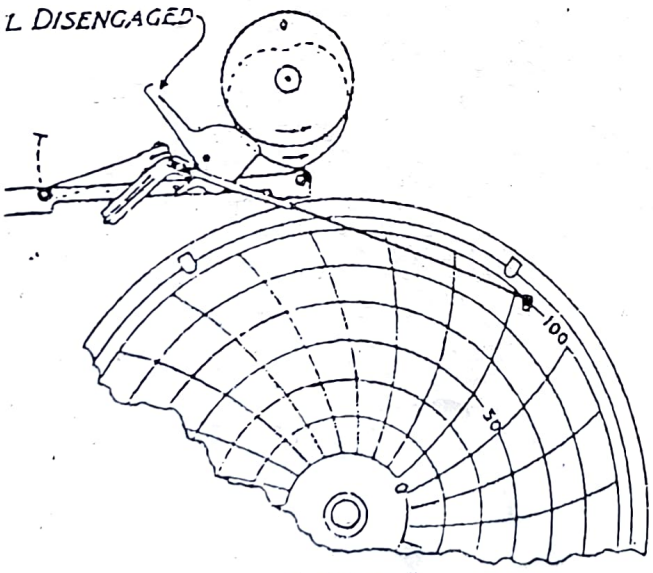
Position 2



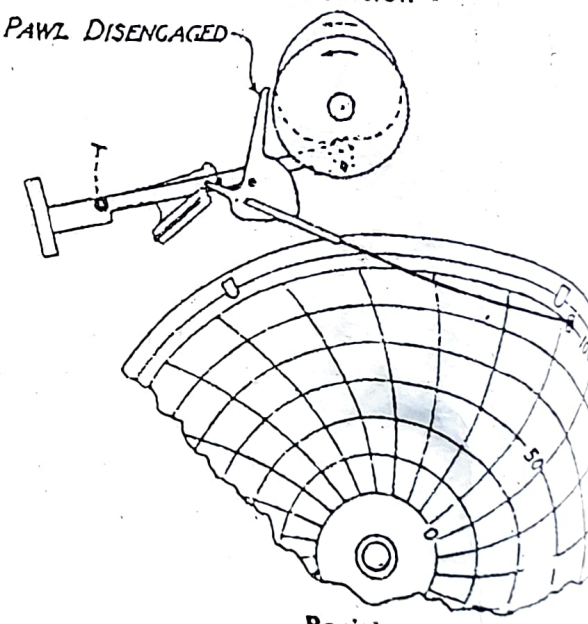
Position 3



Position 4



Position 5



Position 6

FIG ... *Operation of intermittent escapement pattern integrator*

B. ELECTRICAL INTEGRATORS

Electrical flow meters makes use of the kilowatt hour meter's principle of operation.

The normal a.c. electrical kilowatt hour meter, such as is installed in the average home using electricity.

The instrument possesses two operating coils or magnets, one connected to the circuit voltage and the other carrying the circuit current. If a metal disc is placed in a given manner in the field of these two coils, eddy currents are induced, and the inter-action of the fluxes due to the two coils and the eddy currents produces a driving torque T_D which rotates the disc.

There is also a permanent magnet equiped on the meter between whose poles the disc rotates. This magnet produces a decelerating torque which is proportional to the rate of revolution N of the disc.

Gearing the disc, therefore, to a normal type of counter will give us a means of continuous electrical flow integration, since the rate if revolution will be a direct measurement of flow. It will be obious, however, that the effect of variations in circuit voltage and the supply frequency must be reduced to a minimum, otherwise errors will be added to the measurement.

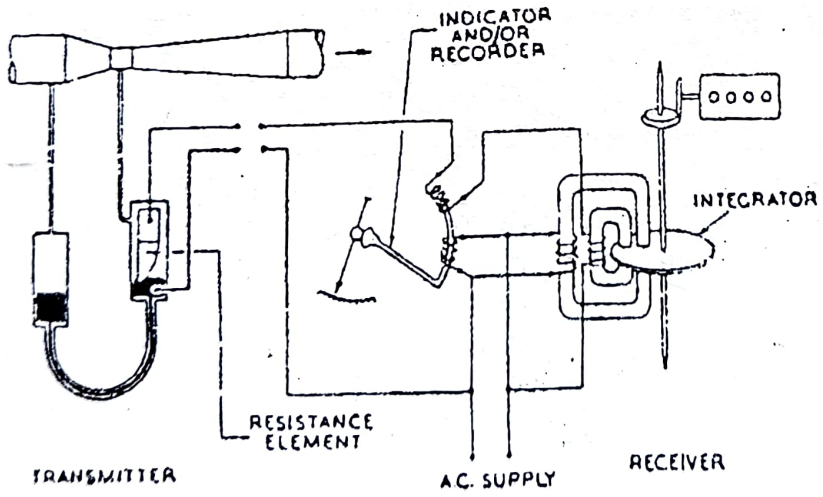
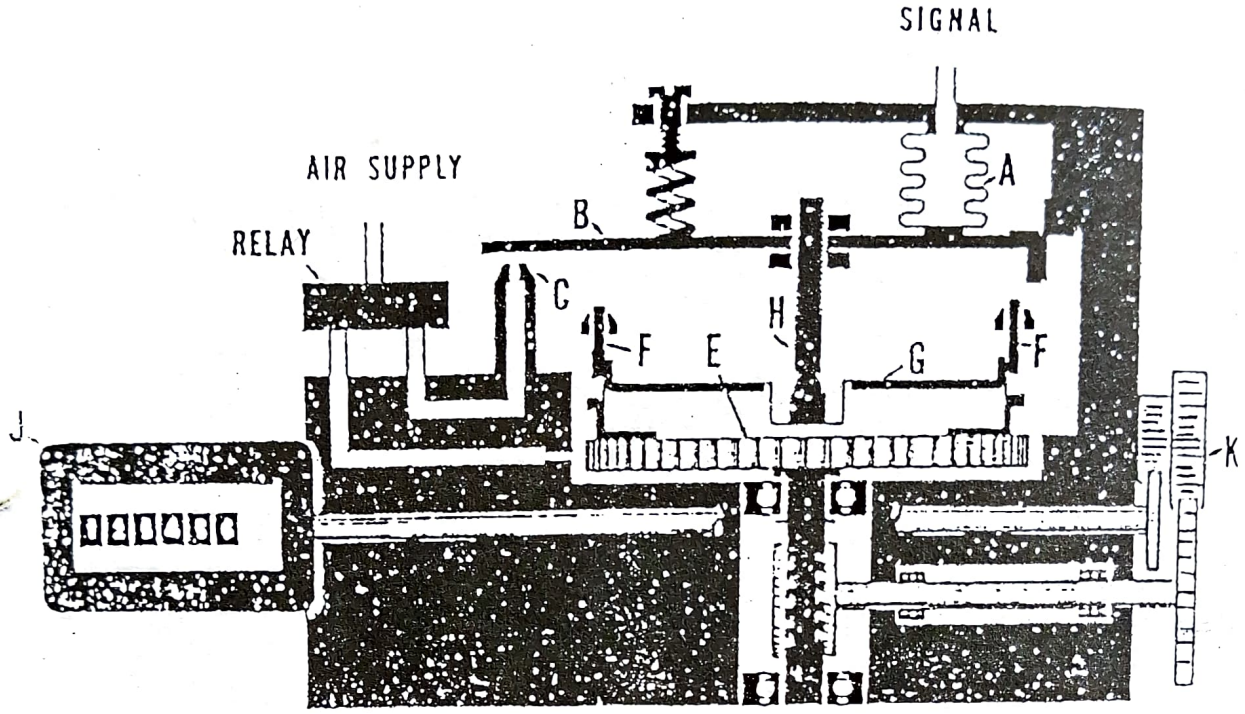


FIG ... KW hour meter integrator

C. PNEUMATIC INTEGRATOR

This type of integrator is a pneumatic instrument whose input is a 20 to 100 kPa signal. The integrator mechanically extracts the square root of this signal and totalizes the result

on an integrally-mounted counter. Since pneumatic signals from head-type flowmet are proportional to the square of the flow, the integrator can read directly in flow un The counter gears can be selected so that the flow is indicated in the desired units.



PRINCIPLE OF OPERATION

A pneumatic signal, proportional to 0 to 100 % of differential pressure from a flow transmitter, is applied to the integrator receiver bellows (A). The force exerted by the bellows positions a force bar (B) in relation to a nozzle (C). With an increase in differential pressure the force bar approaches the nozzle; and the resulting back pressure at the relay regulates the flow of air to drive the turbine rotor (E). As the rotor revolves, the weights (F), which are mounted on a cross-flexure assembly (G) on top of the rotor, develop a centrifugal force. This force feeds back through the thrust pin (H) to balance the force exerted on the force bar by the bellows.

The turbine rotor is geared directly to the counter (J) through gearing (K). Changes in flow continuously produce changes in turbine speed to maintain a continuous balance of forces.

The centrifugal force is proportional to the square of the turbine speed. This force balances the signal pressure which, for head-type flowmeters, is proportional to the square of the flow. Therefore, turbine speed is directly proportional to flow; and integrator count, which is a totalization of the number of revolutions of the turbine rotor, is directly proportional to the total flow.

1.7.3 INFERENTIAL FLOW METERS

Definition

A flow-measuring device which determines indirectly the mass flow by either density or velocity compensation(s) to volumetric flow rate devices.

Inferential flow meters are placed directly into line with the flow and measures the rate of flow. The output of these meters are directly proportional to the actual flow rate.

1.7.3.1 Types of inferential flow meters

A. MAGNETIC FLOWMETER

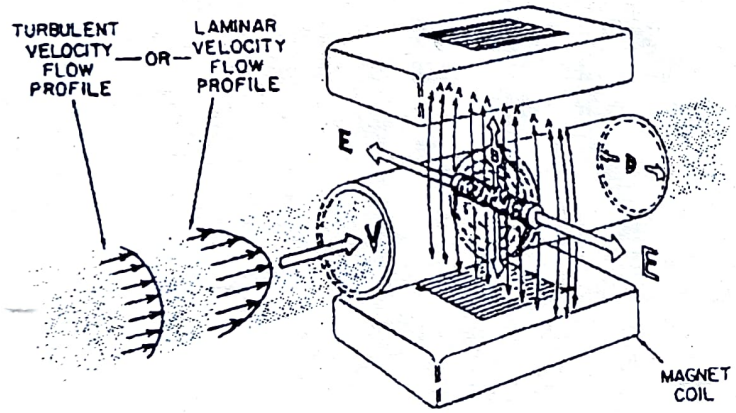


FIG ... Magnetic flowmeter

The electromagnetic flowmeter is capable of measuring the volumetric flow rate of conducting type liquids.

Two electro magnetic coils are used to create a uniform magnetic field. The coils are generally series connected but may be parallel connected if the measured flow velocity is low. The output from electrodes isaon alternating voltage at supply frequency. This low-level voltage is generally between 1 mV and 30 mV at full flow rate. This low level alternating voltage must be measured and converted either into a record or a display.

This signal, typically 4 mA zero and 20 mA at full scale, can be fed into a recorder or controiler.

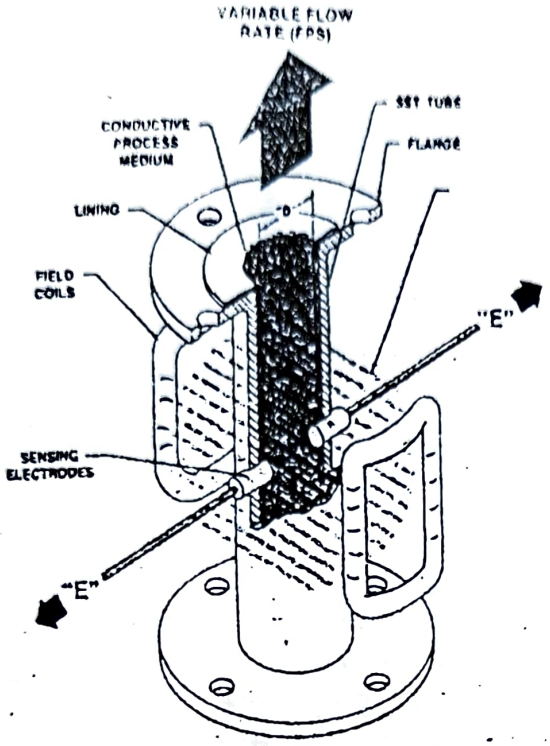
PRINCIPLE OF OPERATION

This meter works on faradays law for magnetic induction: When a conductor cuts a magenetic field an emf is induced in the conductor. The magnitude of the emf depends on the strength of the magnetic field, the length of the conductor, and the speed with which the conductor cuts through the magnetic field. If the strength of the field and the length of the conductor are kept constant, the induced emf then depends on the velocity of the conductor.

A constant magnetic field is provided by a steady direct current flowing through the coil of an electromagnet that surrounds the pipe. The fluid, which must be a conductive material, flows through a nonmetallic, non magnetic section of pipe. The effective length of the conductor corresponds to the inner diameter of the pipe. Since the diameter is constant, the length of the conductor is constant. Thus, with the length of the conductor and the magnetic field both constant, the induced emf depends on the velocity with which the conductor cuts through the field. The induced voltage in the conductor is picked up by a means of two electrodes set in opposite sides of the nonmetallic section of pipe. The rate of flow is determined by measuring that voltage. To determine the voltage the following equation can be used:

Generated $emf = \beta l v$

where β = flux density
l = length between electrodes
v = velocity of fluid (conductive)



USES OF THE MAGNETIC FLOWMETER

The magnetic flowmeter measures volume rate at the flowing temperatures independent of the effects of viscosity, density, turbulence or suspended material. Most industrial liquids can be measured by electromagnetic flowmeters. Water, acids, bases, slurries, liquids with suspended solids and industrial waste are common application. The limitation is the electrical conductivity of the liquid.

Advantages

- Can handle slurries.
- Cap handle corrosive fluids.
- Has very low pressure drop.
- No obstruction in pipe.
- Available in many construction materials.
- Available in large pipe sizes.
- Piping configurations not critical.
- Bidirectional flow measured by reversing connections manually or automatically.
- Measurements unaffected by viscosity, density, temperature or pressure.
- Can measure turbulent or laminar flow.
- Unaffected by conductivity changes as long as the minimum conductivity is maintained.

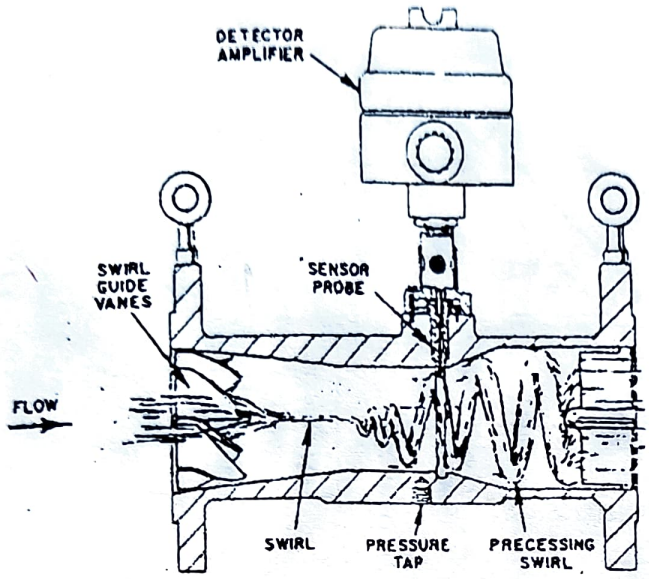
Disadvantages

- Conductivity must be as high as the minimum required by the particular manufacturer - from 0.1 to 20 micro ohms.
- Meters must be full at all times.
- Relatively high in cost.
- In-line mounting required.
- Meters must be explosion proof when installed in hazardous electrical areas (intrisic safe).

B. SWIRL FLOWMETER

The swirl flowmeter is basically a velocity sensitive device that measures the volumetric flow of gases and liquids. The meter introduces a rotary motion into the flowing fluid; thus a vortex flow pattern is formed. A pipe enlargement causes vortex precession. The rate of precession (spinning body round another axis) is proportional to the volumetric flow rate.

Fixed swirl blades impart a swirling motion to the fluid as it enters the meter. As the swirling fluid goes through a throat section in the meter body to an enlargement section, the centre of the swirl becomes displaced from the meter body centre line and precesses about in at angular velocity proportional to the flow rate. Thus the centre of the swirling fluid follows a helical path as it proceeds downstream through the enlargement. The helical path (precession) causes fluctuations in the pressure and velocity of the fluid. Velocity fluctuations are sensed by a velocity sensitive thermistor mounted near the wall in the enlargement section of the flowmeter. The thermistor is electrically heated to a temperature above that of the flowing fluid. As the velocity past the probe fluctuates, thermistor temperature and thus resistance fluctuates, and a square wave voltage output is obtained.



Advantages

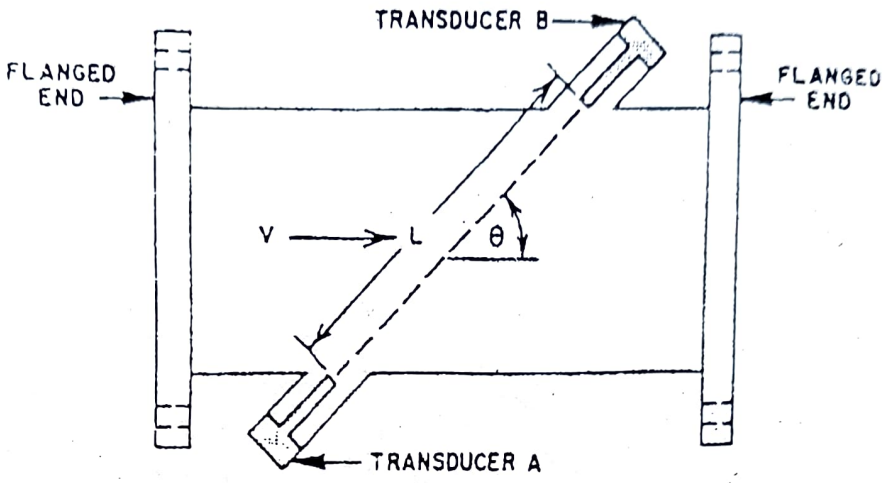
- No moving parts.
- Low pressure losses.
- High accuracy.

Disadvantages

- Expensive.

C. ULTRASONIC FLOWMETERS

For a number of years a great amount of effort has been expended to develop flowmeters that operate on the principle of measuring the velocity of sound as it passes through the fluid flowing in a pipe. The most common approach to this problem is described below:



Piezoelectric crystals (barium titanate or lead zirconatetitanate) are used as transmitte to send acoustic signals through the fluid flowing through the pipe to receivers that a also piezoelectric crystals. The fluid flows through the pipe at a velocity V. the distan between each transmitter-receiver pair is X. The velocity of the sound through the flu is Y. The path of the sound is at an angle ∞ from the pipe wall. The velocity of sou from transmitter A to receiver A (increased by the fluid velocity) is $Y + V\cos\infty$, and frequency is:

$$f_A = \frac{Y + V\cos\infty}{X}$$

where ∞ = sound angle
Y = sound velocity
x = distance between *Tx* and *Rx*
V = fluid velocity

The velocity of sound from transmitter B to receiver B (reduced by fluid velocity) is $V\cos\infty$, and its frequency is:

$$f_B = \frac{Y - V\cos\infty}{X}$$

The difference frequency Δf is:

$$f_A - f_b \text{ or } \Delta f = \frac{2V\cos\infty}{X}$$

Since ∞ and X are constant, the flow velocity can be obtained by measuring this beat frequency.

Advantages

- No pressure loss.
- Can be used with liquids, gasses and solids.
- Bi-directional.
- Very accurate.
- Pipe diameter has no affect.

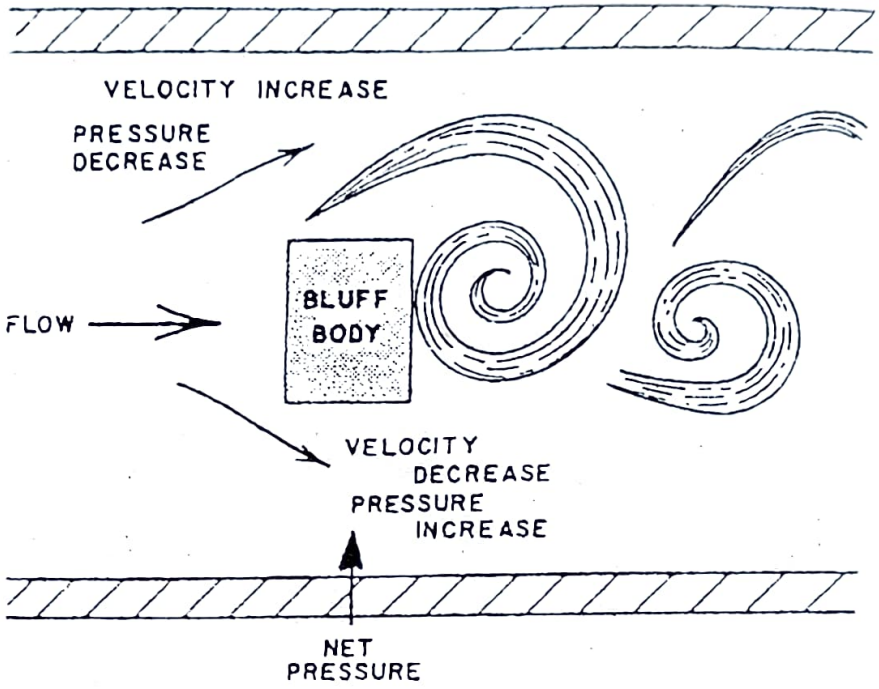
Disadvantages

- Expensive.
- Limited range.

D. VORTEX SHEDDING FLOWMETERS

The phenomenon of vortex shedding can occur whenever a non-streamlined object called a bluff body, is placed in a flowing stream of gas or liquid. The flow will follow the irregular contour at low approach velocities. However, as velocity increases the flow can no longer follow the sharp contours, and the stream becomes separated from the body. Separation takes place because of the frictional forces acting on the fluid boundary layer next to the bluff body. The fluid particles in the boundary layer decelerate because of the loss of energy due to the friction. As the fluid particle decelerates, it stops and reverses to accommodate the pressure feed. The reversing motion forms a vortex or eddy, which grows in size, separates from the surface, and moves downstream.

There is led to the relationship that the fluid velocity approaching the bluff body is proportional to the frequency that the vortices break away from the body multiplied by the diameter of the body. Since the dimensions of the body and flow conduit are constant for a given flow meter, the frequency is proportional to the volumetric flow rate. Sensors for the detection of the shedding frequency are based on the detection of temperature, strain, pressure, or motion.



Advantages

- Excellent rangeability.
- No moving parts.
- Within the linear range density, temperature, pressure and viscosity variations has very little effect on it.
- Very low pressure drop.

Disadvantages

- Limited application data.
- In-line mounting required.
- Limitations imposed on upstream and downstream piping requirements.
- Expensive.

TARGET FLOWMETER

The target meter combines in a single unit an annular orifice and a force-balance transducer. Output is either electric or pneumatic signal proportional to the flowrate.

The target is formed by a circular disc supported in the centre of a tubular section. Flow through the open ring between disc and tube develops a force on the disc proportional to the square of the flow. The disc is mounted on a rod passing out through a flexible seal (diaphragm seal). The force on the disc is measured from the rod outside the seal using a standard force-balance transducer mounted on the flow tube (flapper, nozzle assembly).

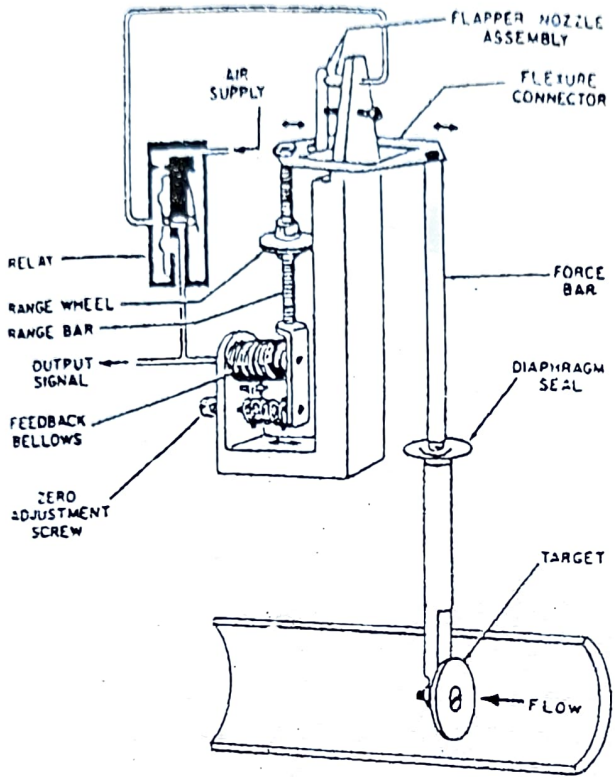
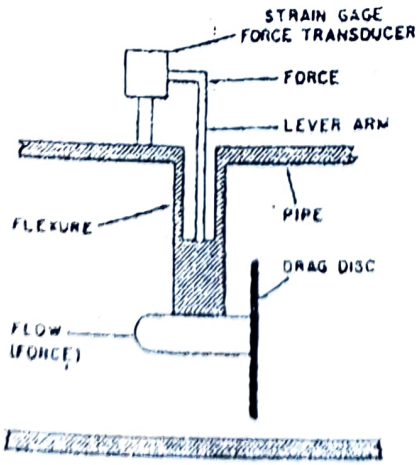


FIG. Target meter

F. DRAG BODY FLOWMETER

Another variation of the target meter is a drag body flowmeter.

A similar operating principle applies to the drag body flowmeter which detects the impact forces by a strain gungy circuitry. The difference between the two is that the drag body flowmeter has only an electrical output signal.



Advantages

- Measures liquids, vapours and gasses.
- Connections doesn't "freeze" (ΔP).
- Can be used for high temperatures.
- Less sensitive to change in Reynolds number.
- Very accurate.

Disadvantages

- Can only zeroed under no-flow conditions.
- In-line mounting required.

G. HEAT TRANSFER FLOWMETERS

Heat transfer flowmeters measures the rise in temperature of the fluid after a known amount of heat has been added to it.

The theory of the heat transfer flowmeter is based upon the specific heat equations.

$$Q = W_{C_p} (T_2 - T_1)$$

BTU - BRITISH THERMAL UNIT

where
Q = heat transferred (BTU/hr)
W = mass flow rate of fluid (kg/hr)
C_p = specific heat of fluid (BTU/kg°C)
T_1 = temperature of the fluid before heat is transferred to it (°C)
T_2 = temperature of the fluid after heat has been transferred to it (°C)

Heat is added to the fluid stream with an electric immersion heater. The power to the heater equals the heat transferred to the fluid (Q) and is measured by a wattmeter. T_1 and T_2 are thermocouples or resistance thermometers. Since we know the fluid, we also know the value of its specific heat. Thus, by measuring Q, T_1 and T_2 we can calculate flow rate (W). T_1 and T_2 do not have to be recorded separately; they can be connected together so that the temperature difference ($T_2 - T_1$) is recorded directly.

We can either take the difference in temperature between the two couples as a function of the flow rate, or the power needed for the heater to keep the temperature constant (at one temperature) as a function of the flowrate.

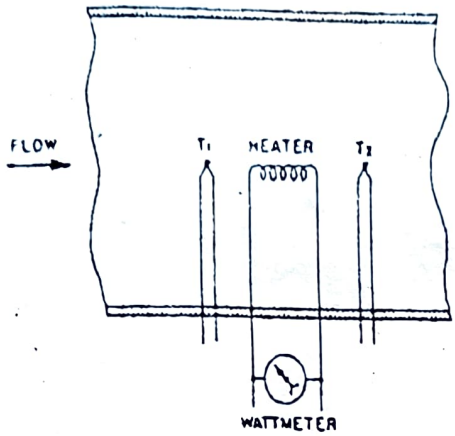


FIG ... *Heat transfer flowmeter*

Disadvantages

A flowmeter of this type of construction has many problem areas. The temperature sensors and the heater must protrude into the fluid stream. Thus, these components (particularly the heater) are easily damaged by corrosion and erosion.

To overcome these problems, the heater and the upstream and downstream temperature sensors can be mounted on the outside of the pipe wall.

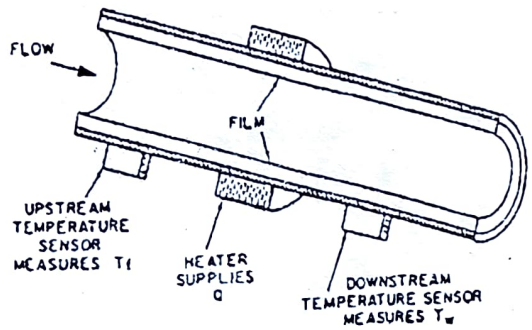
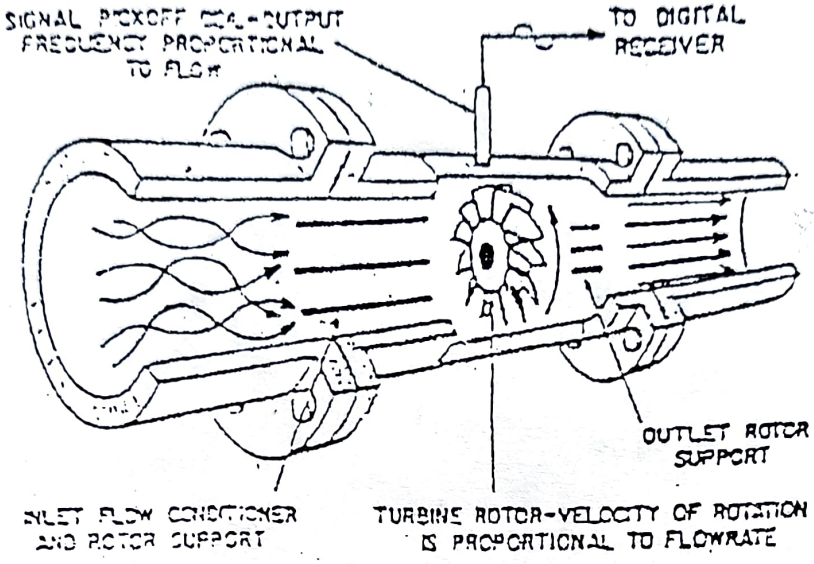


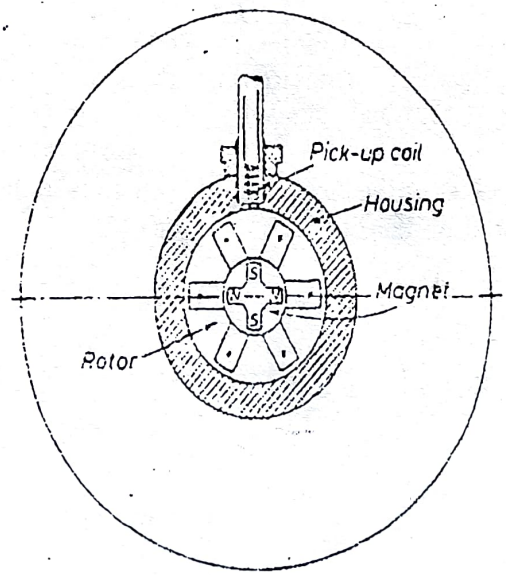
FIG *External elements and heater*

H. TURBINE FLOWMETERS

The turbine flowmeter provides a frequency output signal that varies linearly with a volumetric flow-rate over specified flow ranges with fluids of given kinematic viscosity.

The relationship between rotor speed and volumetric flow-rate is linear within given limits of flow-rate and fluid viscosity.





The pick-off assembly transduces the rotor velocity to an equivalent frequency signal. Variable reluctance type pick-off assemblies are the most commonly used. In this system the meter housing must be non-magnetic. The rotor must be of permeable material. The pick-off assembly consists of a small powerful permanent magnet and a coil winding. The field of the magnet is influenced by the moving blades of the rotor which are of permeable material.

As a rotor blade passes through the field of the magnet, it provides an easier path for the field and the field distorts thus moving across the coil winding. The relative motion between the magnetic field and the coil winding generates an AC voltage, the frequency of which is proportional to flow-rate.

A turbine flowmeter is to develop a precisely known number of pulses for a given volume measured.

APPLICATION INFORMATION

The following information is essential to the manufacturer to properly select a turbine flowmeter for a given application:

1. Process fluid should either be defined by name of its properties. Any trace chemicals and particles in the flow stream must be noted. These dates are necessary to select bearings properly.
2. The viscosity of the fluid over the operating temperature range.
3. The linear range.
4. The temperature and pressure over which the meter will be operated.
5. The process connections.
6. Signal transmission system,
7. The type of totalizer, rate of flow indicator.

SELECTION

The reasons for selecting turbine flowmeters for a measurement are varied and many depending on the particular application or installation, but the significant, recurring reasons are:

1. Ranges up to 150:1 are possible.
2. Performance including repeatability, absolute accuracy and linearity.
3. Retention of calibration.
4. Digital information.
5. Small size of given capacity.
6. Simple installation.
7. Base of maintenance.

The output signal from a turbine flowmeter is a frequency proportional to flumetric flow-rate. Therefore, each pulse generated by the turbine flowmeter is equivalent to a measured volume of liquid. Generally, turbine flowmeter information is tallied by totalizer type instruments. For the totalization to be valid, the value of each pulse must be essentially constant and, therefore, the turbine flowmeter must be linear.

Advantages

- Accurate.
- Repeatable.
- Good rangeability.
- Low pressure drop.
- Easy to install and maintain.
- Can be compensated for viscosity variations.

Disadvantages

- In-line mounting.
- Expensive.
- Limited use for slurry applications.
- Straight runs of pipe (15 diameters) required ahead of the meter.

I. VARIABLE AREA METER (ROTAMETER)

It is a vertical tube of conical shape, the area gradually expanding from the bottom to the top. In the tube is a fluid flowing in an upward direction, and in it is placed a disc, free to move, so that it acts as a sort of float in the fluid. An orifice is set up between the disc and the inside surface of the tube, and a pressure drop exists across it. Certain forces act on the disc and these are in equilibrium when it is at rest. A change in flow will affect the pressure drop. altering the relation between inlet and outlet pressure thus upsetting the equilibrium of the forces acting on the disc. The disc will then move up or down the tube adjusting the area of the orifice (due to the conical shape of the tube) until the pressure drop is at the original value, when the forces are again in equilibrium. The position of the float in the tube is then a measure of the rate of flow.

TYPES OF VARIABLE AREA METERS

a) *Glass Type*

The basic feature of this type of meter is the conical section glass tube. For accuracy, the diameter of this must be maintained at very close limits. Clear glass is used which is highly resistant to thermal shock and chemical action. The tube is normally clamped in a metal frame, the inlet and outlet being sealed into connections.

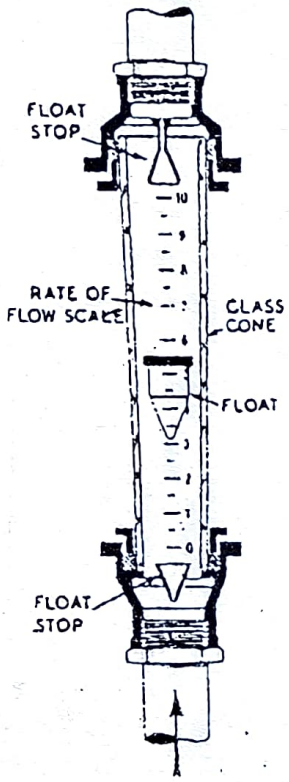
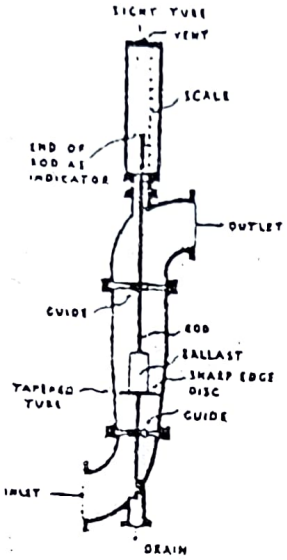


FIG *Diagram of glass body variable meter*

b) *Metal Tube Types*

For larger flows than the glass type tubes can accommodate, a conical metal tube pattern is introduced. Here, the metal body is of gunmetal, cast iron or stainless steel with a stainless steel float. The latter is carried on a rod which moves between two guides, one at the lower end and the other at the upper end of the tube. The guide rod passes through the upper part of the tube into a compartment with a glass scale, the end of the rod acting as an indicator.



FLOAT MOVES ROD WHICH MOVES AN INDICATOR IN A GLASS SCALE

FIG *Metal body variable area meter*

DENSITY VARIATION AND COMPENSATION

The density of a fluid can vary, e.g. with temperature changes, and errors will be introduced into the flow measurement.

The equilibrium of the float will be upset, and it will change to a new position. Errors will thus be introduced into the flow measurement. We require, therefore, that the change of rate of flow with density be zero.

The density of the float may be made very large, i.e. many times that of the fluid. This results in reducing the density errors to reasonable proportions.

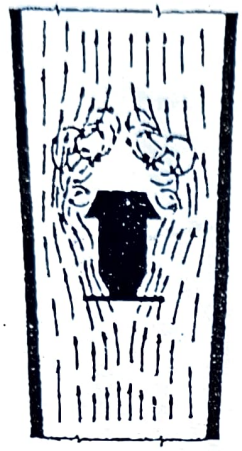
VISCOUS FLOW

At low flow rates, laminar or viscous flow conditions may exist. We must reduce the effect of viscous drag force to a minimum. SInce this is dependent on the effective length of the float, the reduction of this length should result in a lessening of the viscosity effect.

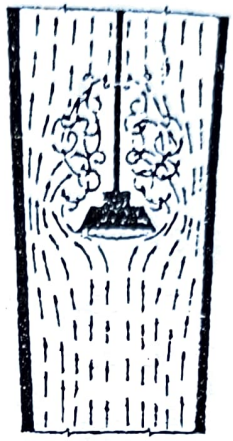
The effective orifice part of the float has been reduced to a sharp edged disc, i.e. with the minimum length along the direction of flow.

FLOAT SHAPES

Bullet shaped disc



Mushroom-shaded disc



Plug shaped disc

Float shapes depends on the type of fluids to be measured. If the fluids are very sticky of nature, spiral groves can be machined into the floats to cause a rotation. This rotating action will discard the sticky fluids from the float by centrifugal force.

Advantages

- Good rangeability.
- Relatively low cost.
- Good for metering small flows.
- Low pressure drop requirement.
- Can be used in some light slurry services.

Disadvantages

- Glass tube types subject to breakage.
- Not good in pulsating services.
- Must be mounted vertically.
- Generally limited to small pipe sizes (unless bypass rotameter is used).
- Limited to relatively low temperatures.

B. QUANTITY METERS

POSITIVE DISPLACEMENT FLOWMETERS

Positive displacement (PD) meters have the nature of separating streams into known volumes or segments, thereby measuring rated and/or total quantities. They are

classified as mechanical meters and have moving parts that can be damaged easily and that require some degree of maintenance.

Several types of displacement meters are available, including the notating disc, the reciprocating piston and the four bellows meter.

TYPES OF POSITIVE DISPLACEMENT FLOWMETERS

NUTATING DISC FLOWMETER

A movable element made up of a disc mounted on a ball, from the top of which is a shaft pedicular to the plane of the disc. A radial partition in the disc separates inlet from outlet and divides the metering chamber into four volumes, two above the disc and two below. The shaft is held in an inclined position by the disc, which is in contact with the chamber bottom on one side of the ball and the chamber top on the disc, causing the disc to nutate. For each cycle, the meter displaces a volume equal to the volume of the measuring chamber minus the volume of the disc. The axial shaft moves in a circular motion and operates the output shaft through a bevelled gear.

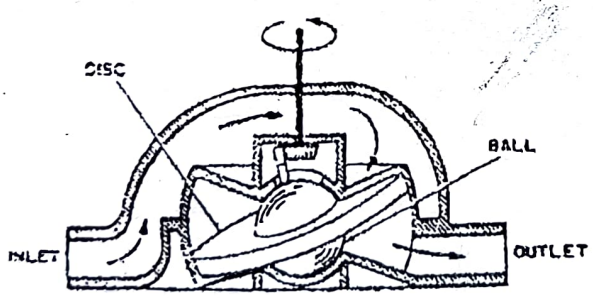


FIG *Nutating disc meter*

Advantages

- Relatively low cost.
- Acceptable pressure loss.
- Several construction materials available.

Disadvantages

- Limited to pipe size and capacity.
- Not 100 % accurate.
- Only clean fluids can be measured.

RECIPROCATING PISTON

One or more cylinders are alternately filled and emptied. One meter arrangement utilizes a double-acting piston and a slide valve. As the piston reaches one end of its stroke, the incoming liquid is directed by the slide valve into the cylinder. The pressure of the fluid forces the piston to the opposite end of its stroke as it fills the cylinder. Simultaneously, liquid in the opposite cylinder is forced out the outlet. Upon reaching the opposite end of the stroke, the slide valve moves to permit the opposite cylinder to fill from the inlet port, reversing the procedure. A crank arm actuated by the reciprocating action of the piston drives the output shaft.

The reciprocating piston meter is the most complicated and thus the most expensive liquid positive displacement meter. It requires a high-pressure drop and operates in noncorrosive, lubricating liquids. It is the most accurate positive displacement meter.

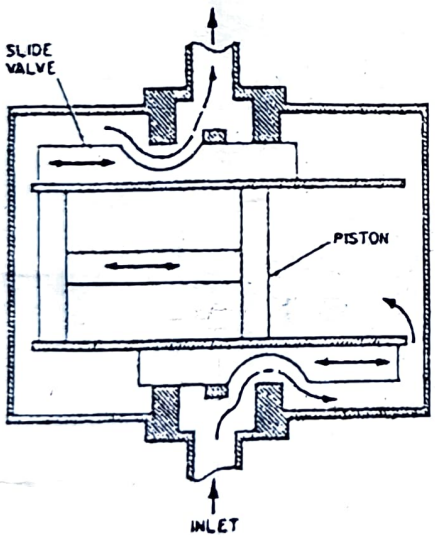


FIG *Reciprocating piston meter*

Advantages

- Very accuracy.
- Construction materials not limited.

Disadvantages

- Relatively high cost.
- Subject to leakage.
- Dirty fluids present problems.
- Maintenance costs may be high.
- Restricted to moderate flow rates.

The bellows meter consists of a sheet metal, gas-tight housing divided into two sealed compartments. Each sealed compartment is again divided into two sealed chambers by a flexible gas-tight elastomeric bellows. This configuration has lead to its designation as a dry positive displacement four-bellows meter. Slide valves are used to direct flow into and out of the four chambers.

The slide valves are moved to one of three positions by linkages which maintain their synchronism with the bellows. If the inside and outside of the bellows in one compartment are designated chambers 1 and 2, then the slide valve can admit inlet gas to chamber 1 and exhaust chamber 2 to outlet, admit to 2 and exhaust 1, or close both ports. An identical slide valve performs the same service for the second compartment. Thus the four chambers can be filled and emptied in sequence, using the pressure of the inlet gas to evacuate the outlet gas. The area of the bellows causes the meter to operate at very small pressure differences or a very small pressure drop.

The bellows meter can be used for any gas that does not attack the flexible bellows.

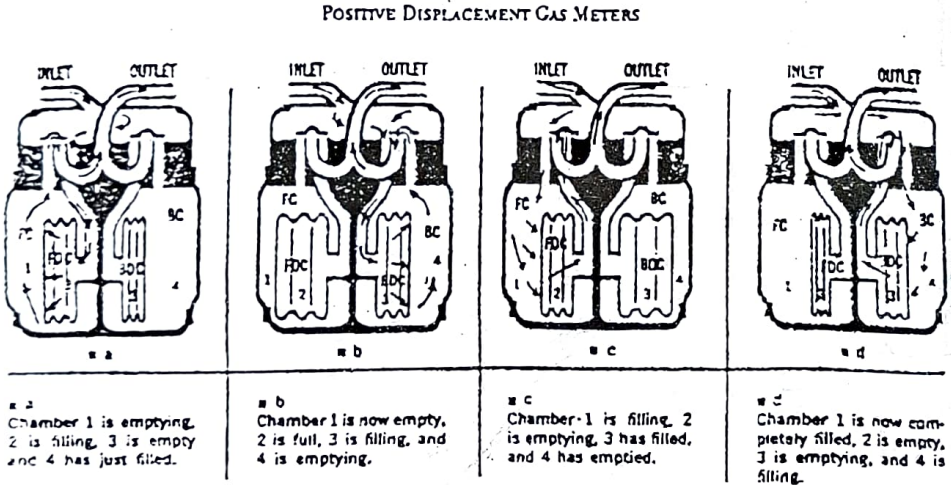


FIG *Four bellows meter*

Advantages

- Easy calibration.
- Linear readout.
- Very accurate.

Disadvantages

- High pressure drop.
- In-line mounting.
- High cost.

- Limited range.

FLOW MEASUREMENT IN OPEN CHANNELS

Devices used for measurement purposes fall into two classes: weirs and flumes. Both involve the production of a head of liquid proportional to the rate of flow.

WEIRS

Rectangular Weir

This type of weir comprises a metal plate with a rectangular slot cut in it in the manner of Fig ... The slot has sharp edges and the plate is installed at right angles to the direction of flow. The slot is narrower than the channel.

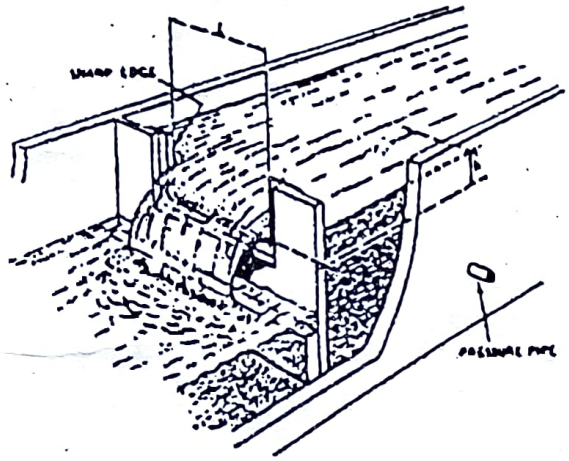
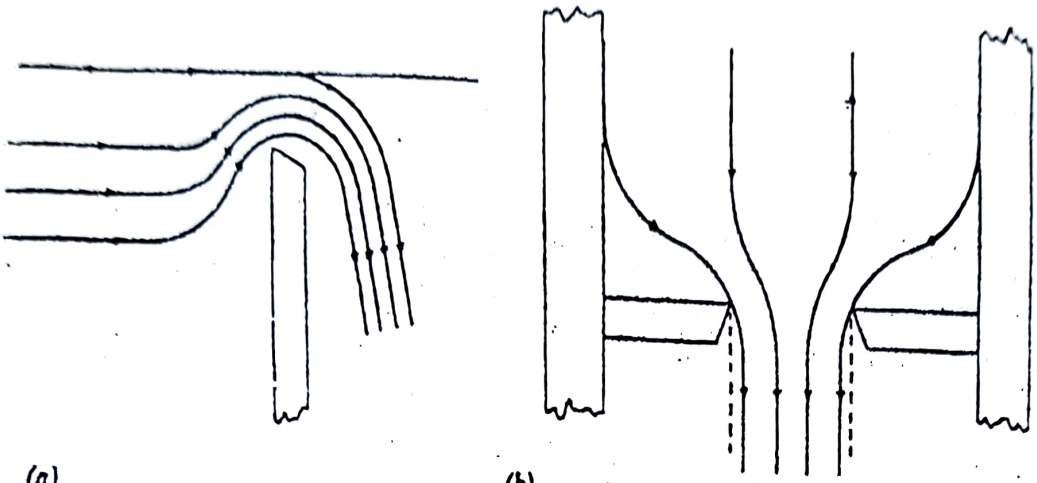


FIG Rectangular weir

In such a weir, the flow suffers reduction. There is a top and bottom reduction as indicated in (a) and there is, in addition, a side reduction as shown in (b) where the jet width after it leaves the weir is less than the weir opening. These factors necessitate the inclusion of a discharge coefficient C.



(a) (b)

CALCULATION OF RATE OF FLOW

To calculate the basic rate of flow we use the following equation:

$$Q = \tfrac{2}{3} BC\sqrt{2gH^3}$$

where Q = rate of flow (m^3/hr)
B = width of weir slot (m)
C = discharge coefficient
H_1 = the head over the weir (m)

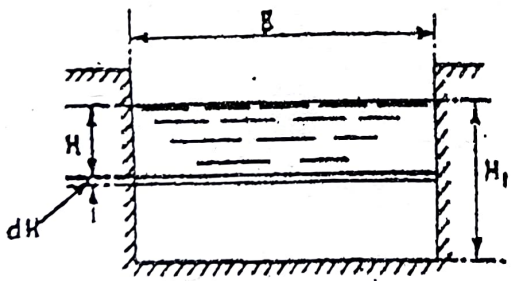


FIG *Method of calculating rate of flow for a rectangular type weir*

EXAMPLE

Calculate the rate of flow in a rectangular type weir with a width of 2m and a head of 750 mm. Take the discharge coefficient as 0,76.

$$Q = \tfrac{2}{3} BC\sqrt{2gH_1}$$

$$= \tfrac{2}{3}.2.0,76.\sqrt{2.10\ (0,75)^3}$$

$$= 2,943\ m^3/hr$$

V-NOTCH

In this form of weir, a metal plate with a sharp edged V-notch cut in it is installed at right angles to the direction of flow.

The formula may be derived, as in the case of a rectangular weir, by considering an element at depth H and integrating over the flow section. The basic formula is:

$$Q = \frac{8}{15} C \tan \frac{\Theta}{2} \sqrt{2gH^3}$$

Θ is normally between 35° and 120°.

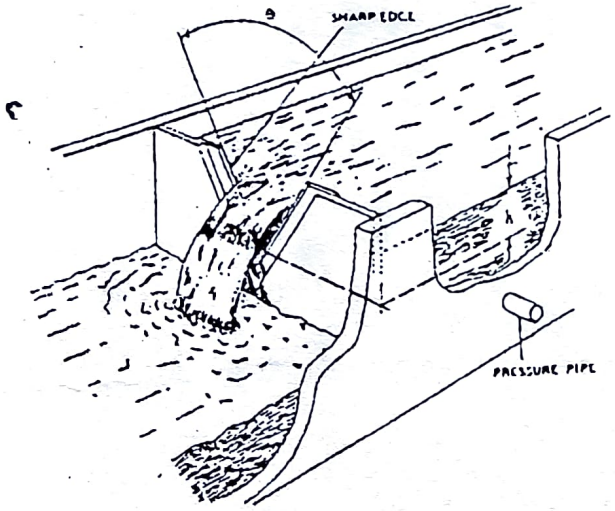


FIG V-notch weir

TRAPEZOIDAL OR CIPOLETTI WEIRS

In an effort to reduce the effect of side reduction the trapezoidal or Cipoletti weir is sometimes used. The sides are made sloping.

The angle Θ is obtained from the relation $\Theta = 2 \tan^{-1}(¼)$ from which Θ is approximately equivalent to 14°. A discharge coefficient is still necessary, however, to take care of other parameters and the relation between flow rate and head is given by:

$$Q = \frac{2}{3} CB \sqrt{2gH^3}$$

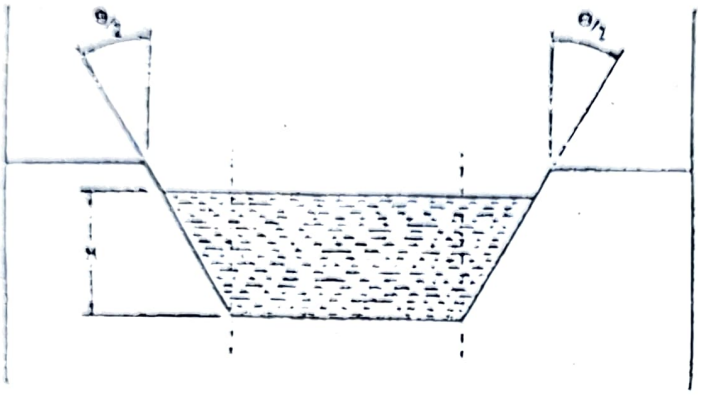


FIG Trapezoidal or Cipoletti weir

SUPPRESSED WEIRS

The suppressed weir comprises a plate with a sharp edge and extends the whole width of the channel. As its name indicates, it is entirely submerged or suppressed beneath the surface of the liquid. Obviously with such a design there is no side contraction effect.

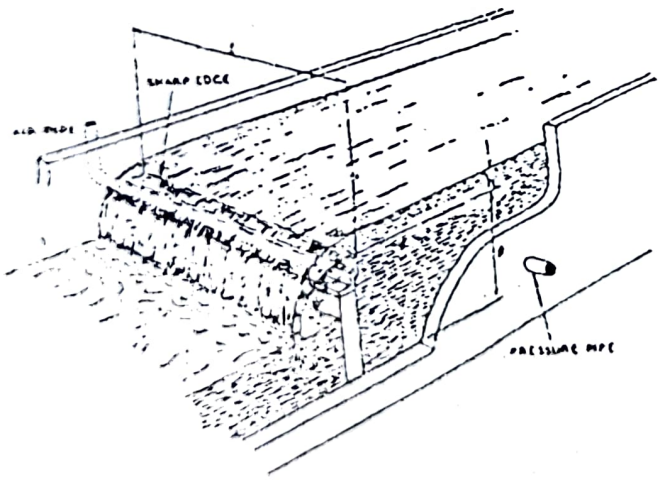


FIG Suppressed weir

OTHER TYPES OF WEIRS

Another type of weir is the hyperbolic one in which the sides are shaped to produce a linear relation between flow rate and head. Compound weirs are employed in many

cases where the installation of one particular type would not meet the operating conditions.

PRECAUTIONS TO BE OBSERVED IN INSTALLATION OF WEIRS

- To assist in uniform velocity distribution, the upstream channel should be relatively wide and deep.
- The upstream edge of the notch must be square and sharp.
- A free overfall must be maintained. This means that the level of liquid on the downstream side of the weir plate must be some distance below the crest of the weir or bottom of a V-notch. The bed and the sides of the channel downstream should not be less than 150 mm from the weir and the water level downstream should be at least 75 mm below the weir base.
- The measurement of the head over the crest of the weir or bottom of a V-notch must be made at some distance upstream from the weir plate. If H is the maximum, value of the head, then measurement should be made at a distance of not less than 6H from the upstream face of the weir.
- The underside of the nappe of the weir must be at atmospheric pressure. This means aerating for suppressed weirs.
- For suppressed weirs, the sides of the channel must be vertical, smooth and in a plane at right angles to the plane containing the weir for some distance upstream and downstream of the weir.
- The conditions of flow may necessitate the installation of a screen upstream from the weir to prevent the piling up of litter at the weir face.

FLUME TYPES OF DEVICES

VENTURI FLUMES

The Venturi flume is virtually a rectangular section Venturi tube with the upper surface removed. Thus we have the inlet contraction, throat and outlet expansion.

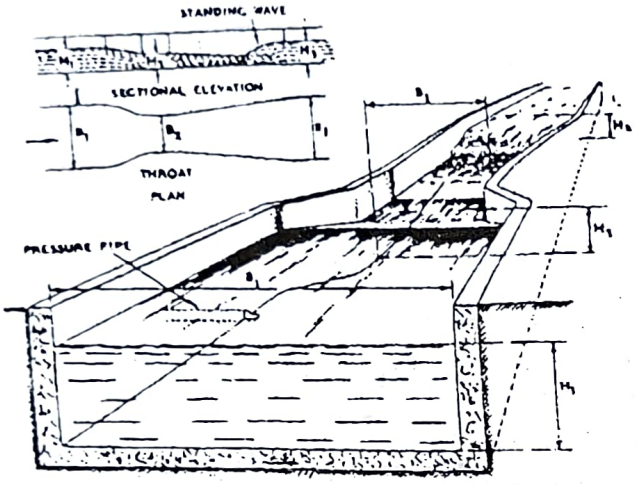


FIG *Rectangular section Venturi flume*

If Q is the rate of flow of the liquid through the flume then:

$$Q = KH_2 (\sqrt{H_1 - H_2})$$

where $$K = [C.B_2 \sqrt{2g} \sqrt{\frac{E^2}{E^2 - \alpha^2}}]$$

$H_1 \alpha H_2$ = hydraulic depths
C = discharge coefficient
B_1 = inlet width
B_2 = throat width
$E = \frac{B_1}{B_2}$
$\alpha = \frac{H_2}{H_1}$

If the value of E is made large enough then the last term becomes constant over a fairly wide range of flow rates. Then we can say:

$$Q \alpha H_2 \sqrt{H_1 - H_2}$$

Another type of flume is the U-section Venturi flume.

It is not necessary for the cross-section of a Venturi flume to be rectangular. U shaped sections are not uncommon, and trapezoidal forms have been used.

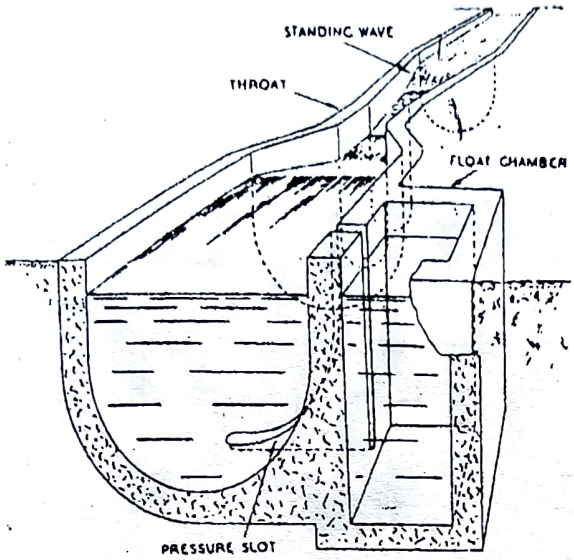


FIG U-section Venturi flume

CRITICAL FLOW CONDITIONS

A rough guide for the establishment of critical flow conditions in the throat is that H_3 / H_1 must be a maximum value of 0,8.

$$\frac{H_3}{H_1}$$

When the critical value of the ratio H_1 is exceeded, the flume is operating under "drowned" conditions.

A vertical contraction is sometimes necessary when the rates of flow in the lower part of the range are such that "drowning" will occur. The introduction of the hump ensures free flow conditions.

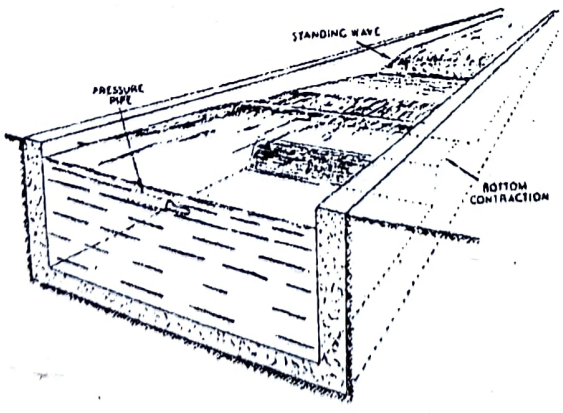


FIG ... Venturi flume with vertical contraction

2. DENSITY

2.1 INTRODUCTION TO DENSITY

Density can be defined as mass of a unit volume of a substance. In SI units it is the mass in kilograms of one cubic meter of the substance.

$$\text{e.g.} \quad \text{Density} = \frac{\text{Mass of Substance}}{\text{Volume of Substance}}$$

As we already know temperature has an effect on density. Liquid density decreases with the increasing of temperature and to a much smaller degree increases with pressure. The following formula can be used to determine the real density from the initial density:

$$\text{Real Density} = \frac{\text{Initial Density}}{1 \alpha 8 (T_1 - T_2)}$$

where T_1 = final temperature
T_2 = initial temperature
α = co-efficient of cubical expansion

2.2 SPECIFIC GRAVITY

Specific gravity or relative density is the number of times the mass of any value of a substance is greater than the mass of the same value of water under the same temperature and pressure.

$$\text{Specific gravity} = \frac{\text{Mass of any volume of substance}}{\text{Mass of the same volume of water}}$$

2.3 LIQUID DENSITY MEASUREMENTS

Many methods are available of which some are variations on those used for liquid level measurement on the basis that, if the level is kept constant, the measuring system will then be influenced by changes in density of the fluid. These include:

a) Buoyancy systems where a float or displacer is immersed in the liquid.

b) Hydrostatic systems of which the air bubbler method is widely used.

c) *Radiation systems*

Ore, one of the most important measurements accomplished with nuclear radiation, is the determination of the density of liquids and semi-liquids such as slurry and slime.

2.4 BUOYANCY METHODS

2.4.1 MEASUREMENT OF RELATIVE DENSITY AND DENSITY BY MEANS OF HYDROMETERS

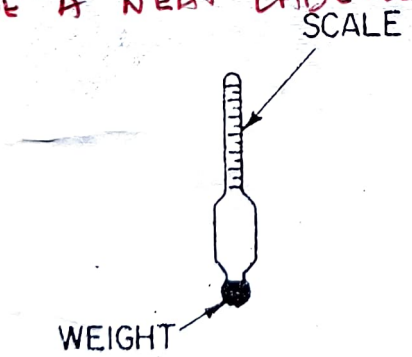
HHH Aug 2019

Hand held hydrometers are the most convenient and widely used devices for determining the specific gravity of relatively transparent liquid.

EXPLAIN WORKING PRINCIPLE OF A HHH (6) Aug 2019

The hydrometer displaces a volume of the unknown liquid until an equilibrium is reached between upward buoyant force and downward hydrometer weight force. By suitable graduation, calibration with known fluids and the addition of lead-pellet weights, the scale beading is calibrated to read either specific gravity or some related unit.

MAKE A NEAT LABELLED SKETCH OF A HHH (4) Aug 2019



2.4.2 FOTO ELECTRICAL HYDROMETER

The feed electrical method operates on the same principle as the hand held method.

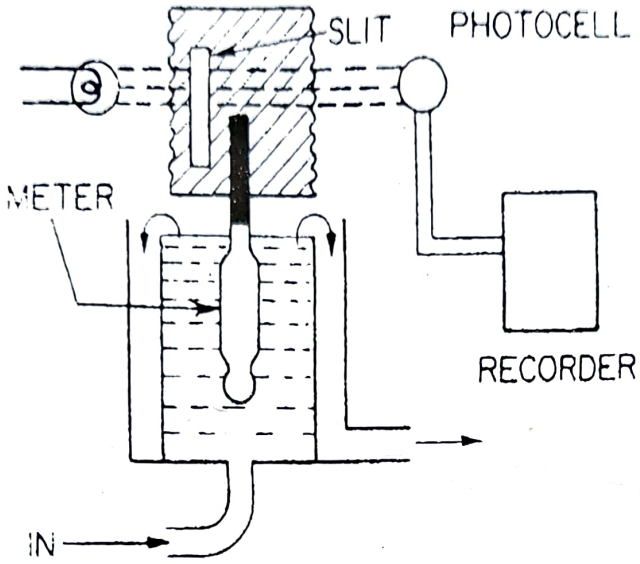


FIG Foto electrical hydrometer

A hydrometer with a dull stem is placed in a continuous flow vessel. The liquid must be kept at the same level. The liquid with a specific density will move the hydrometer to a certain level. The blacked-out stem will then block off a certain amount of light from the light source. These light rays will be picked up by a feed cell which in turn will send a signal to the recorder. The amount of light received by the fotocell is directly proportional to the density of the liquid.

2.4.3 INDUCTIVE BRIDGE HYDROMETERS

The inductive bridge method operates on the same principle as the previous methods.

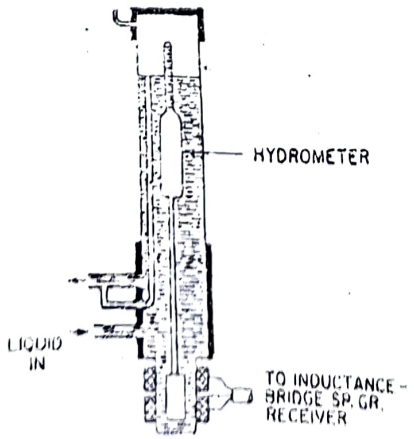


FIG Inductive bridge

A hydrometer with a iron-core fitted to the lower side of the hygrometer is placed in a continuous flow of liquid. The liquid with a specific density will move the hydrometer to a certain level. The iron core will be positioned between the two coils thus changing the inductance of the two coils accordingly. This changed inductance will be send to a inductive bridge which in turn will be send to a recorder. The inductance will be directly proportional to the density of the liquid.

2.4.4 CHAIN-BALANCED FLOAT

This instrument uses a submerged plummet which is self-centering, operates without friction, and is not affected by surface contamination.

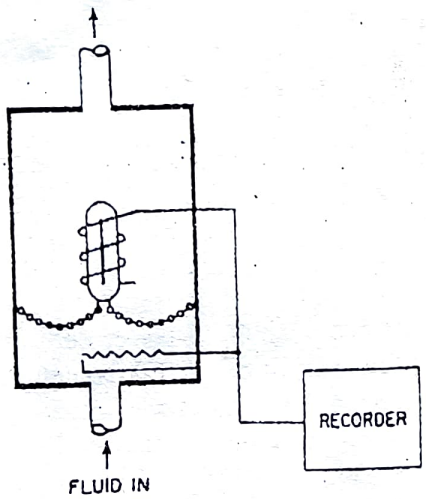


FIG Chain-balanced float

The volume of the body is fixed and remains entirely under the liquid surface. As the plummet moves up and down, the effective chain weight acting on it varies; and for each density within the range of the assembly, the plummet assumes a definite equilibrium point.

In order to transfer these changes in density, the plummet contains a metallic transformer core, whose position is measured by a pickup coil. This voltage differential as a function of plummet displacement, is a measure of specific-gravity change.

Measuring bridge notes the temperature change and affects a voltage across the recorder which is equal and opposite to the voltage transmitted to it by the pickup coil due to the temperature-induced density change.

2.4.5 DISPLACEMENT-TYPE HYDROMETER

Displacement-type operates on the same principle as the chain-balanced float method.

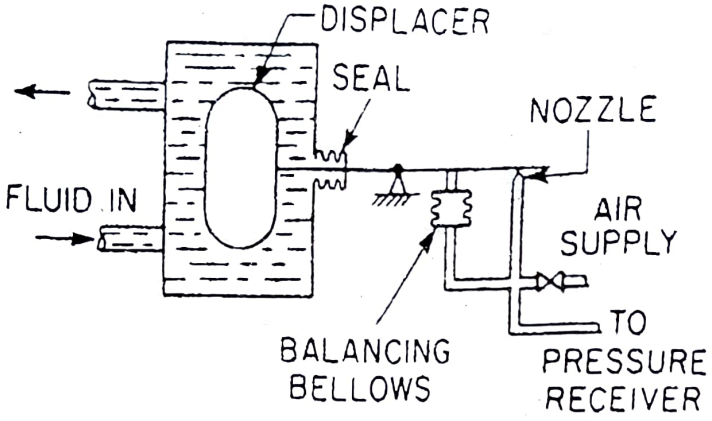


FIG Displacement hydrometer

In the displacement meter liquid flows continuously through the displacer chamber. An upward force acts on the balance beam because of the volume of liquid displaced by the float. A pneumatic system, similar to the one shown, balances this upward force and transmits a signal proportional to the density of the liquid. Liquids with specific gravities 0.5 and higher can be measured with this equipment as long as suitable materials are used to prevent damage from corrosion. If the temperature of the flowing liquid changes, a thermostatic heater may be used to hold it constant.

2.4.6 HYDROSTATIC METHOD

2.4.6.1 Hydrostatic Head

This method is suitable for a number of industrial processes in the system that continuously measures the pressure variations produced by a fixed height of liquid.

2.4.6.2 Principal of operation

The difference in pressure between any two elevations below the surface (A and B) is equal to the difference in liquid head pressure between these elevations. This is true regardless of variation in level above elevation A.

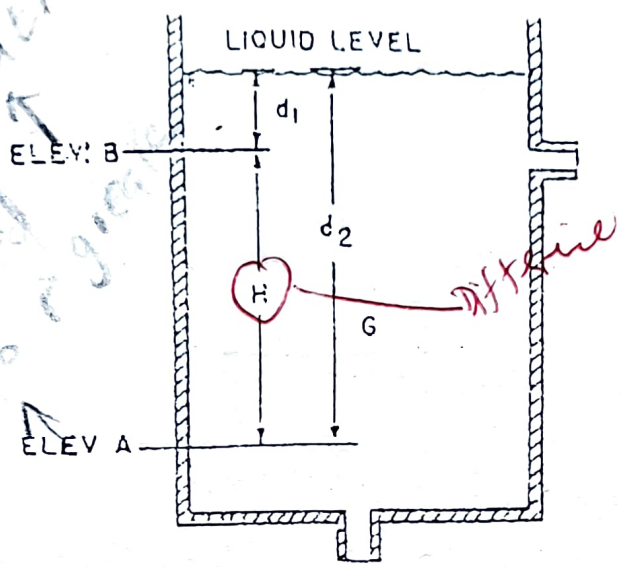


FIG ... *Fixed height of liquid for density measurement*

This difference in elevation is represented by dimension H. Dimension (H) must be multiplied by the specific gravity (G) of the liquid to achieve the difference in head in meters of water, which are the standard units for instrument calibration.

The resultant reaction between the different heads and pressures are collected by the differential pressure transmitter whose output may therefore be calibrated in density.

2.4.6.3 Types of Hydrostatic Head Hydrometer

DIFFERENTIAL HEAD HYDROMETER

The differential head type operates with two tubes placed at different levels inside a container in which air is passed through.

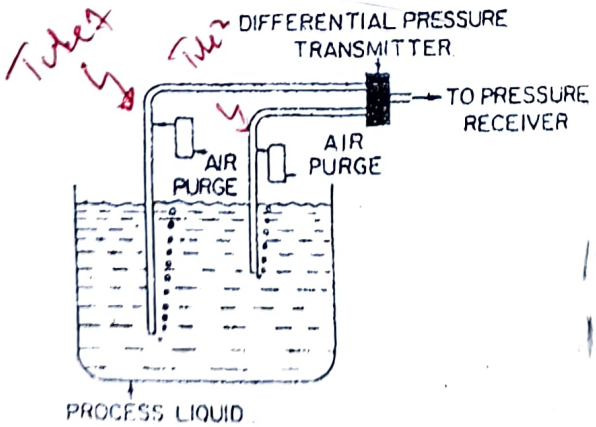
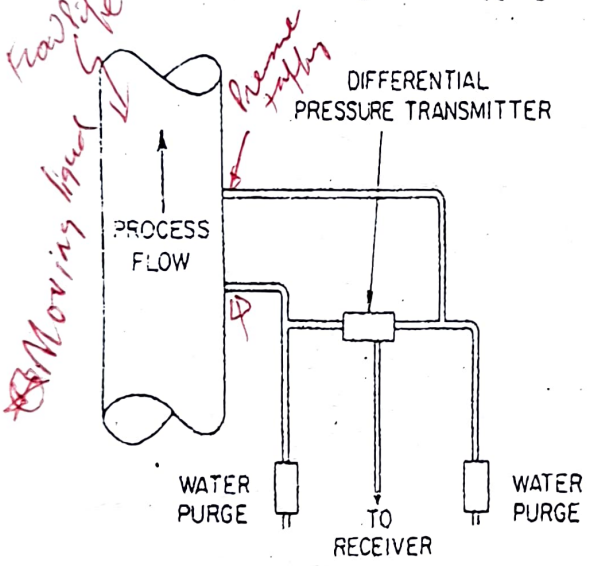


FIG ... *Differential head hydrometer*

A constant air is supplied to both the tubes. The pressure at P_1 will be higher than the pressure at P_2 thus more pressure is needed at P_1 for bubbles to form than at P_2. When the bubbles becomes visible the pressure put forth at the end of the tube by the liquid will be almost equal to the applied air pressure. The difference in pressure will then be directly proportional to the sum of a constant volume at each tube. This brings us back to the density equation, thus we can say that ΔP is directly proportional to the density.

The same principle is used to measure moving liquid's density. The flow pipe must be installed vertically with two pressure tappings at different heads.



DIFFERENTIAL HEAD RANGE HYDROMETER

This hydrometer operates on the same principle as the differential head hydrometer. The difference is that this hydrometer's range can be adjusted.

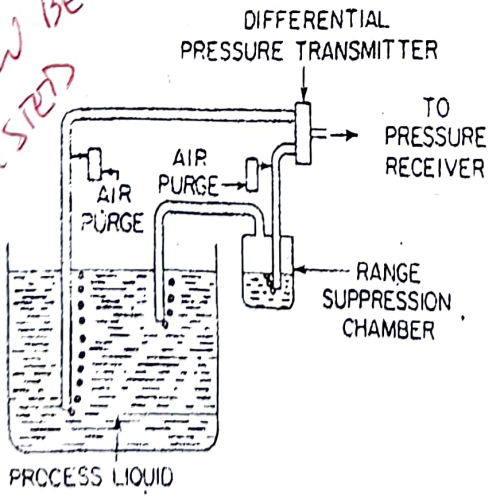


FIG ... Differential head range hydrometer

In the previous hydrometer pressure P_2 will start from 0 Pa upto a maximum. When the range suppression chamber is added to the system, the pressure at P_2 will not start from 0 Pa, but from the pressure obtained from the suppression chamber. The range can be adjusted by varying the height of the tube in the chamber for example, if the height of the tube in the suppression chamber is adjusted to obtain 20 Pa the P_2 pressure will start from 20 Pa up to maximum.

REFERENCE TUBE HYDROMETER

This hydrometer measure density of an unknown liquid by referring it to a known density, normally water. The difference in pressure will then be directly proportional to the unknown density

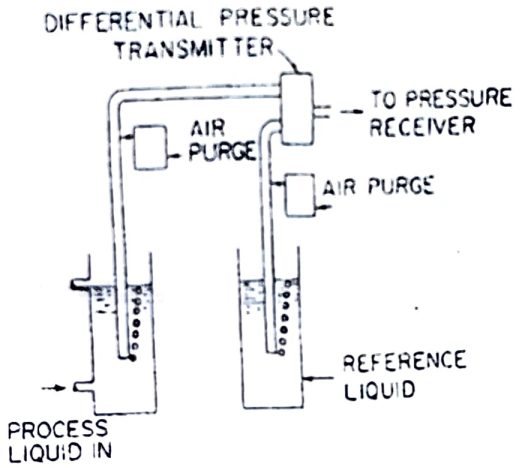


FIG ... *Reference tube hydrometer*

Both tubes must be placed at the same head, because P_1 must equal P_2 when the densities are the same. When a unknown density must be measured the pressure at the same head will differ from the reference density.

If P_1 is higher than P_2 the bubbles will appear at different times because more air pressure is needed at P_1 to overcome the density pressure. This differential pressure will be directly proportional to the unknown density.

SATURATION TEMPERATURE HYDROMETER

Saturation temperature is the temperature at which a liquid starts to boil. The boiling point depends on the specific gravity and the density of the liquid. The temperature at which a liquid boils must be compared with the temperature at which water boils at the

same pressure. The difference in temperature is then directly proportional to the density.

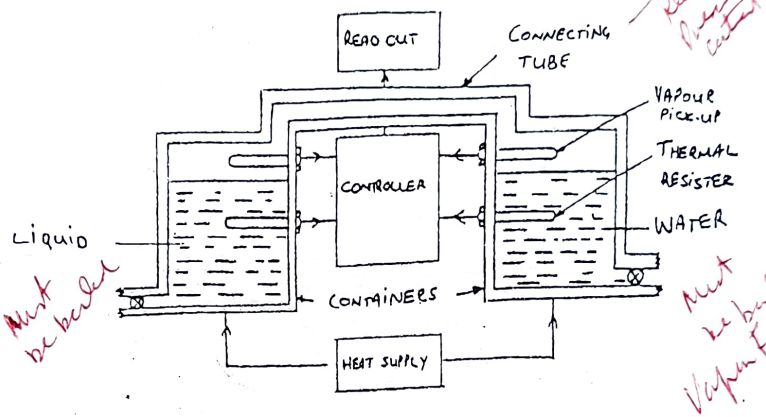
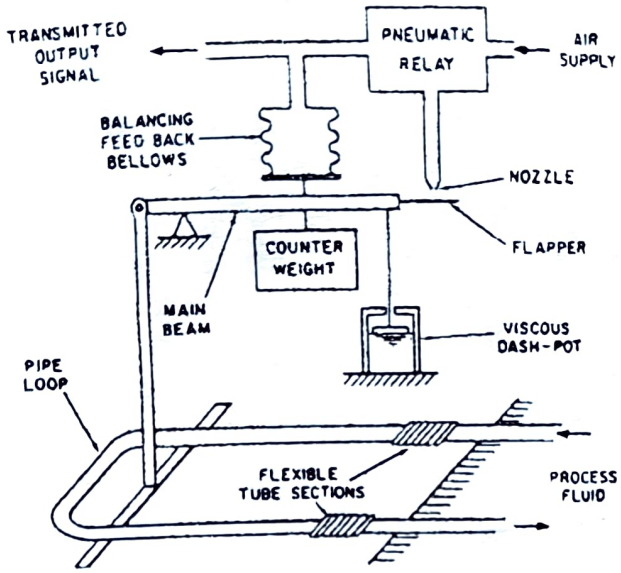


FIG ... *Saturation temperature hydrometer*

The liquid and the water must be boiled. When the liquid and the water starts to boil, vapour will be formed. This vapour will be picked up and send to the controller to record the temperature measured with a thermal resister. If the liquids temperature at boiling point differs from the water's boiling point, the difference in temperature will be directly proportional to the density of the liquid. The pressure of both containers is kept constant with a connecting tube, connecting the containers.

WEIGHING METHOD

This method is based upon continuous weighing of a container of known volume through which the liquid passes. In case of slurries and other difficult liquids where acceleration and lack of homogeneity might cause errors, the vessel may take the form of a tube of about 30 mm bore in a the shape of a hairpin some 3m overall suitable pivoted and counterbalanced. The liquid is pumped through the tube by way of flexible connections at a velocity sufficient to prevent settling out of solids. Variation in weight is detected by an electric or pneumatic system.



When the weight of the U-tube increases the force beam will be pushed in a clockwise direction by the coupling bar. This motion will increase the gap at the nozzle and more air will escape. The pressure will drop in the feedback bellow which will cause the force beam to rebalance at a new output pressure. This pressure will be directly proportional to the density of the liquid. This system can be calibrated by a counter balancing weight and the motion is damped by a silicon filled damper which only allows a few millimetres per second. The balancing of the beam will be continuous so that the density will be measured continuously.

BALANCED-FLOW VESSEL

A fixed-volume vessel is used, through which the measured liquid flows continuously. This vessel is weighed automatically be a scale, spring balance, or pneumatic force-balance transmitter. SInce the weight of a definite volume of the liquid is known, the instrument can be calibrated to read directly in specific gravity or density units. Either open or closed vessels with flexible connections can be used. High-accuracy measurement results, this is especially useful for automatic density control.

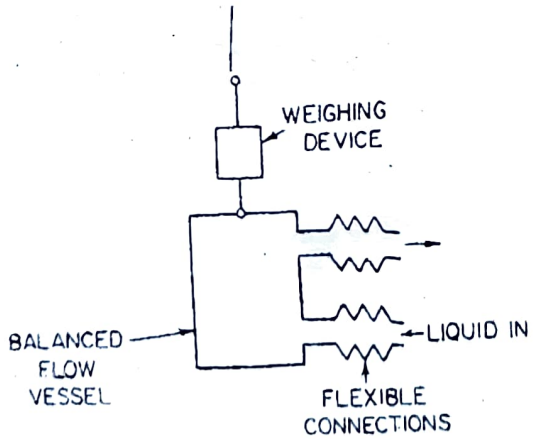


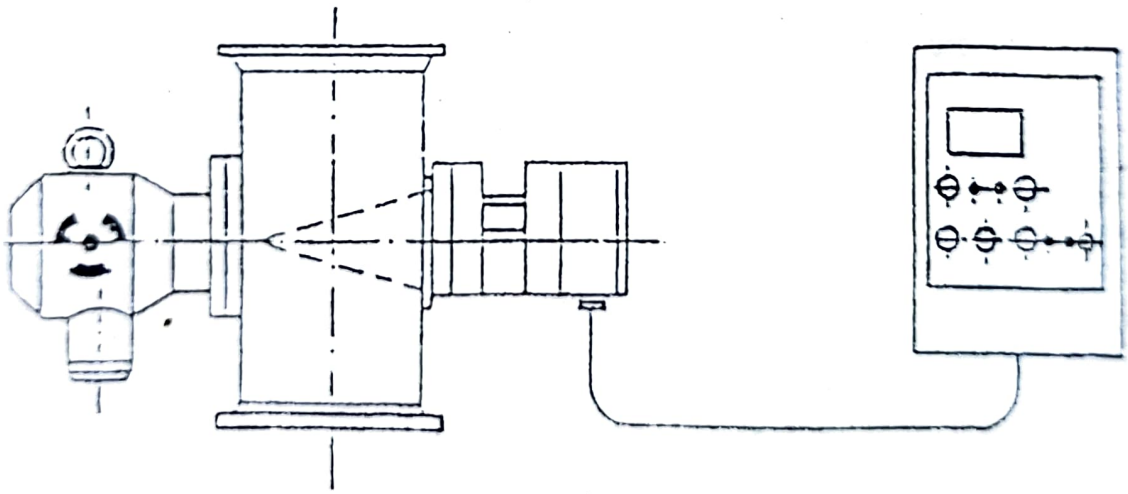
FIG ... Balanced flow vessel for measuring specific gravity and density

RADIATION SYSTEMS

Radiation is the most important kind of nuclear radiation for the use of many kinds of measurement. X-rays do not contain substance, they are electromagnetic waves without electric charge, which means that they will not be deflected by magnetic or electric fields. These electromagnetic waves are produced by an isotope. Atomic and species of the same atomic number (same chemical behaviour) but having a different mass (because of different number of neutrons) are called isotopes. Certain isotopes, especially atoms with large mass, integrate naturally forming new stable elements with other chemical qualities. As this transformation takes place, nuclear radiation is emitted. This nuclear radiation consists of both substance and energy.

DENSITY MEASUREMENT USING GAMMA RADIATION

The adsorption of gamma-radiation is a function of the mass of the material situated between the gamma-ray emitting source of detector. If the measured medium is a solution or suspension in liquid or semi-liquid form and is contained in a pipe with a defined and fixed wall thickness and density, the radiation detector will only respond to the density variation of the medium within the pipe and its signal output is a function of the density measured.



The complete measuring system consists of a radioactive source built into a lead shielded source container (a), a measuring section (normally the process pipe) (b) and the radiation detector (c) and finally the measuring converter or amplifier (d) which converts the signal received from the detector into a standard process signal.

3. HUMIDITY

3.1 INTRODUCTION

A knowledge of the concentration of water vapour in air and other gases is of considerable importance in a variety of industries. Two well-known examples are tobacco and textile manufacturers. The concentration is measured in terms of relative humidity, absolute humidity, dew point and weight concentration.

3.2 TERMS USED IN HUMIDITY

DEFINITIONS

Absorption - The taking in of a fluid to fill the gaps in a solid.

Adsorption - The adhesion of a fluid in extremely thin le7ers to the surfaces of a solid.

Saturated solution - A solution which has reached the limit of solubility.

Saturation pressure - The pressure of a fluid when condensation (or vaporization) takes place at a given temperature. (The temperature is the saturation temperature.)

Saturation temperature - The temperature of a fluid when condensation takes place.

Partial pressure - In a mixture of gases, the partial pressure of one component is the pressure of that component if it alone occupied the entire volume at the temperature of the mixture.

Relative humidity - The ratio of the mole fraction of moisture in a gas mixture to the mole fraction of moisture in a saturated mixture at the same temperature and pressure. Or the ratio of the amount of moisture in a gas mixture to the amount of the moisture in a saturated mixture at equal volume, temperature, and pressure.

Specific humidity - The ratio of the mass of water vapour to the mass of dry gas in a given volume.

3.3 ABSOLUTE HUMIDITY

This is defined as the weight of water vapour present in unit volume of moist air or gas at a given temperature.

Dew point - Dew point is defined a the temperature at which the saturation vapour pressure equals the partial pressure of water vapour in the atmosphere or other gas. Alternatively the dew point can be regarded as the temperature to which the medium must be reduced for water to commence condensing from it. The meaning will become clearer in the following descriptions of dew point instruments. If the dew point is below 32°F (0°C) it is termed to hoar frost point.

Hygrometer - An apparatus that measure humidity.

Hygroscopic material - A material with great compatibility to moisture.

Sampling system - This is a clean presentable sample which is delivered to a measuring element at the correct pressure, temperature and flow rate.

ANALYZING THE DEW POINT IN GAS

WET AND DRY BULB HYGROMETER

When water changes from the liquid state to the vapour, i.e. as it evaporates, it requires heat in the form of *latent heat*. The supply of heat by the water tends to lower its temperature. The rate of evaporation is dependent on the amount of moisture in the atsmophere. Upon these basic facts it is possible to construct a humidity measuring instrument known as the wet and dry bulb hygrometer

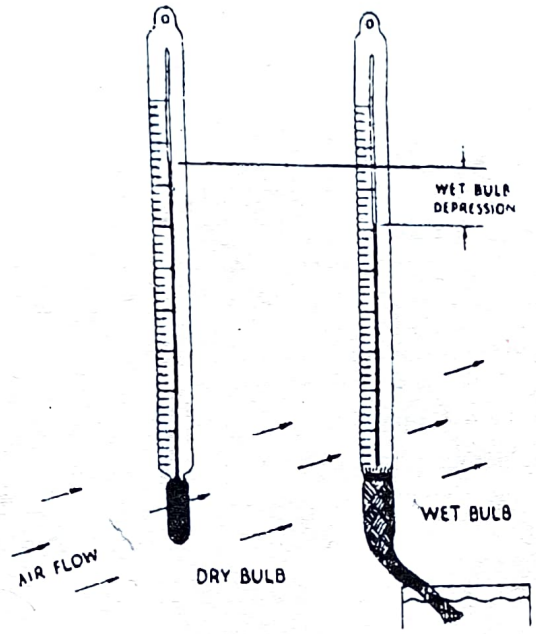


FIG *Mercury-in-glass type of wet and dry bulb*

It consists of two thermometers and a simple design is shown above. Round the bulb of one thermometer is a tight fitting cotton wick which dips into a vessel of water. This is termed the wet bulb thermometer. The second thermometer, in a adjacent position, is standard and known as the dry bulb thermometer. The action of the water evaporating from the wick of the wet bulb thermometer coils the bulb, and its temperature. T_w falls below that of the dry bulb, T_D by an amount depending upon the relative humidity of the air. A true theoretical explanation of the process has been sought but to some extent the equation relating the various factors remains experiential.

$$p = p_s - AP\,(T_D - T_W)$$

where
- p = the vapour pressure due to the vapour actually present in the air
- ps = the saturation vapour pressure at the wet bulb temperature T_w
- A = a constant. It is found to be dependent on the capacity of the wet bulb if the velocity of air steam past the bulb is to ensure that the air is in movement at this or greater values, if accurate readings are to be obtained.
- P = the atmospheric pressure
- T_D = the dry bulb temperature
- T_w = the wet bulb temperature

Tables based on the above formulae are issued by various manufacturers and the Meteorolotical Office. These are normally in terms of the wet bulb depression $(T_D - T_W)$ and the dry bulb temperature TD, i.e. the temperature of the air at which the humidity requires to be known. Usually the table values are caluculated for P = 760 mm of mercury, but alterations can be made for other values of atmospheric pressure. Unless the change is large, errors are not serious.

Two developments of the wet and dry bulb hygrometer in which an adequate flow of air past the wet bulb is ensured are described below.

Sling Hygrometer or Whirling Hygrometer

Two mercury-in-glass thermometers are securly fastened in a frame pivoted on a handle. The wet bulb assembly with its wick is so constructed that it cannot spill during operation. The frame is rotated by hand, in a very similar fashion to a sports rattle, the motion of the thermometer bulbs through the air producint a relative velocity equal to or greater than the value necessary to keep A, in equation constant. The thermometer readings are noted and the relative humidity must be found from the thermometer readings and tables.

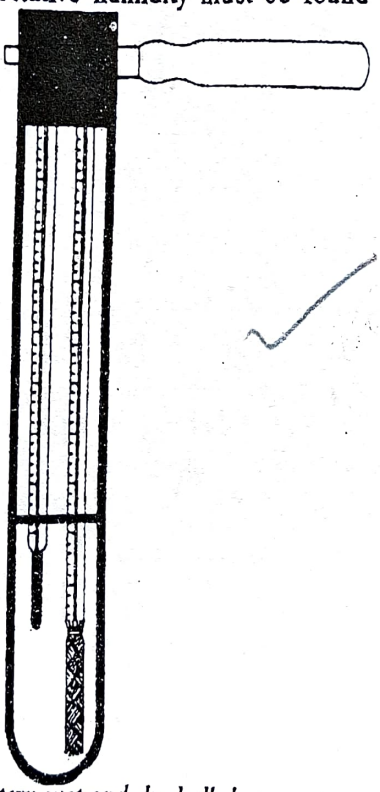


FIG *Sling or whirling pattern wet and dry bulb hygrometer*

Assmann Hygrometer

In the design, the mercury-in-glass bulbs are situated in two small metal tunnels which form part of a small channel. At the top of the channel is a fan driven by a clockwork or electric motor. In operation, the fan draws air up through the small tunnels, past the bulbs, and exhausts at the top of the channel.

Again the two thermometer readings must be observed and the relative humidty found from tables.

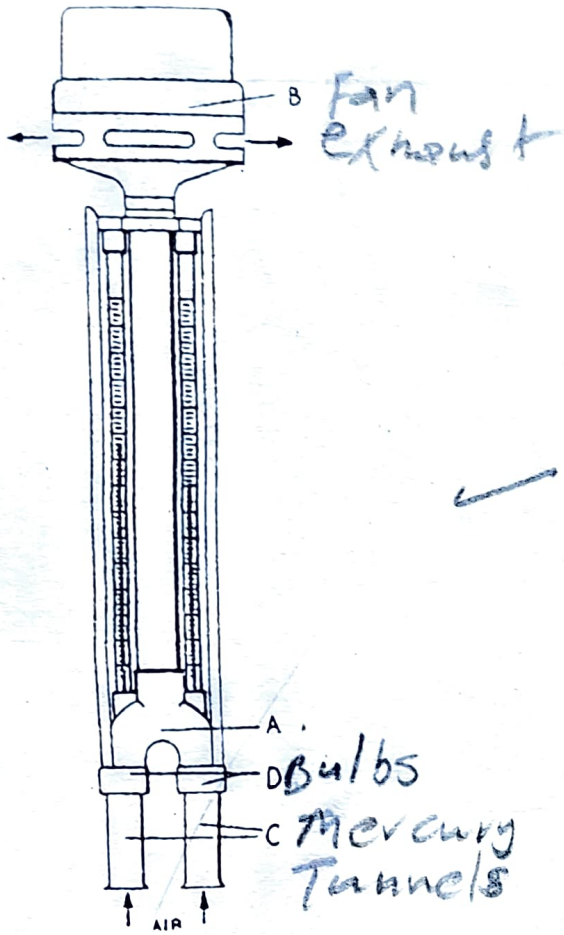


FIG *Assmann pattern wet and dry bulb hygrometer*

CALCULATION OF RELATIVE HUMIDITY

To calculate relative humidity we must amek use of steam tables and equations.

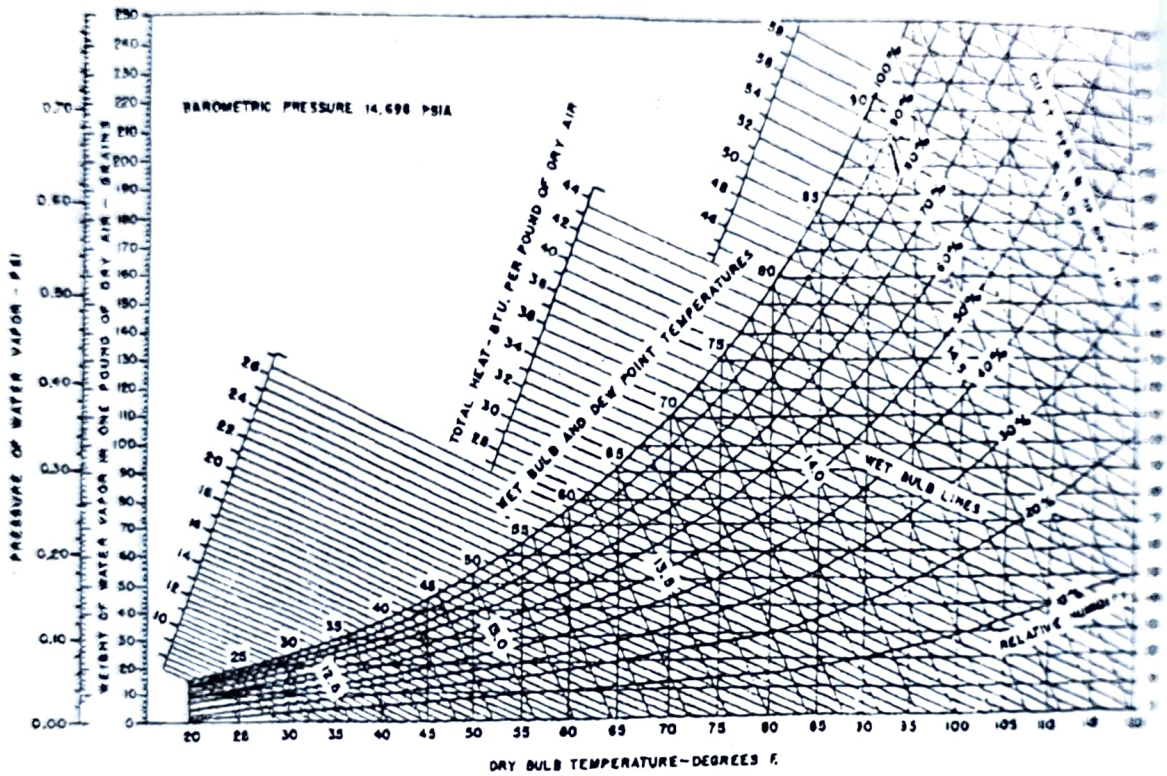
$$\text{Relative humidity} = \frac{\text{Partial pressure of water vapour}}{\text{Saturation pressure}}$$

$$\therefore \quad \text{Relative humidity \%} = \frac{P_{vl}}{P_{sl}} \times \frac{100}{1}$$

However, there is another method of determining relative humidity by means of a psychometric chart.

We will look into the latter method.

Consider the following chart.



EXAMPLE

Calculate the percentage relative humidity if the dry bulb and wet bulb thermometer readings are 100°F and 85°F, respectively.

From chart:

Relative humidity % = 58 % (dot on graph) obtain the following data from the chart as well:

a) Total heat;
b) Cubicmetres per kilogram dry air;
c) Mass of the water vapour; and
d) Pressure of the water vapour.

From chart:

a) Total heat = 50.8 Btu/lb dry air (from 80°F to the left)
b) Cu.ft/lb dry air = 14.65 Cu ft/lb dry air (from 58 % point across)
c) Water vapour = 146 grains (from 58 % point right)
d) Water vapour pressure = 0.545 Psi (from 58 % point far right)

HAIR HYGROMETERS

The principle of measurement ~~made~~ use of this device is the change of length of certain organic and synthetic fibres when they are exposed to a moist atmosphere. A mechanical linkage is used to amplify the element movement for readout.

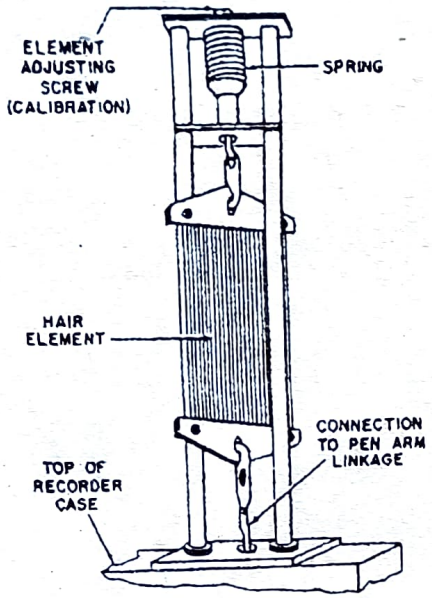


FIG *Hair hygrometer element*

The hair hygrometer consisting of several fibres mounted together to form a band; one end of the bank is fixed while the other end is free to move, driving a pen arm through a linkage. Hair hygrometers, similar to wet and dry bulb hygrometers, require good circulation of the measured gas.

MOUNTING PLACES FOR A HAIR HYGROMETER

In channels there is generally adequate gas velocity to insure a dependable measurement; however, if the instrument is to be mounted in a room, the location must be carefully chosen. The instrument should not be mounted near doors or other openings where it will be exposed to artificial drafts; flush mounting on a panel should be avoided because the atmosphere in the back of the panel is motionloess. The hair element can be mounted on top or on the back of the instrument case, depending on the installation. The element can also by mounted on an extension in the back of the instrument so that the sensing portion is in the room or compartment where relative humidity is to be measured while the readout device is surface mounted on the wall outside. Recorders are generally available as two-pen instruments with the second pen recording temperature.

SOLUTION CONDUCTIVE HYGROMETER (DEWCELL)

All solids condense and absorb small quantities of water on their surfaces. If the solid is soluble, a layer of saturated solution is formed ow which the vapour pressure is lower than that of pure water at the same temperature. Some solids form saturated solutions in water of which the vapour pressure is less than the partial pressure of water vapour in the atmosphere contacting the solution. When that is the case, water vapour will continue to condense until the vapour pressure of the solution equals the partial pressure of the water vapour. On the other hand, water vapour will evaporate from the solution should the water vapour partial pressure decrease below the vapour pressure of the solution. Under steady-state conditions, solution vapour pressure equals water vapour partial pressure, and knowledge of the former can be used to determine the moisture content of the atmosphere.

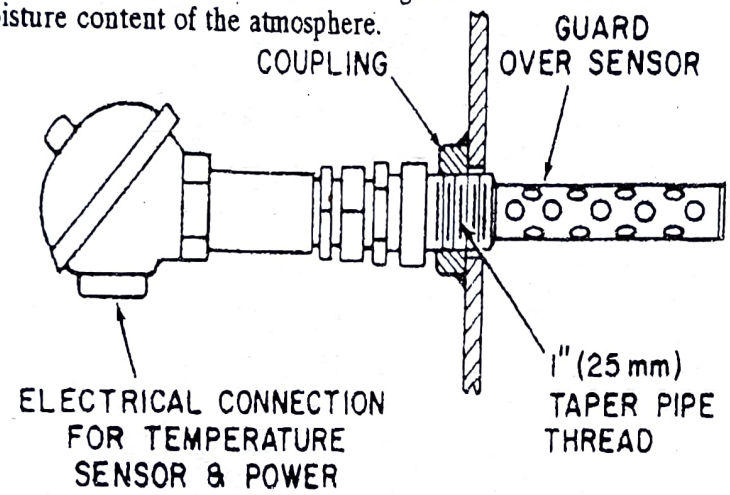


FIG *Solution conductivity probe with resistance temperature element*

The sensing element consists of a thin-walled, hollow-metal socket wrapped with tape filled with salt crystals (lithium chloride). Two wires are wrapped over the tape and connected to a regulated, alternating current voltage source; the electric circuit between the wires is completed by the salt crystals. When the sensing element is exposed to the sample atmosphere, water condensing on the crystals forms an ionic solution which allows an electric current to flow between the wires. This current in turn heats the solution and raises its vapour pressure. As more water condenses, more current flows, resulting in a further increase of the solution vapour pressure until equilibrium is reached. On decreasing moisture content of the measured gas, water evaporates from the element, decreasing the current flow and resulting in a new equilibrium at a lower solution vapour pressure. A temperature sensor inside the hollow socket is used to detect sensor temperature and to provide a signal for readout.

The output calibration can be in terms of dew-point temperature, water vapour pressure or specific humidity.

Installation

The instrument does not require a high velocity sample to perform satisfactorily; in fact, sample velocity in excess of 0.3 m per second will result in a poor measurement and shortened element life. The reason is evident when one considers the measurement principle involved, namely the change of solution vapour pressure and temperature on the sensing element by heating to establish equilibrium with the sample. At high sample velocities, convective heat losses drench the measurement signal and operation at higher power causes a reduction of element life. The sensing element must, therefore, be located in a relatively motionless zone or must be protected from direct influence of the sample. In channels, a sheet metal hood installed over the element and open on the downstream side is adequate. In pipelines, the element can be installed through the side outlet of a tee or can be mounted separately in a sampling chamber piped to the process line. The latter installation is preferred for samples under pressure because the element can be serviced without shutting down the pipeline.

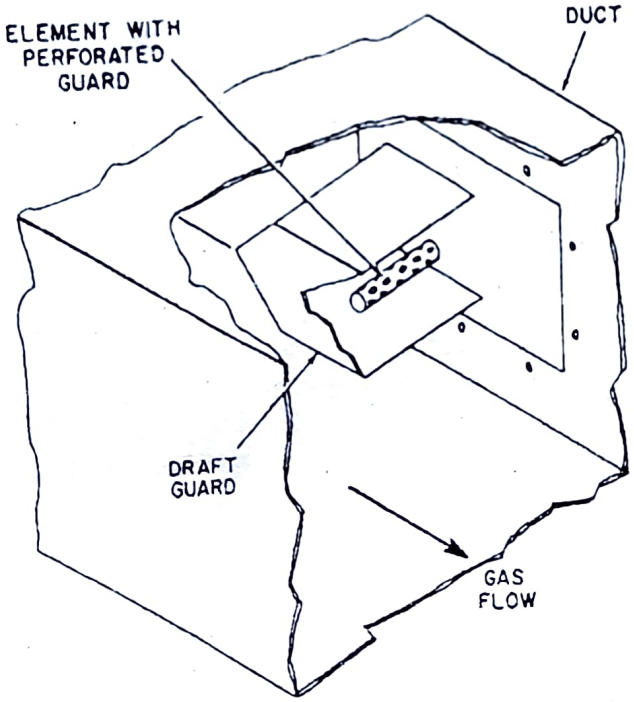


FIG *Installation of a dew cell*

Sampling System

A sampling system is required when:

a) the sample temperature is above the operating limit of the sensing element [operating limit approximately 200°F (93.3°C)], or

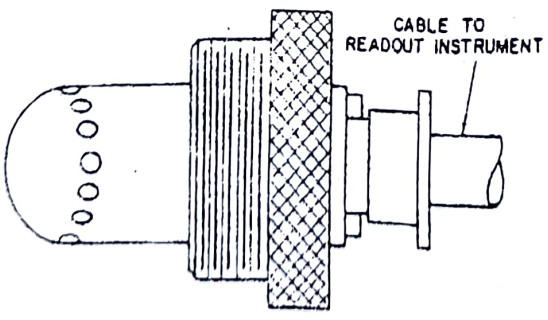
b) the process line cannot be shut down for maintenance of the element.

Limitations

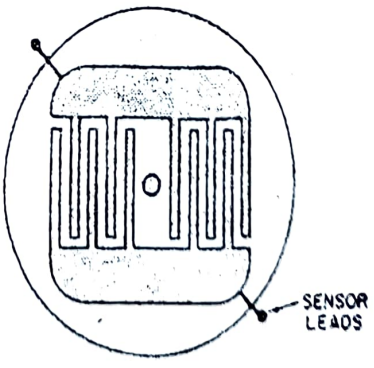
The instrument can be used on all non-corrosive gases that do not react with the salt crystals or water. Specifically, such gases as ammonia, sulphur dioxide, sulphur trioxide, and chlorine react with water and this instrument is therefore not applicable for such use. The instrument is not suitable for sample containing diethylene glycol or triethanolamine.

SURFACE CONDUCTION HYGROMETER

Every object in a moist atmosphere has water molecules on its surface; the concentration on these molecules is related to the temperature of the object and the dew point of the atmosphere. If the temperature of the surface is above the dew point of the atmosphere, the thin layer of molecules is invisible, however, as the surface is cooled to the dew point, the density of water molecules at the surface becomes so much that water condenses on the surface and dew can be seen. At surface temperatures above the dew point, the moisture density at the surface can be detected electrically although the water vapour is invisible. This water vapour will permit a current to flow on the surface of even an excellent insulator. This current flow is a function of the surface material and moisture density at the surface. For a given material and fixed applied potential, the current will increase logarithmically as the surface temperature decreases to the dew point of the atmosphere. Below the dew point, the logarithmic relationship does not hold, and thus surface conductivity can be related to a dew point. Of course, dew formation can also be observed optically, however, visual observation is more difficult since dew will form at or below the dew point temperature while conductivity is a continuous function of temperature.



Sensor



Sensing surface

The measuring element consists of a highly polished immobile surface inlaid with an intermeshed gold grid and a thermocouple imbedded in the surface. A fixed potential is maintained across the gold grids, and the current flow is compared to the reference

current flow at dew point. This signal is amplified and used to regulate a cooler to maintain the surface at the dew point of the sample. The cooler is a bismuth-telluride crystal that pumps heat away from the sensor when electric power is supplied to it.

From the thermocouple imbedded in the surface, the output is normally presented as dew-point temperature; however, readout in terms of relative or specific humidity is also possible.

Sampling System

A sampling system is required where the sample gas is outside the operational limits of the instrument. At high temperatures, a sample cooler is required to lower the sample temperature below the operating limit of the sensor. The sample cooler, however, must be chosen so that the sample will never be cooled to its dew points under any possible operating or ambient condition. At high sample pressures, a pressure-reducing valve is required. For measurement of motionless atmospheres, low-pressure samples, or closed systems, a pump may be required to draw the sample over the sensor.

Limitations

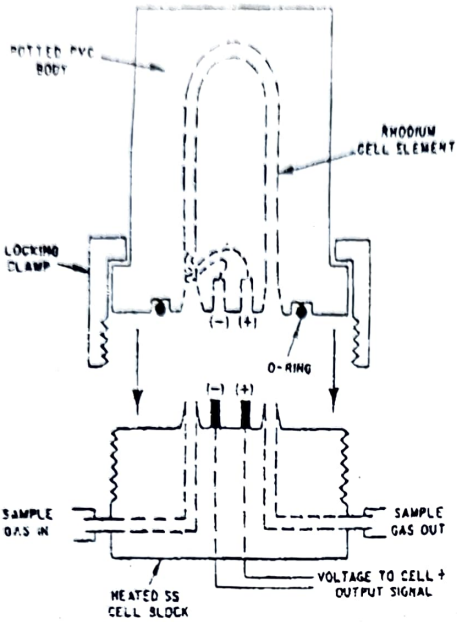
The instrument can be used with all the common atmospheric gases such as nitrogen, oxygen and carbon dioxide. The instrument is not suitable for samples containing components that react with water such as chlorine or sulphur dioxide and components with a dew point higher than that of water.

HUMIDITY MEASUREMENT IN LIQUIDS AND GASSES

ELECTROLYTIC HYGROMETER

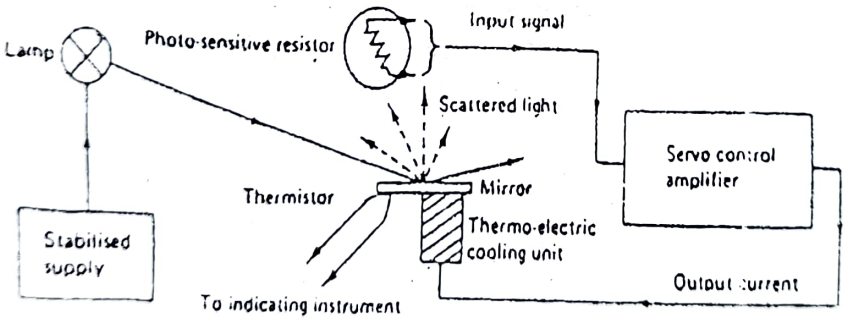
The principle of measurement utilized involves the electrolysis of water into oxygen and hydrogen. Since two electrons are required for electrolysis of each water molecule, the electrolysis current is a measure of the water present in the sample. If the volumetric flow rate of sample into the electrolysis cell is controlled at a fixed value, the electrolysis current is a function of water concentration in the sample.

The hygrometer cell sees only a mass flow of water - number of water molecules electrolyzed per unit time. In order to obtain an output in the form of moisture content, the sample flow rate must be a known, constant value. Furthermore, the accuracy of the output can never exceed the accuracy with which the sample flow is controlled.



THERMAL-ELECTRICAL DEW POINT HYGROMETER

The lightbeams of radiated light will move in a different direction when it passes from one medium to another. If we pass light through water the beam will be reflected. This reflected light can be picked up by a light sensitive device.



This thermal-electrical dewpoint hygrometer consist of a thermo-electric cooling unit which will cool the air passing over a mirror. A light beam from a lamp will be directed in the direction of the surface. A photo-sensitive resistor is placed at a point where the radiated light beam will fall after passing through the dew drops formed on the mirror due to the cooling of the mirror. If no dew drops are present the light beam will not be picked up by the photo-sensitive resistor and the more the dew drops the more reflected light will be picked up. This signal will be send to the cooler to switch of so that the temperature will alternate around the dew point. A thermocouple is used to determine the temperature of the mirror. This temperature will be the dew point temperature of the substance.

The electrolytic hygrometer cell consist of a small chamber containing two noble metal electrodes which support a thin layer of desiccant. Moisture in the sample is absorbed by the desiccant and electrolyzed by means of a voltage-regulated power supply connected to the electrodes. Units are available for use in nonhazardous areas with the sample flow control, electrolysis cell, and electronics packaged as a single unit. When used in hazardous areas, the cell and flow controller are housed in an explosion-proof pipe and the electronic circuitry is remote mounted.

Recombination Effect

Although this instrument will operate satisfactorily with a variety of samples, a phenomenon called the recombination effect introduces large errors at low moisture levels in hydrogen-rich or oxygen-rich samples. Recombination is the returning to water of the electrolysis products; it introduces an error into the measurement when the recombined oxygen and hydrogen are re-electrolyzed. Apparently all electrodes catalyses this reaction, although some electrode materials do so more than others. The use of rhodium as the electrode material has been found to minimize recombination.

When monitoring very low moisture level in oxygen-rich or hydrogen-rich atmospheres, the recombination produces a large error even with the best choice of electrode materials. For such an application two sensors are used: One measures at sample flow rate X and the other at sample flow rate 2X. Since the error due to recombination is constant, subtraction of the two sensor outputs yields a signal that is not controlled by recombination.

Cell Limitations

The cell will perform satisfactorily with a variety of samples, however, there are a number of sample materials which will cause problems or should not be monitored with this instrument. Gases that cause problems are hydrogen and oxygen. These gases have been discussed under the recombination effect above. Gases for which this instrument is unsuitable are unsaturated monomers, alcohols, amino, ammonia, hydrogen fluoride, and $CHClf_2$ (Freon) refrigerant.

Alcohols are seen by the cell as water, aminos, and ammonia react with the desiccant, and hydrogen fluoride destroys the internals of the cell.

Generally, the instrument should not be used with samples of which the components may deposit in the cell. When a cell becomes contaminated, it will show a memory for polarity; that is, the outputs under forward and reverse flow through the cell will not be equal.

A cell will lose sensitivity when exposed to moisture levels of a few parts per million over a period of weeks. This sensitivity loss is due to the elution of desiccant with the sample.

However, this process occurs over a long period of time and the cell can be recoated fairly easily in the field during periodic maintenance.

Lifting Samples

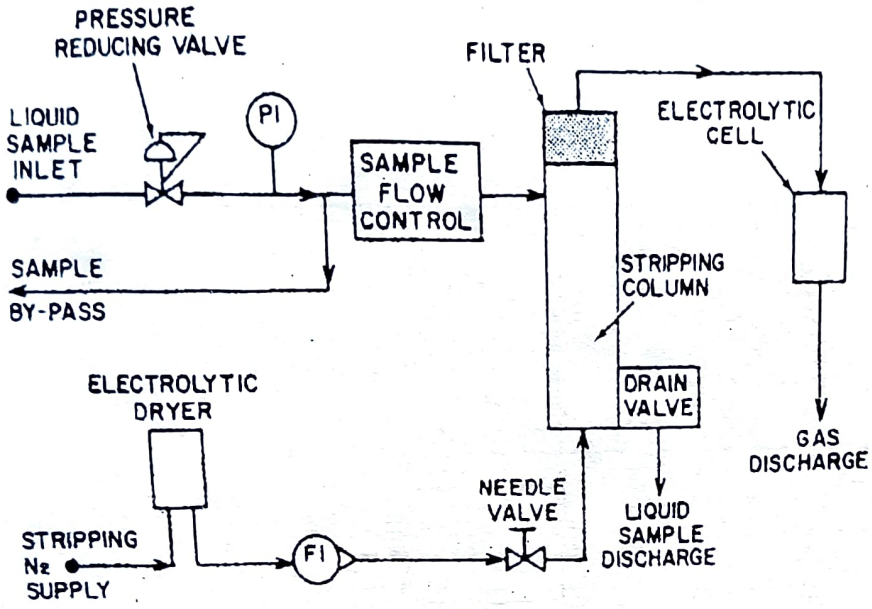
The electrolytic hygrometer cell can be exposed only to gases. When the process samples is in the liquid state, there are two methods available to effect a measurement using the electrolytic hygrometer.

1. *Sample Vaporization*

   If the liquid sample has enough high vapour pressure, the sample can be vaporized, and the measurement is then made as a gas sample. The sample is vaporized by means of a vaporizing regulator placed as close as possible to the sample take-off point to minimize lag. The remainder of the sampling system is the same as for gas samples.

2. *Sample Stripping*

   If the sample cannot be vaporized, this method can be used to obtain a gas sample at the cell. The moisture is stripped from the liquid sample in a falling film column. Liquid sample is continuously metered into the top of the column where it descends as a thin film. Dry nitrogen, metered into the bottom of the column, ascends and removes moisture from the descending film of liquid sample. A filter placed at the top of the column removes any droplets entrained in the nitrogen before the gas is passed through the measuring cell. A drain valve at the bottom of the column facilitates removal of stripped sample. The nitrogen is dried before entering the column by passing through an electrolytic dryer, similar to the measuring cell, where moisture is removed by electrolysis. The supply nitrogen moisture level should be fairly low, less than 500 ppm, to prevent overloading the dryer.



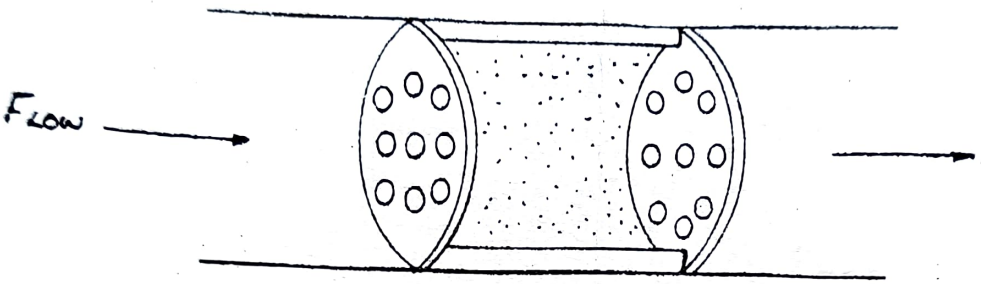
Liquid sample stripping

Sampling System

As pointed out previously, the electrolytic hygrometer is sensitive only to the mass flow of water into the cell, the concentration of moisture is inferred from the known sample flow rate. Therefore, the accuracy of the readout is dependent on the accuracy of the sample flow control, and the accuracy of the moisture measurement can never exceed that of the sample flow control. This fact should be kept in mind when choosing a sample flow controller.

CAPACITANCE HYGROMETER

The principle of measurement utilized is the change of capacitance associated with a change of the sample dielectric constant between capacitor plates. The value of capacitance is a function of a plate area, plate spacing, and the dielectric constant of the material between the plates. The dielectric constant of a material has a unique value for each substance and is related to the polar characteristics of the molecules. This property can, therefore, be used to detect the presence of a specific substance in a pipeline stream.



Capacitance Hygrometer

The measuring cell of the hygrometer consists of two electrically insulated concentric metal cylinders to form the measuring capacitor plates. The annulus between the cylinders is filled with alumina desiccant. Two porous metal discs support the cylinders and retain the desiccant in the annulus. The sample is allowed to flow through the annulus and in the process water is absorbed or desorbed by the desiccant which remains in equilibrium with the sample in terms of percent saturation. The moisture content of the sample is thus amplified by the desiccant because of the saturation level of the desiccant is much higher than that of the sample. The measuring capacitor is part of an electrical circuit which includes a reference capacitor. This circuit is powered by a 15-kilocycle, fixed-amplitude sine wave. The measuring and reference capacitors are switched alternately into the circuit, in such a way that its output voltage is a function cycle sine wave whose amplitude varies with the measuring and reference capacitors as they are switched into the circuit. This difference in amplitude is related to the measured capacitance which in turn is a function of the moisture content of the sample.

It was pointed out previously that the desiccant in the measuring cell is in equilibrium with the sample in terms of percent saturation. For this reason it is necessary to maintain the sample at constant temperature. The sample is maintained at constant temperature by passing it through a coil immersed in a constant temperature bath immediately upstream of the measuring cell. Temperature control of the sample is recommended even if the process stream is temperature controlled since ambient conditions will affect the sample temperature.

Limitations

The instrument is not suitable for polar materials such as alcohols because these become conductive at the 15-kilocycle operating frequency and short the measuring capacitor. Data on instruments operating at higher frequencies are not available.

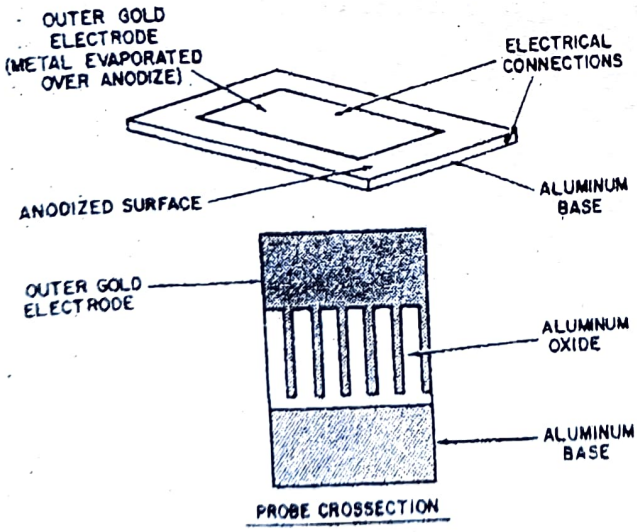
Free sulphur and iron oxide concentrate on the desiccant and short the element.

At sample viscosities above 500 Saybolt Seconds Universal (SSU), moisture cannot come in contact with the desiccant, and the measurement becomes meaningless.

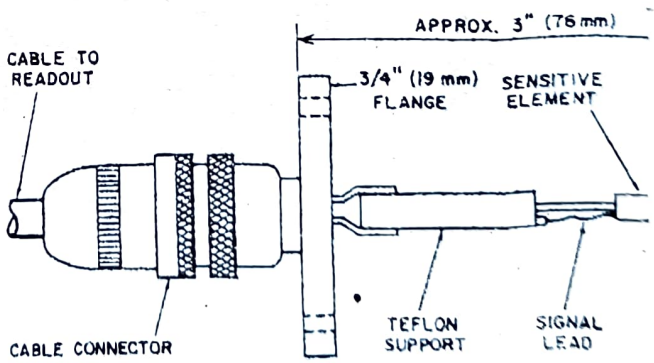
The essence of the desiccant is a function of solids in the stream which pass through the filter. While the desiccant can be replaced very readily, the short life of the desiccant excludes the use of this instrument on sample streams containing large quantities of fines.

IMPEDANCE HYGROMETER

This instrument measures the water content of a sample by means of a probe whose electrical impedance is a function of the vapour pressure of moisture in the fluid. Impedance is the apparent opposition to the flow of alternating current.



Impedance-measuring sensor



Flanged probe

The probe consists of an aluminium strip, which is anodized to form a porous layer of aluminium oxide. A thin coat of gold is applied over the aluminium oxide. Leads from the gold and aluminium electrodes of the probe connect the sensing element to the measuring circuitry.

Water vapour penetrates the gold layer and equilibrates on the aluminium oxide. The number of molecules absorbed to the aluminium oxide is the function of the water vapour pressure in the sample. Each water molecule absorbed contributes a distinct addition to the total conductivity of the aluminium oxide. The total probe impedance, the reciprocal of probe conductivity, is thus a measure of water vapour pressure in the sample. Water vapour pressure of a gas sample uniquely determines the dew-point temperature and moisture content of the sample. The output is normally calibrated to these units since they are more convenient to use than vapour pressure. In the case of certain liquid samples, the moisture content can be measured through the application of Henry's law.

In terms pertinent to this measurement, Henry's law states that at constant temperature the mass of water vapour dissolved in a given volume of liquid is in direct proportion to the partial pressure of water vapour in the sample. Henry's law can be restated in simple form to read that the weight concentration of moisture in the sample is equal to the partial pressure of water vapour times a constant. However, Henry's law holds only for liquids with moderate solubility for water vapour such as pure hydrocarbons. The instrument cannot be used on liquid samples with high solubility for water such as the alcohols because the relationship expressed by Henry's law does not hold.

Sampling System

Normally a sampling system is not required on gas samples. On liquids, sample temperature must be held constant, as mentioned above.

Limitations

The instrument can be used on all gases not corrosive to the probe and which will not combine spontaneously on contact with the probe materials. On liquids, the instrument is limited to those fluids with a moderate solubility for water vapour, thus the instrument is not suitable for measurement in polar liquids such as the alcohols.

PIEZOELECTRIC HYGROMETER

Piezoelectric quartz crystals have a number of uses: in communications, to control frequencies and, in industry, to measure temperature and thickness of metal films as well as to generate ultrasonic waves. In moisture measurement, advantage is taken of the oscillating crystal's sensitivity to deposits of foreign material on its surface. Commercially available crystals will show a frequency change of 2000 cycles per second (cps) per microgram of material deposited. For moisture measurement, the quartz crystals are coated with a hygroscopic material and exposed to the sample, water from the sample is absorbed by the crystal coating, thus increasing the total mass and decreasing the oscillating frequency of the crystal. In order to measure changes of

standard hydrogen electrode is subtracted from or added to the measured voltage.

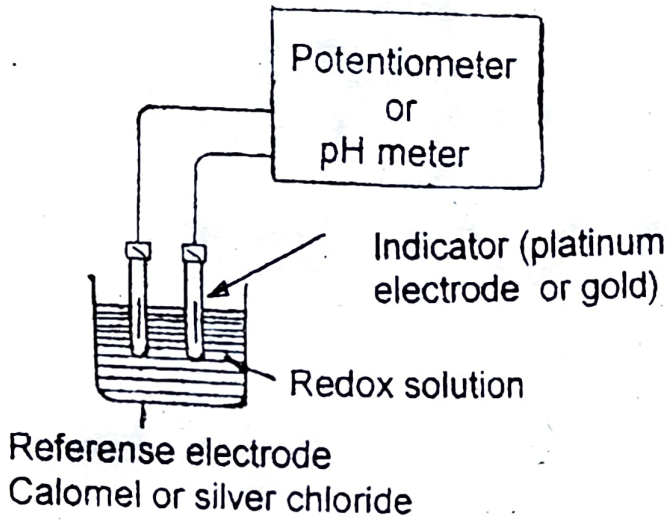
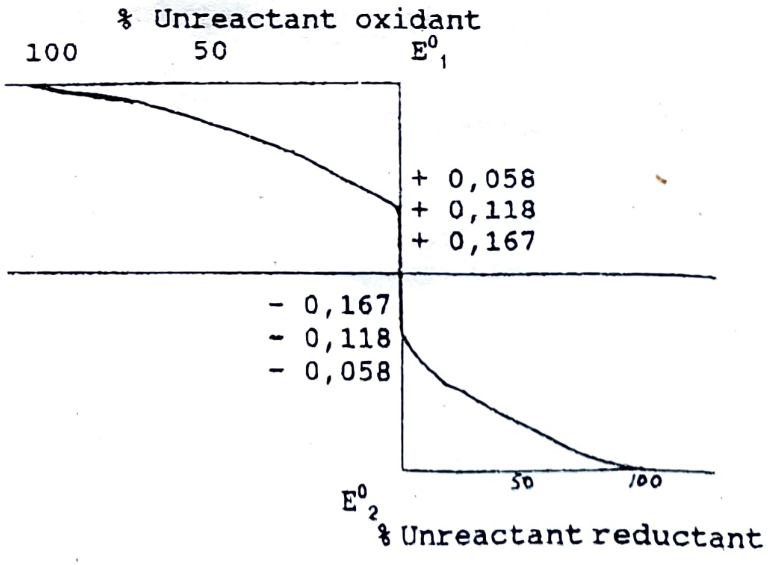


FIG *Reduction potential measurement*

2. Amount of oxidant or reductant present, determined by titration. The oxidant and reductant concentrations appear as a ratio of quantities and when this ration changes by a factor of 10, the electrode voltage changes 0,0591 volt. Electrode voltage is independent of quantity of oxidant or reductant and is determined by concentration ratio.



Potentiometric titration curve the actual equivalence voltage is determined by the equilibrium constants of the particular oxidant and reductant used,

Since various substances in solution may be either oxidants or reductants with respect to hydrogen, solution potential measurements must recognize "sense" or polarity. Although confusing conventions are in common use, the preferable system involves reduction potentials whose polarity is negative for agents that are more effective reductants than hydrogen and positive in sign for agents that are more effective oxidants than hydrogen.

By combining solution potential measurements with titrate step, it is possible to determine both the identity and concentration of a given oxidant or reductant in solution.

ELECTRODES

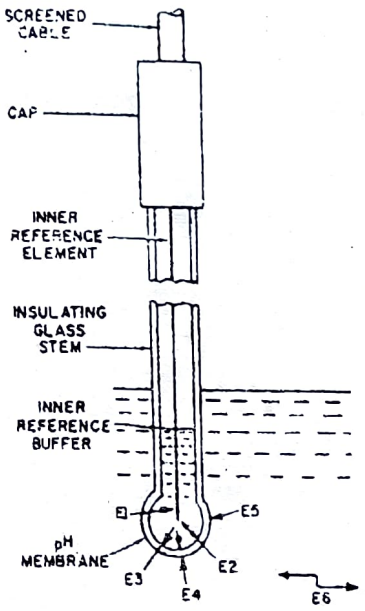
Consider initially what happens when a metallic electrode is placed in a solution. There is a tendency for positive ions of the metal to enter the solution resulting in the solution becoming positively charged. But if the solution already contains ions of the metal there is a tendency for ions to be deposited on the electrode, giving a positive charge. The electrode ultimately reaches an equilibrium potential relative to the solution. The value and sign of the potential depend on the concentration of metallic ions in the solution.

Two common examples are zinc, which assumes a negative potential relative to solutions of its salts, and copper which reaches a positive potential relative to solutions of its salts. In passing it may be noted that these two electrodes and solutions of their salts form the basis of a Daniell cell.

Hydrogen, although a gas, can be made to form the basis of an electrode.

GLASS MEMBRANE ELECTRODE

The glass electrode in generalized form consists of a pH-sensitive glass membrane fused to a glass tube. A silver wire coated with silver chloride protrudes into the thin-walled glass bulb. The bulb contains a internal filling solution, generally a buffered solution containing chloride ions.



pH Glass membrane electrode

For a glass membrane electrode to function, both surfaces of the membrane must be hydrated. Hydration occurs by absorption of water by the membrane interface in contact with the solution. There is also an exchange of univalent cations of the glass for hydrogen ions from the solution. At both surfaces of the membrane there will be a boundary potential that will be a function of the hydronium ion activity in the solution at the interface. Because hydronium ion activity for the internal filling solution is constant, the potential will be a function of the hydronium ion activity of the external solution. This potential is:

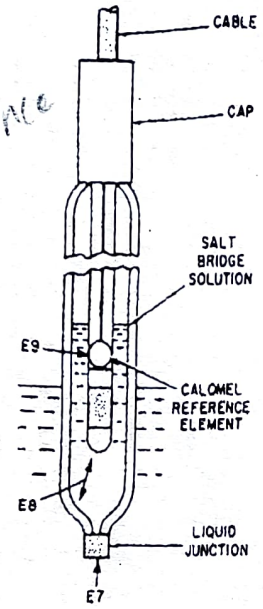
$$E = k - 0{,}059 \log [\alpha H_3O^+] \text{ at } 25°C$$

where k includes such factors as the activity of hydronium ions in the internal solution and the asymmetry potential of the membrane. Asymmetry potential arises from differences in the characteristics of the two surfaces of the glass membrane and results from such things as mechanical stresses, deterioration of the external surface by dissolution of the glass, chemical attack on the glass, dirty surfaces and so on. Thus, the potential E will not be the same from electrode to electrode or for a given electrode with relation to time for a constant concentration of active hydrogen ions. For this and other reasons, a pH system must be standardized or adjusted periodically for its entire useful life.

REFERENCE ELECTRODES

The function of a reference electrode is to complete the electrical circuit with the glass membrane electrode. It must provide a stable potential that is relatively invariable, despite changes in either the chemical composition or the physical properties of the process stream.

We will discuss a reference electrode most widely used namely the calomel electrode.



A typical calomel reference electrode

A inner tube contains calomel (mercurous chloride), potassium chloride paste and mercury. For electrical connection purposes, a platinum wire extends down the tube and makes contact with the mercury. An outer tube contains a saturated solution of potassium chloride and terminates in a porous plug. The latter can be of glass, ceramic material or asbestos fibre. The potassium chloride acts as a "salt bridge" and seeps or diffuses through the porous plug to make an effective liquid junction with the solution whose pH is to be determined.

Alternative designs have a small hole near the base of the outer tube and this is covered with a ground glass sleeve or a silicone rubber sleeve. These sleeves permit a faster rate of diffusion than with the asbestos fibre plug pattern generally. On a broad basis, the sleeve pattern is suitable for solutions of high viscosity or turbidity and with slurries, oil emulsions and soap solutions. It may be necessary after each measurement to lessen the sleeve and flush the tip of the electrode.

The choice of potassium chloride solution in the electrode is made for several reasons, but the main one is that the potential between this substance and many other solutions is extremely small and can be made very stable.

A reference electrode must be substantially stable with regard to EMF. At room temperature, the calomel electrode is reasonably satisfactory in this direction.

Duel Electrodes

Under certain conditions it may be advantages to use a dual electrode, that is, a combined glass membrane and reference electrode. This type of electrode is eminently suited to applications where the temperature difference between the two electrodes must be reduced to a minimum. Such electrodes can have an accuracy of ± 0,1 pH for a temperature variation of ± 20°F within the range 4-10 pH. Dual electrodes also have the advantage that only one aperture is required instead of two.

Calomel Electrode

The calomel electrode has been made by bringing mercury into contact with a solution saturated with mercurous chloride and containing potassium chloride at a fixed concentration.

Best results are obtained when mercurous chloride and mercury are mixed to a paste with Kc1 solution and placed in contact with liquid mercury. A platinum wire immersed in the mercury is used as a lead wire.

Indicator Electrodes (Platinum - Bright)

Platinum-wire electrodes are suitable for general application as indicator electrodes.

These electrodes have been used to indicate the voltage of oxidation-reduction couples (such as the ferrous, ferric couple) and to form the conducted contact for a gas or liquid electrodes (such as the hydrogen gas electrode or bromine electrode). Platinum is, however, catalyst for some reactions and will anomalous voltage readings when used as an electrode in such systems.

Hydrogen Electrode

The hydrogen electrode is formed by immersing a portion of a platinum electrode, preferably coated with a platinum black, in a solution containing hydrogen ions and maintaining an atmosphere of hydrogen gas in contact with the exposed portion of the electrode.

This electrode is responsive to changes in hydrogen-ion concentration in the solution. When the pressure of hydrogen gas is maintained at 1 atm and the hydrogen-ion concentration at 1 mole per litre, the electrode becomes a standard hydrogen electrode (see reference electrodes, pH).

Iodine Electrodes

The iodine electrode is formed by immersing a platinum or gold electrode in a solution saturated with iodine.

Silver-Silver Chloride Electrode

This electrode has been growing in popularity in recent years because of its good stability, particularly at higher temperatures. Electrodes may be prepared by electrolyzing silver wire in a solution of hydrochloric acid with the wire as anode. Silver ions are formed at the metal surface and react with the chloride to form an adherent coat of silver chloride on the electrode. This electrode immersed in a solution of potassium chloride develops a stable voltage. It is important that the electrode wire be free from strains and that precipitated chloride by thoroughly free from electrolyte. When these precautions are observed, the silver-silver chloride electrode may be used as a primary standard.

Two-phase Electrodes

Gases such as hydrogen and oxygen, liquids such as bromine, and solids such as iodine dissolve in water to reach an exchange equilibrium with ions already in solution, producing the characteristic solution potential across the half-cell so formed. Electrodes may therefore by produced employing the gas, liquid or solid as the responsive element by making suitable contact with the solution.

Quinhydrone Electrode

The compound quinone may be electrolytically reduced to hydroquinone. The voltage of this couple is very reproducible. These compounds form a stable equimole compound called quinhydrone which dissolves to give equal concentrations of quinone and hydroquinone.

It is also very useful as a reference electrode in calibrating pH meters for the measurement of oxidation-reduction potentials A quinhydrone electrode, however, cannot be used in solution where the pH value is above 8,5.

Tungsten Electrode

The tungsten electrode has been found very useful as a reference electrode in oxidation-reduction titrations where a platinum or other noble metal indicator electrode is used. The voltage of a tungsten electrode remains relatively constant through the end point of a titrate using strong oxidizing reagents, while the voltage of the indicator electrode changes sharply at the end point.

e tungsten electrode is one very useful as an indicator electrode in measuring idation potentials, since its voltage usually drifts slowly over a period of contact with ectrolyte and it must therefore be frequently calibrated against a calomel electrode or equivalent as a primary standard.

IATERIALS

All materials likely to come into contact with the process liquid must be chemically esistant at all measurement temperatures. This applies both to the assembly and the sealing materials. The ideal housing will have resistance to corrosion, strength against thermal and mechanical shock and in addition, for measurement at high temperatures, impermeability to water vapour. Because permeability increases with temperature, water vapour will destroy the electrical insulation.

Both metals and plastics are used. The exact choice of materials depends upon the characteristics of the process liquid and the conditions under which the measurement is made. Metals are strong and impermeable but have limited resistance to corrosion, while offer resistance to corrosion but have an indifferent water vapour permeability.

Metals

Many metals, especially soft metals and amalgams, are specific electrodes for the measurement of the concentration of their ions in a solution. Silver, copper, mercury, antimony and others have been used. In general, metal electrode may be made by simply mounting wire rod or plate electrodes in suitable insulating mounts. A portio of the metal is immersed in the solution. Mercury is usually placed in a well or cup so that its upper surface contacts the solution. Contact with the pool of mercury is made by means of a platinum wire usually entering from the bottom. Platinum wire or foil may be electroplated with mercury to produce useful electrodes. These require frequent replacing in order to ensure reliable voltage measurements.

Platinum (Surface Active)

Platinum black is often deposited on platinum electrodes to increase surface area where the electrode is used with a gas as in the hydrogen electrode. Voltage stability is improved by this procedure. Platinum black may be deposited by electrolysis from a 3 percent solution of chlorplatic acid. Electrodes must be thoroughly rinsed after treatment. Last traces of chlorine may be removed by the generation of hydrogen on the electrode. Platinized electrodes are sometimes heated to redness to convert the black to a grey deposit and reduce catalytic activity.

Gold

Gold-wire or -foil electrodes are also suitable for general application as indicator electrodes and are to be preferred over platinum for some systems where platinum is catalytically active.

SUSPENSIONS AND DEPOSITING CONDITIONS

The problems arise when the nature of the solution interferes with the measuring function of the electrodes. Where the liquid junction device is in contact with a highly charged sediment, a spurious lowering of the pH can occur. Serious potential effects are rarely involved in measurements on the dispersed suspension and the main precaution necessary is to prevent blockage of the liquid junction of the reference electrode. Sols may also give rise to residual junction potentials and gels sometimes cause sluggish glass electrode response. Sols and suspensions normally cause difficulties only if they are allowed to settle. Constant agitation of the process bath prevents this, and a reservoir-fed reference electrode minimized the flow of particles into the junction.

Although coagulated solids do not usually cause glass electrode problems, their presence and the formation of colloids can present serious difficulties. The material deposits on the glass electrode membrane, and depending on the nature of the precipitate and the temperature of the solution, inhibits the electrode response.

Film formation on the glass electrode surface is the principal problem. Where oils are dispersed in water, the globules may coagulate on the glass and inhibit its response. Where the measurement is on viscous materials such as glue or latex sluggishness is generally experienced. This is partly due to lack of water and glass electrodes requiring less hydration for their functioning are generally more satisfactory.

The most serious problem is loss of response due to adherence of dispersed oil on the membrane. This is more likely to occur with fine emulsions than with coarse ones where the oil tends to separate out. When measuring slowly flowing coarse emulsions, the electrodes should be situated high in the solution of dense oils and low in the solution for oils that float on water.

WAYS OF REDUCING THE DEPOSITION RATE

1. Careful choice of locating the electrode assembly, with a compromise between the greatest distance from the precipitation region and the desirability of close proximity to minimize process control lags.

2. To incorporate the assembly into the liquid circulating pipeline from the deposition tank.

3. Bleed arrangement which is used on effluent treatment plants.

4. Flow systems can be utilized in cases where finely divided particles can exert a cleaning action.

A number of mechanical devices can also be used to keep the electrodes clean, but eventually solvent cleaning is necessary for all types of deposit.

ELECTRODE CLEANING AND CLEANING DEVICES

In order to maintain their speed of response and accuracy of determination, the electrodes should be cleaned periodically. The frequency of cleaning and the method which should be applied are dependent on the nature of the process liquid.

In general, washing with water and wiping with cotton wool soaked in dilute hydrochloric acid will suffice, but in severe cases the electrodes may require soaking in solvent. Inorganic solvents are preferable for removing greasy and oily deposits. Alcoholic and acetone are commonly used but care must be taken that another deposit just as immiscible is not on the electrode. Organic solvents have a dehydrating effect on the membrane and only brief immersion in the solvent followed by soaking in hydrochloric acid should be employed.

Electrode cleaning frequency may be considerably reduced by the fitting of an in site cleaning device. One such device provides for a met of solvent to be periodically directed at the electrodes. Another technique, providing continuous cleaning without the introduction of solvents, employs an ultrasonic transducer attached to a thin plate and operated by a remotely mounted generator.

Leads

Screening is necessary for the insulated cable attached to the glass electrode. Polythene-insulated standard wire enclosed in metal braid is normally used with an outer sheath of PVC. At temperatures above 140°F, the hot solution itself acts as a screen for the immersed cable, but a covering of polypropylene or Teflon is used to insulate the cable core.

The length of the lead is limited by its capacitance, for this governs the reading response of the measuring instrument. The actual length that can be tolerated depends upon the capacity of the instrument being used and on the glass electrode resistance.

Bridges

It is often advisable to separate the two half-cell of a system to be studied and still maintain electrical contact. The salt bridge is used for this purpose. Such a bridge is

positioned to dip into each electrolyte and provide a conductive path with very slight diffusion of the electrolyte into either half-cell.

Important design criteria for salt bridges are:

1. A positive pressure between the salt solution in the bridge and the solution outside the bridge so that the direction of flow opposes diffusion into the bridge. Contaminants allowed to reach the half-cell through the bridge usually lead to unreliable measurements.

2. A barrier at the termination of the bridge such as a capillary, agar plug, round glass stopper, etc. provides high flow velocities at small volumes to avoid excessive dilution and contamination of solutions being measured.

This arrangement is effective in isolating the solution from contamination, reducing liquid junction potentials, and in isolating solutions where the temperature of the sample solution may be different from that of the reference electrodes.

MEASURING CIRCUITS

The emf from the pH electrode system is d.c. and the measuring instruments fall into two broad classes:

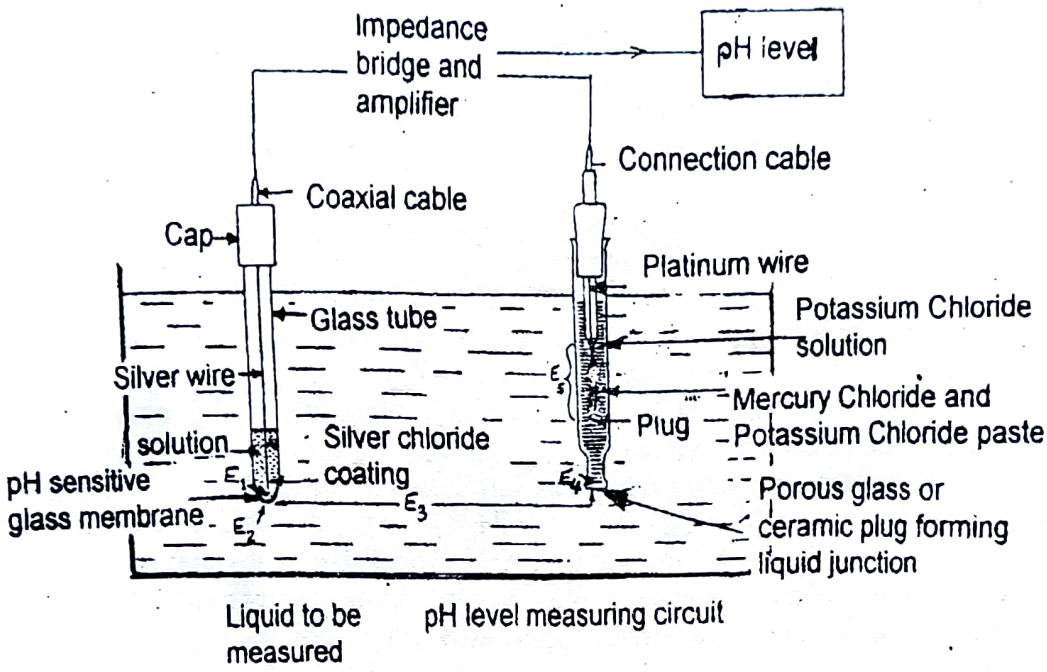
1. Those which amplify directly the electrode emf so that the output can operate a standard indicator, recorder and controller. This class involved a dc amplifier with very stable characteristics.

2. Those which compare the electrode emf with a known standard emf. The self-balancing potentiometer is the obvious instrument here, but it may involve the interposition of some form of dc-ac converter unit which transposes the dc emf signals to corresponding ac ones.

Whichever class is used, the following general requirements should be met:

1. The instrument to be substantially free from errors caused by supply voltage or frequency variations.
2. The zero reading to be stable over long periods.
3. The current taken from the electrode system to be extremely small to avoid polarization effects.
4. The design to include compensation for variation in characteristics of components or be such that the output is independent of such variations.
5. Temperature compensation to be included, either manual or automatic.
6. Facilities to be included for easy checking of calibrations.

TYPICAL MEASURING CIRCUIT FOR pH CONTENT OF A LIQUID



pH Level measuring circuit

The circuit consist of two electrodes placed in a liquid of which pH level we determine. To complete the circuit the liquid must be of conductive substance and a impedance bridge is used to determine the voltage drop between the two electrodes.

A stable voltage E_1) will be induced by the internal solution of the glass membrane electrode. This will be a known constant value. Due to the ionization taking place at the pH sensitive glass membrane caused by the amount of hydronium ions present another voltage E_2 will be induced.

If the reference voltage induced by the calomel electrode, E_4 and E_5, differs from the value at E_2 an potential difference, E_3, will exist between the two electrodes. This potential difference, E_3, will be increased or decreased by the amount of hydronium ions present at the glass membrane.

A impedance bridge is used to determine this potential difference, E_3, and amplify the signal to be send to a measuring instrument calibrate in a pH scale.

POTENTIAL MEASURING INSTRUMENTS

Potentiometer-Galvanometer Systems.

The potentiometer-galvanometer system affords a simple means of measuring oxidation-reduction potentials.

It is essential that measurement be made with no appreciable withdrawal of current in order to avoid polarization at the electrodes.

Vacuum-tube Voltmeter.

The mosfet voltmeter as a pH meter is available in many commercial models. These instruments incorporate potentiometer and galvanometer circuits and in some models can be switched from pH scale readings to millivolt scale readings. (Oxidation-reduction potentials are measured on the millivolt scale: pH meters for Redex measurement.) Measurements of oxidation-reduction potential is a very simple operation with the pH meter. Commercial electrodes are available from the various manufacturers for use with their instruments. Special electrodes may be made if required.

Potentiometric Titrations

The direct-reading pH meter is most convenient for the determination of titration end points. Electrode potential measurements are made as above as reagent is added to the sample and the electrode potential is plotted against the volume of reagent added. For simple titractions of univalent, strong electrolytes the end point of the titration is at the maximum slope of the curve, which is the mid-pint of the steep portion.

If a titration is carried out in such a system and at the end point the reversible reaction is no longer possible, one electrode becomes polarized and the indicator current suddenly stops flowing. This is called a "dead-stop" end point and is extremely sensitive.

Automatic Titration and Control

Potentiometric measurement of oxidation and reduction potentials, as applied both to direct measurement and to the determination of titration end points, is a procedure well adapted to automatic operation. Automatic instrumentation for titration and control of both batch-type and continuous flow processes in the field of pH is well advanced and the growth of this phase of instrumentation in the oxidation-reduction field is in progress.

Factors Affecting Accuracy of Measurement

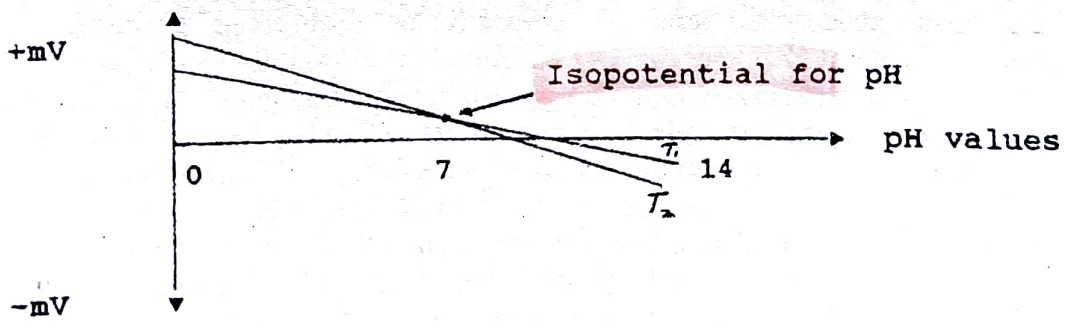
1. Temperature
2. Mixtures and reaction rates.

3. Electrode surface conditions.

Temperature

The main requirement is to correct for scale length changes, which can be accurately calibrated and compensated for by altering the resistance of the feedback circuit. This change in resistance may be effected by means of a resistance thermometer, directly measuring the sample temperature, or by means of a manual control which may be set to the expected mean sample temperature. It is also possible to compensate for zero shift by arranging that the current in the feedback circuit is zero at the isopotential value of the electrodes. This is effected by means of a variable resistor known as the isopotential control.

For certain applications where little variation is expected in the sample temperature, and where standard electrodes of a chosen isopotential value are used, it is possible to dispense with the methods of temperature compensation described above. Instead, a pre-set temperature control is used which may be set at a value about which a change of no more than ± 1 is expected. The accuracy of such a system is greatest when the pH value being measured is close to the isopotential value of the electrodes used, but should not be greater than 0,1 pH at any point over the whole range of 0-14 pH.



Variation of electrode emf with temperature

It can be calculated that potential changes at a rate of 0,186 mv/°C. The slope mv/pH for several temperatures is given below:

Temp °C	*mv/pH*
0	54,20
25	59,14
50	64,08
75	69,02
100	74,00

Commercial pH analyzers generally provide manual or automatic temperature compensation or both. In general, this is done by changing the gain or sensitivity of the instrument. For example, an automatic temperature compensator consists of an element in contact with the process stream which changes in resistance with temperature. At the isopotential point the electrodes will be self compensating.

A manual thermocompensator is calibrated variable resistor. It must be pointed out that aqueous solutions, even buffers solutions, change in pH with temperature. This is especially true of highly alkaline solutions that exhibit quite dramatic changes in pH with temperature, whereas acid solutions change only to a minor degree. These changes are very real and no attempt is made, or should be made, to affect them with thermocompensation.

MIXTURE AND REACTION RATES

The combination of the oxidized and reduced form of a given substance is called an oxidation-reduction couple. When two or more such couples are present in the same solution, a reaction must take place between them until equilibrium is reached. At equilibrium the oxidation potential calculated for the solution from the ion concentrations using the Nernst equation will be the same for either ion couples. The infinitely high voltage called for by the Nernst equation when the ratio of the ion concentrations approached infinity is obviously never observed in actual solutions. Substantially pure oxidizing agents, for example, are always able to react with minute concentrations of impurities or with water to produce small amounts of the reduced form of the couple. Drifting electrode voltages are obtained only when the concentration of the ion present in smaller amount reaches a level of about 10^{-5} mole per litre. In most practical systems one ion is usually reduced to a concentration below this limit. The oxidation potential is determined by the other ion couple. Thus in the calculation of titration end-points potentials, data for ion concentrations of one couple are used on one side of the end point and data for the other couple on the other side.

When oxidizing or reducing agents which react slowly are present in solution, electrode potentials are unstable and sometimes drift slowly over long periods of time. Solutions or reducing agents which are exposed to oxygen often show this behaviour.

ELECTRODE-SURFACE CONDITIONS

Metallic electrodes are subject to poisoning by trace quantities of impurities in electrolytic solutions and in some cases by exposure to the atmosphere. Precautions must therefore be taken in preparing or storing electrodes and in following procedures developed for producing reproducible surface conditions in order to obtain reliable readings on various systems. Treatment with strong oxidizing agents such as hot chromic acid or nitric acid solution has been found effective for preparing platinum, gold and

tungsten electrodes for use. Thorough rinsing is required following treatment. Cleaning with fine abrasive is sometimes required for metal electrodes. Electrode activity is generally more stable when electrodes are stored under water.

Some difficulties in making precise measurements with glass-electrode systems are encountered because of electrical leakage due to moisture on the electrode bodies. The use of silicone coatings on the electrodes - particularly glass electrodes - has been found effective in eliminating this source of trouble.

AMPLIFIERS

The relatively high electrical resistance of glass electrodes imposes stringent design requirements on pH meters, since without amplification, satisfactory discrimination will not be achieved in the measurement. Also, the measurement must be made in open-circuit conditions to avoid polarization of the electrodes; therefore an amplifier unit of very low and stable leakage current is required.

Direct Coupled Amplifiers

The first stage in the amplification circuit consists of a semi-electrometer valve with a grid current of less than 10^{-11} amperes. The potential difference between the electrodes is applied to his valve, the potential of the glass electrode going to the grid. Careful stabilization of the internal power supplies is essential to care for the wide variation in external supplies which may occur.

A well-designed pH meter using this type of amplifier should maintain a zero stability within 0,02 pH for periods of more than 24 hours.

Vibrating Condenser and Contact Modulated Amplifiers

Both of these types use on input system which converts the DC signal from the glass electrode into AC permitting the use of more stable AC amplifiers in the succeeding stages, in the vibrating condenser amplifier, the electrode potential difference is applied to the fixed plate of the condenser while the other plate is driven by a solenoid to produce a change in capacitance at a fixed gas-filled and sealed to prevent deterioration. The zero stability of pH meter using a vibrating condenser amplifier will remain within 0,005pH for periods or more than 24 hours.

In the contact modulated or low frequency chopper amplifier, a high grade condenser is alternately charged by the electrodes and switched every second to the input of a self-balancing potentiometer via an amplifier. If a change in pH occurs, an amplified pulse is fed to the galvanometer and the slide wire adjust itself until the potential across the slider is equal to the voltage presented by the condenser. This arrangement has the

advantages that the input resistance is solely limited by the physical insulation of the capacitor and switch mechanism, and the zero stability is independent of the input amplifier and is governed only by the accuracy of the self-balancing potentiometer.

Receivers (any receiver can be connected to the pH measure)

pH meters are built in various configurations to suit particular industrial requirements. Most instruments are designed for either wall or panel cases, which may be hosed down without risk of damage. Accessibility is important of servicing, and instruments are designed so that they may easily be withdrawn form their cases.

Increasing use is being made of modular systems, particularly where the measurement of pH is incorporated as part of a continuous process. In such cases a pH amplifier, and indicating meter and one or more control conduces are assembled in an industrial instrument case.

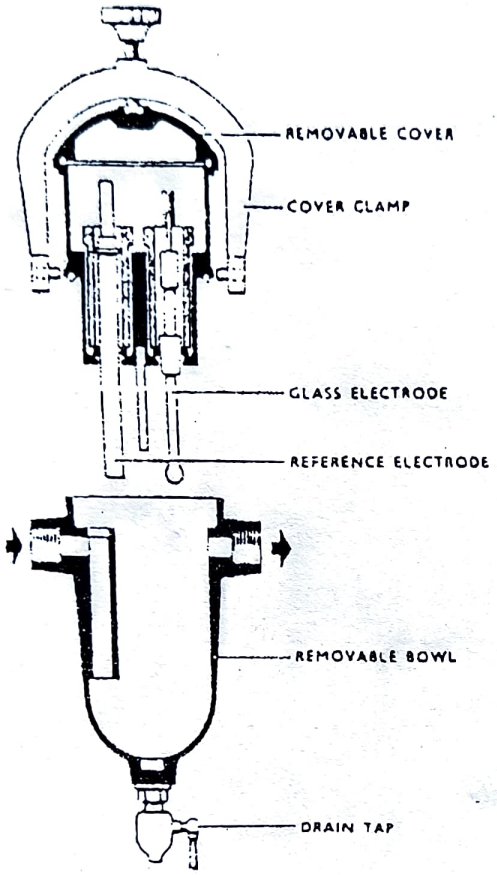
Depending on local requirements, a pH measuring system may incorporate a recorder in addition to, or instead of, an indicating meter. Monitoring of pH at a number of pints may be carried out using a single measuring instrument in conjunction with an electrode switching box and a multi-pint recorder. The switching box connects the electrodes to the measuring instrument in synchronism with the stepping mechanic in the recorder. Simple two-way manual switching boxed are also available.

When measurements are to be made in hazardous areas, equipment certified as intrinsically safe must be used. If this is the case, then the electrode assembly is incapable of seeing sufficient energy to cause ignition and therefore only the readout device requires explosion proof housing or being mounted remotely in a non-hazardous area.

ELECTRODE SYSTEMS FOR INDUSTRIAL PROCESSES

It will be realized that the electrodes shown in their simple state may not be suitable for industrial processes without further protection and it is proposed to give brief details of two typical industrial electrode systems, the continuous flow and total immersion type.

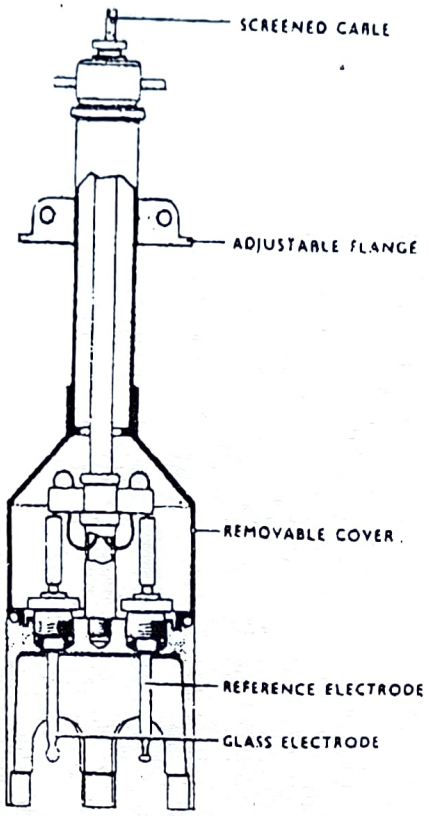
THE CONTINUOUS FLOW TYPE



Flow type pH electrode assembly

In it the measuring glass electrode and the reference electrode are fixed in a removable cover assembly, but dip down into the lower bowl when in use. The process liquid flows from left to right round the electrodes.

THE TOTAL IMMERSION TYPE (DIPPING TYPE)



Dip type electrode assembly

The measuring glass electrode and the reference electrode are fixed to the assembly with a cover to allow excess to electrodes. The assembly will then be totally immersed into a tank.

APPLICATIONS OF SOLUTION POTENTIAL MEASUREMENT

Solution potential measurements have important applications in such widely varied chemical processed as biological systems and dyestuffs, sewage treatment water purification and chemical process determination of water in organic materials by the Karl Fischer titration. Another recently developed application of considerably importance is the determination of chloride ion. There are not simple rules by which any possible application of solution potential principles can be tested because of the wide variety of electrode systems and such practical considerations as electrode maintenance.

AUTOMATIC CONTROL

OPEN LOOP CONTROL (Manual control)

In an open loop there is no feedback signal to a controller to reset the process. Any deviation in the process must be corrected by hand. Fig .1. shows such a control loop.

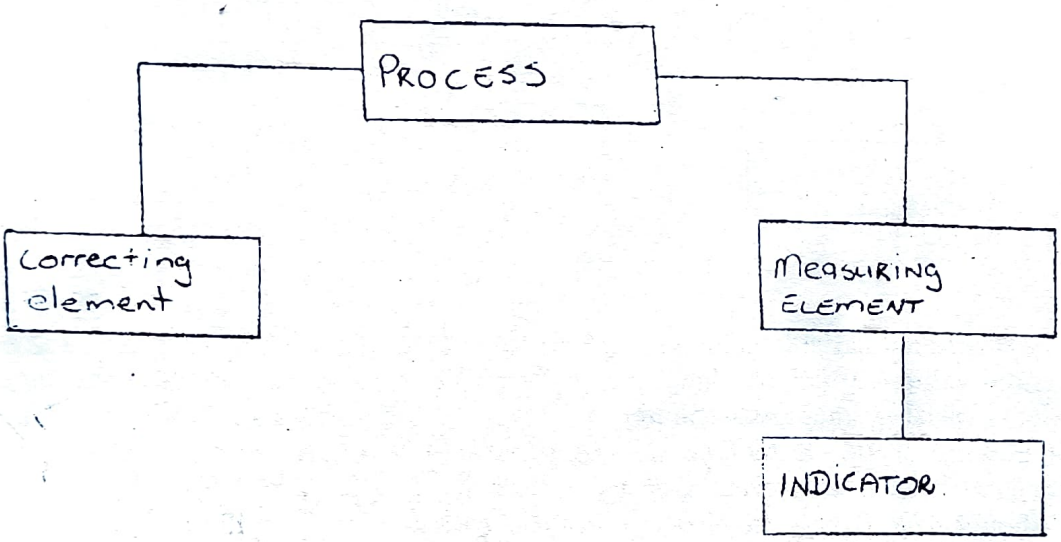
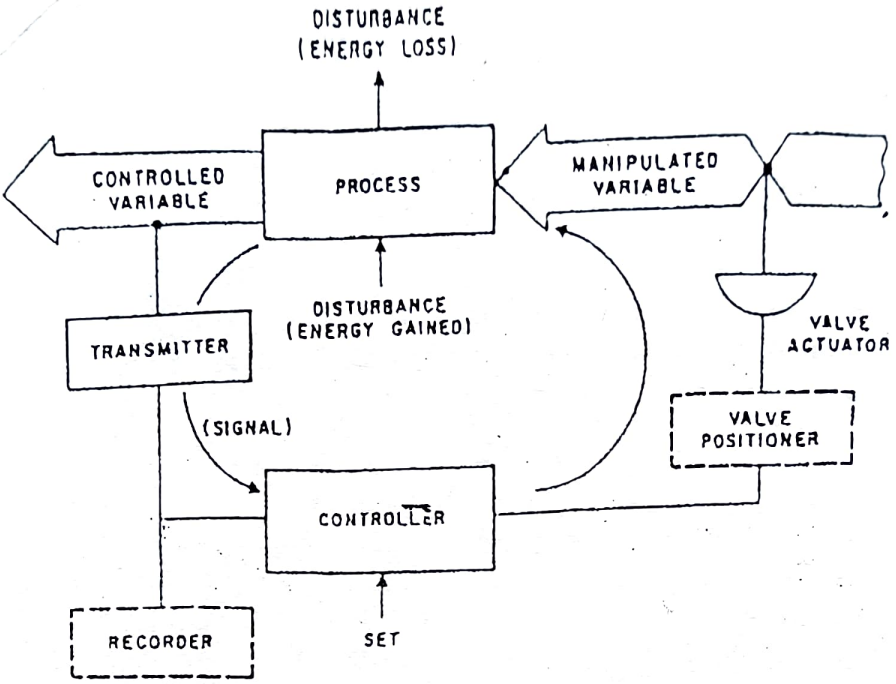


FIG.1

CLOSED LOOP CONTROL (Automatic Control)

This is the most common type of control mechanism used. Any process in which the process variable under control is measurable allows the use of such control strategy.



The controlled variable is the process variable which we are trying to maintain at some desired value (set point). Industrial processes are characterized by the many types of control variables that are encountered, e.g., temperatures, flows and levels. The function of the transmitter is to quantify this variable in terms of signals, which could be pneumatic, electric, hydraulic, or just a mechanical output like the position of a lever. It should be remembered that not all these measurements are linear, i.e., the output signal is not necessarily linear with respect to the controlled variable or, even if it is linear, is not a direct reading of the controlled variable, e.g., the controlling flow based directly on the pressure drop ΔP across an orifice meter. In all these cases, linearization can be achieved, thereby making the transmitter output a direct indication of the controlled variable.

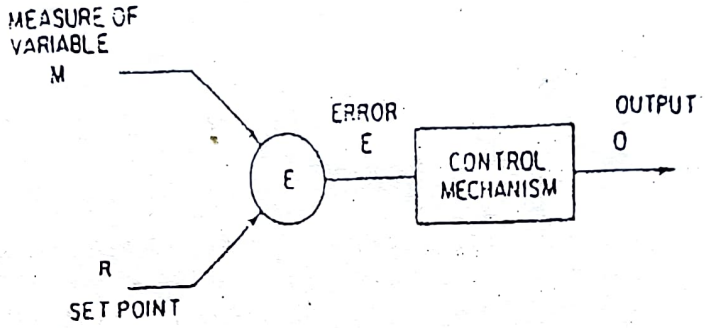
The manipulated variable is that which the controller varies in its efforts to maintain the controlled variable at set point. The controller output is a signal to, say, a valve actuator, causing the valve to move to a position which would depend upon the value of the signal, type of valve, and the process conditions under which it is operating. The valve positioner causes a fixed relationship between the controller signal and valve position.

PNEUMATIC CONTROLLERS

The basic functions of a pneumatic controller include:

1. *receiving* a measure M of the value of the variable being controlled;
2. *comparing* that value to a reference or desired value dv;

3. *determining the magnitude and direction* of any deviation or error E; and
4. *providing an output O* as some function of the deviation.



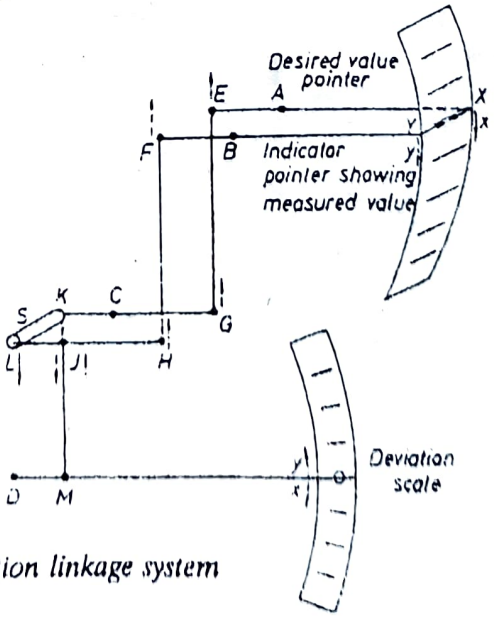
DISCRIMINATION

When we want to control a process we must be able to discriminate between what we want (set point) and what we got (measure of variable). Any difference between these values are known as an error or deviation. This error must be corrected by means of control, actions and if the measure variable doesn't reach the set point after control action has occurred, we call the value "off-wet".

There are two methods of discrimination most commonly used namely position balance and force balance.

POSITION BALANCE

The output pressure is a function of the position of the set point and measure variable. A discrimination linkage system will explain the position balance controller.



Discrimination linkage system

Let us assume the set point to be adjusted to x % value. If the measured value y is equal to x the error will be zero. If we now change the set point by adjusting the x value, the y value will not be at the same point thus creating a error condition. The controller will rectify the error until y value equals x value.

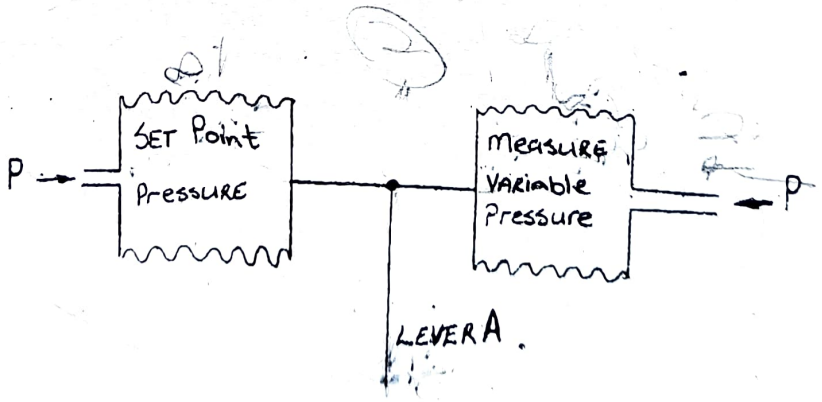
The action taking place is as follows:

When x is moved down point 1 will move up around A. Point 2 will move up accordingly and cause spacer S to down around C. This will cause point 4 to indicate a deviation down. This motion can be used to open or close a flapper nozzle system. Point 3 will be stationary because no error is yet detected until the deviation will change the output. After a time point y will be changed by the measured value until the error is nil again. The same action occurs with a change in load condition is sensed by the measured value and moves y up.

When x is not at the same position as y the linkage will be deformed and a unbalanced condition will induce a force. This force can be used to rectify the error with aid of an flapper nozzle system until x equals y.

FORCE BALANCE

The output pressure is a function of the force acted out between set point and the measure variable. To explain this type of balance we must look at two types of pressures acting on each other.



We can see that set point pressure must equal measure variable pressure to balance the system. When an error exists, the measured pressure will change and cause the bellows to increase or decrease. This in turn will change the position of lever A. Again we can connect a flapper nozzle system to the levers to rectify the deviation. An adjustable coil spring could replace a set point pneumatically signal.

converts mechanical movements into a pressure signal

PNEUMATIC RELAYS

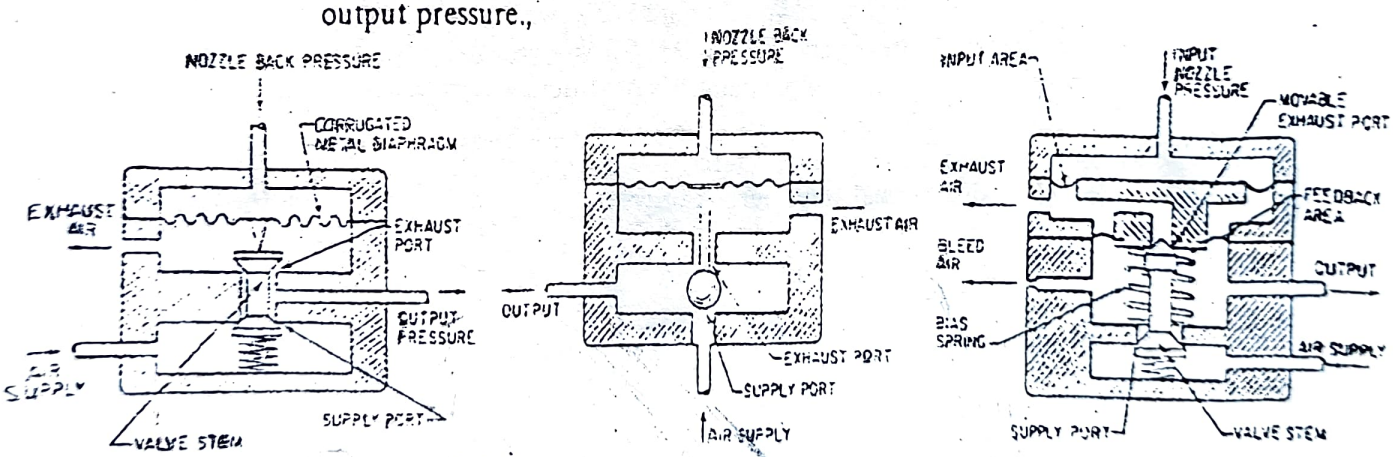
Other than the discrimination systems there is also a Pneumatic Relay present in the controller. The relay has very little flow resistance and can quickly fill the control valve motor under most circumstances. We can refer to a relay as a pneumatic amplifier.

Three types of relays are available:

1. Continuous bleed type

   - ♦ Direct acting - an increase in nozzle back pressure resulting an increase in output pressure.
   - ♦ Indirect acting - an increase in nozzle back pressure resulting an decrease in output pressure.

2. Non-bleed type - an increase in nozzle back pressure resulting an increase in output pressure.,



Direct acting *Indirect acting*

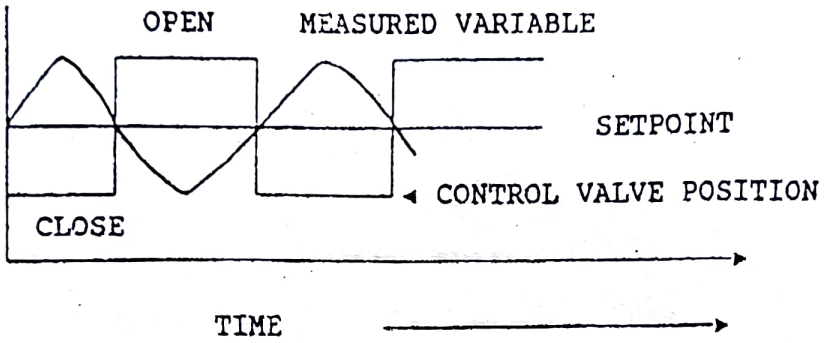
Continuous bleed type *Non-Bleed type*

Five types of control are commonly used namely:

1. On-off or two step control.
2. Proportional control (one term control).
3. Proportional and integral control (two term control).
4. Proportional and derivative control (two term control)
5. Proportional and integral and derivative control (three term control).

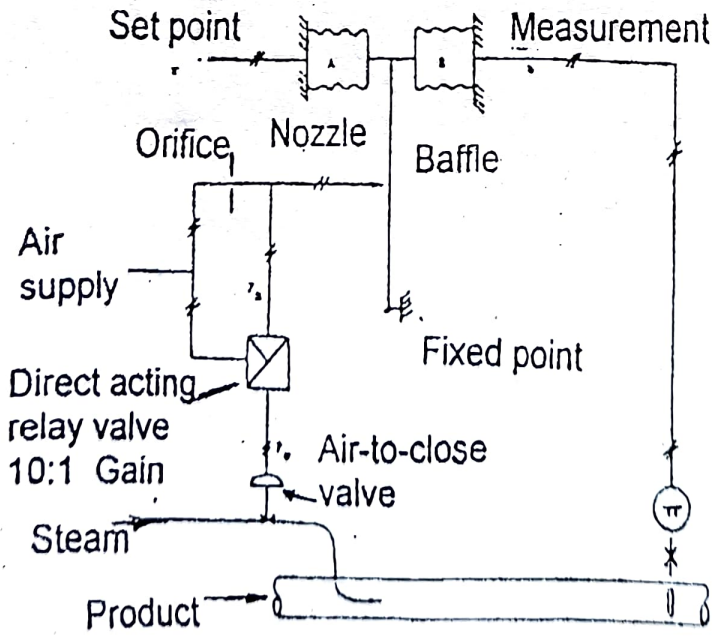
On-off Control

On-off control is the simplest form of control. The correcting unit is either in the open or the closed position.



As soon as the measured variable exceeds the control point, the control valve is closed. It will remain closed until the measured variable drops below the set-point value. At this point the control valve will fully open. The measured variable will oscillate about the set-point value with an amplitude and frequency that is dependant on the process,

This type of control is obtain by using a flapper nozzle and relay system. These components acts like a differential amplifier. A force balance pneumatic heat controller will be used for an example.



On-off pneumatic controller

The controller has no feedback and operates indirectly to the deviation. When the error increases the flapper is moved towards the nozzle. The pressure will build up and cause the relay to de-amplify the error to the control valve which in turn will shut the air supply down to rectify the deviation. The error will decrease pass the set-point and move the flapper away from the nozzle resulting a drop in pressure again. The relay will amplify the error to the control valve which will open up the control valve to rectify the deviation. This action will be repeated and a oscillation above and below the setpoint will occur.

PROPORTIONAL CONTROL

It is important to realise that with a proportional controller each valve position is related to a definite position of the measurement. The controller may be adjusted so that the valve is positioned to bring the measurement to the desired value, but the measurement will be maintained at this point only as long as there is no change in load or other variables.

As soon as a change in load take place, the controller will try to bring the measured value back to the setpoint value, but the controller won't be capable of doing so. There will be a certain off-set between the setpoint value and measured value. The magnitude of this off-set depends on the proportional band setting and the magnitude of the load change.

The effectiveness of this type of control is limited to small load changes.

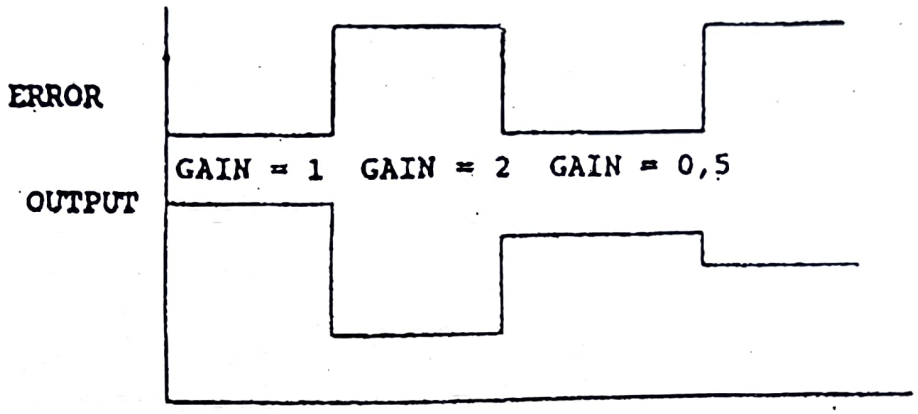
The narrow band through which the input must change to cause the output to change from the bottom to the top of its range, is called the proportional band. Continuous cycling of the controller output and of the controller variable may occur - indicating that the controller gain is too high.

The smaller the proportional band setting of a controller, the smaller the off-set will be. One must keep in mind that although the proportional band must be as small as possible, the setting must be such that the system is stable. With a total loop gain of one, the system will oscillate at a constant amplitude. If the total loop gain is greater than 1, the amplitude of the oscillation will increase. With a total loop gain of less than 1, the oscillation will decrease until it smooths out completely.

Most industrial controllers have a gain adjustment that is expressed in percent proportional band:

$$Gain = \frac{100}{\%\ Proportional\ band}$$

From that we can see that if the proportional band set at 100 % , the gain of the controller is 1.

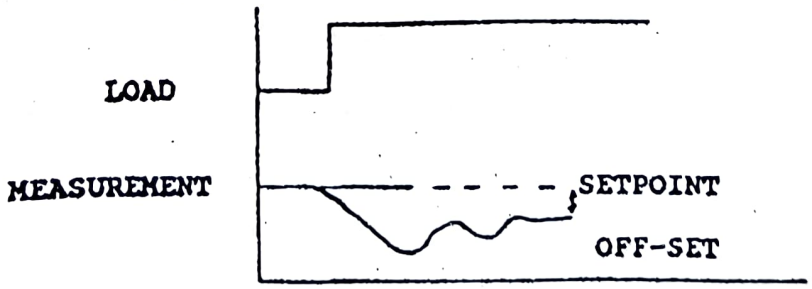


Open loop gain response

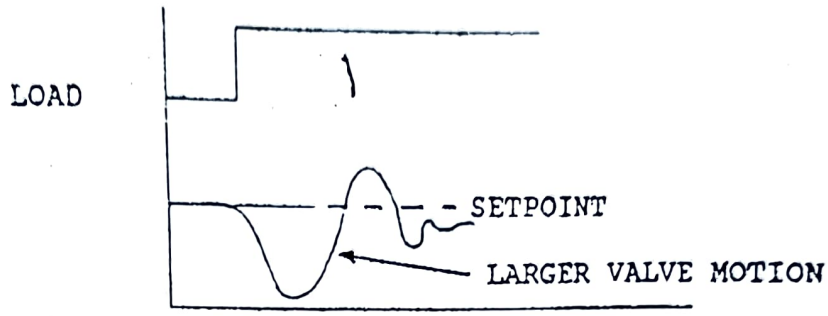
When a deviation occurs the control valve must move a certain distance to rectify the error. These motions are given in %, errors are given in a negative (direction smaller than set point) and positive (direction larger than set point) percentage while valve motion is given from zero (valve fully closed) to hundred (valve fully open) percent.

If, for example, a error represents a decrease in fluid flow (- % error) the valve must open to rectify the error. The fluid flow will increase (+ % error) resulting a decrease in error but the valve will close. A new smaller error exists (- % error) which will open the valve again and decrease the error even more and increase the fluid flow (+ % error). We can see a oscillation taking place until the operation smooths out. This oscillation will cause the measured value to deviate from the set point.

This small deviations is called off-set.



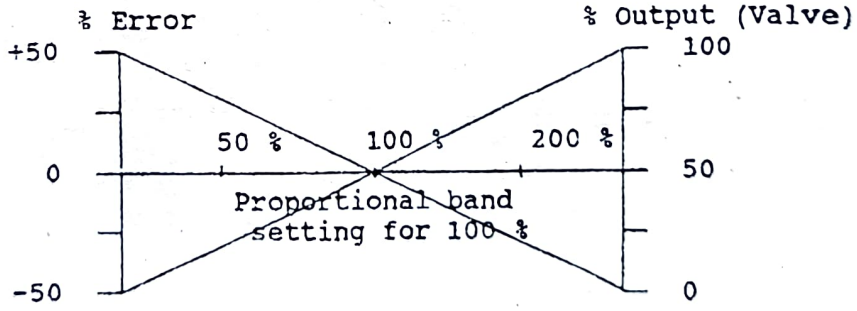
To solve the problem of off-set we can open the valve more for the same error to start the oscillations over the set point.



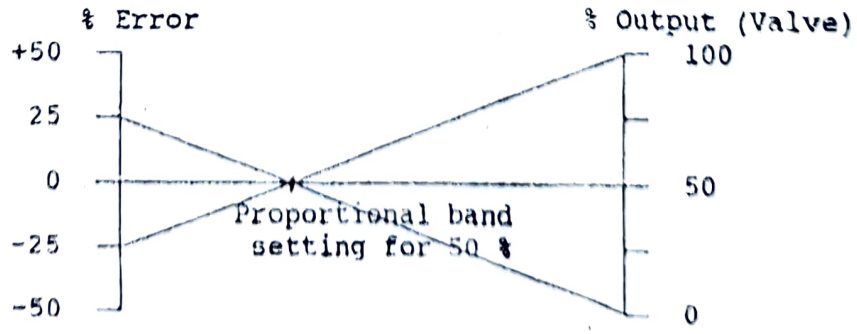
This type of control is called proportional control. To obtain a different valve motion we have to adjust the proportional band setting.

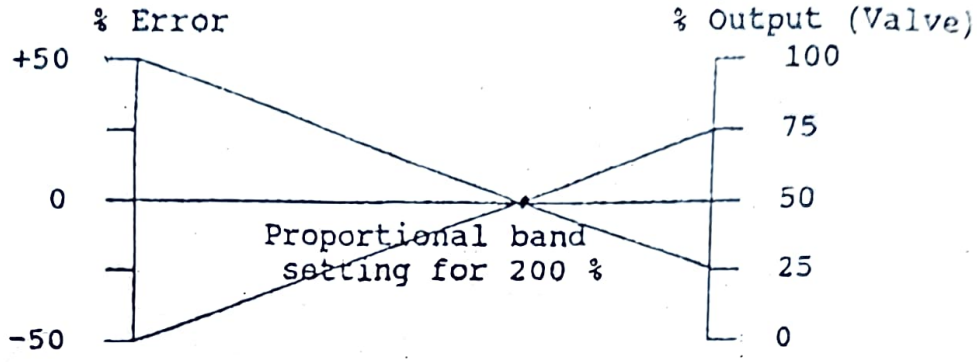
The proportional band of a controller is defined as being to percentage of the scale through which the measurement must move in order to produce a travel of the valve from one extreme to the other. Thus a 100 % proportional band controller, is one where 100 % scale movement of the measurement is required to give 100 % valve movement.

A bias is usually provided to allow the output to be set at approximately 50 % whenever the measurement equals the setpoint as the measurement equals the setpoint. As the measurement deviates from the setpoint, the output will change from 50 %.



If the proportional band is 100 % it means that 1 % error will adjust the valve 1 %. When we move the proportional band adjuster to the left we decrease the band and 1 % error will adjust the valve 2 % for a 50 % setting

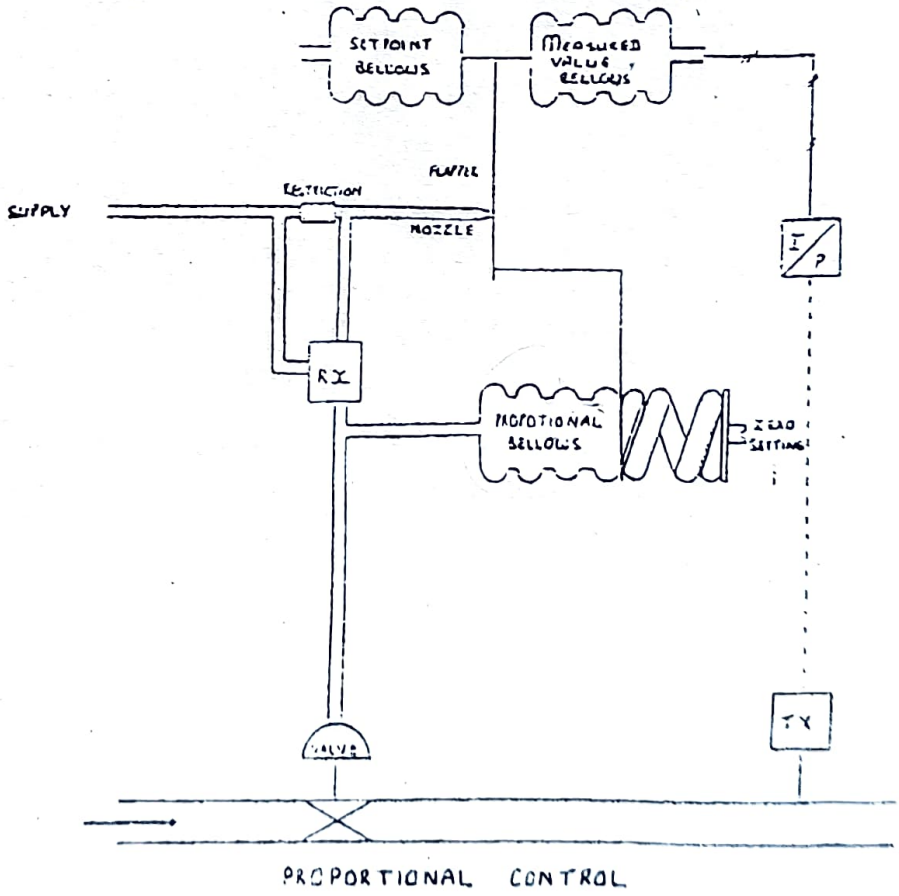




PROPORTIONAL CONTROLLER ACTION

In proportional controller action the controller output signal is proportional to the deviation. The on-off controller has a big disadvantage. The output is independent of the magnitude of the deviation. It is either a maximum or minimum value. To overcome this problem we make use of feedback. The proportional controller has got negative feedback and act directly opposite to the magnitude of the deviation depending on the gain setting (band adjust) of the controller.

Let us consider the pneumatic force balance controller (proportional action alone).



PROPORTIONAL CONTROL

Each controller must have an error-detecting division that determines the difference between the measured variable and reference or set point. This function is performed

by the A and B bellows. Any change in measurement or set point causes a flapper movement which is sensed t the nozzle. The combination of nozzle, flapper and restrictor is the heart of any pneumatic controller. Its counterpart in other controller systems is a high-gain transducer or amplifier. By proper selection of nozzle and restrictor diameters, for example, sensitivity or gain may be as high as 240 kPa/mm. With this high sensitivity it is only necessary to have a change of 30 μm at the flapper to change nozzle pressure Pa from 20 to 100 kPa.

Thus, it is easy to see how small measurement changes at bellows B could cause an on-off response in P_n. Although a high-gain or on-off controller could be made by connecting P_n directly to the control valve, this is seldom done. Most high-gain transducers or amplifiers are signal rather than power devices, and as such perform poorly under direct loading. In the nozzle-baffle-orifice combination, for example, air flow to inflate the control valve motor would be limited by the orifice.

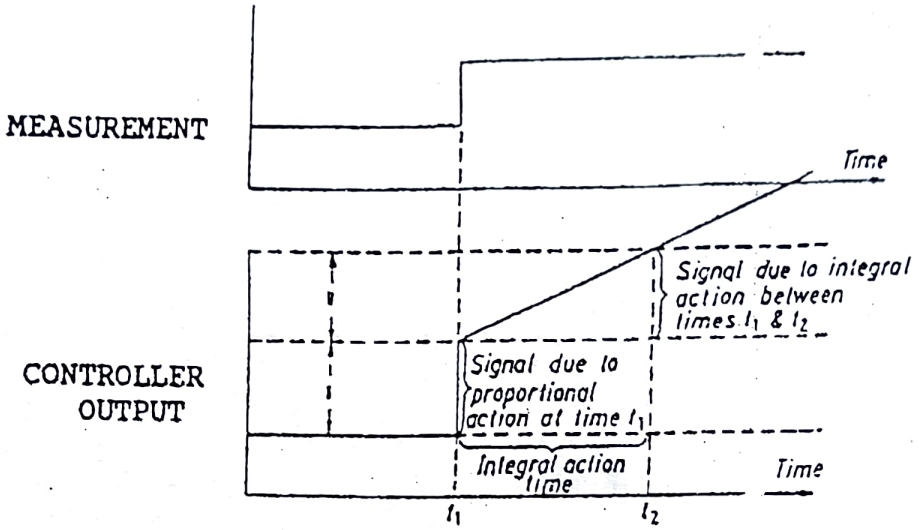
In pneumatic controllers the power amplifier is usually referred to as a relay valve. The relay valve is a direct-acting amplifier with a gain of 10. Thus, a change of ? in P_n causes a change in output pressure P_o. The relay valve has very little flow resistance in comparison to the orifice and can quickly fill the control valve motor under most circumstances.

With a gain of 10 in the relay valve, a flapper movement of only 3,8 μm will cause P_o to vary from 20 to 100 kPa to obtain proportional control, it is fairly obvious that some technique must be found to reduce this high gain. This is achieved through a feedback circuit. Since controller output is a function of flapper motion, reducing overall flapper motion should reduce output and provide reduced controller gain.

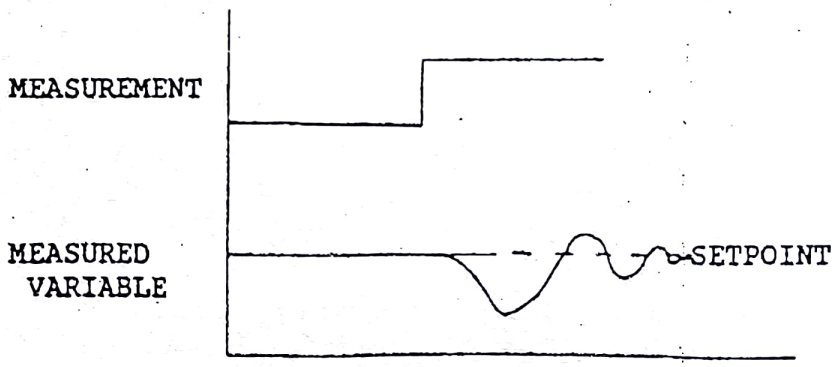
When more accurate control is required a proportional plus integral controller must be used to overcome the problem of off-set with large varying loads.

PROPORTIONAL PLUS INTEGRAL CONTROL

If offset cannot be tolerated, integral controller action must be added. Integral action will integrate any difference between measurement and setpoint; Till this difference is zero.



INTEGRAL TIME
OPEN LOOP RESPONSE

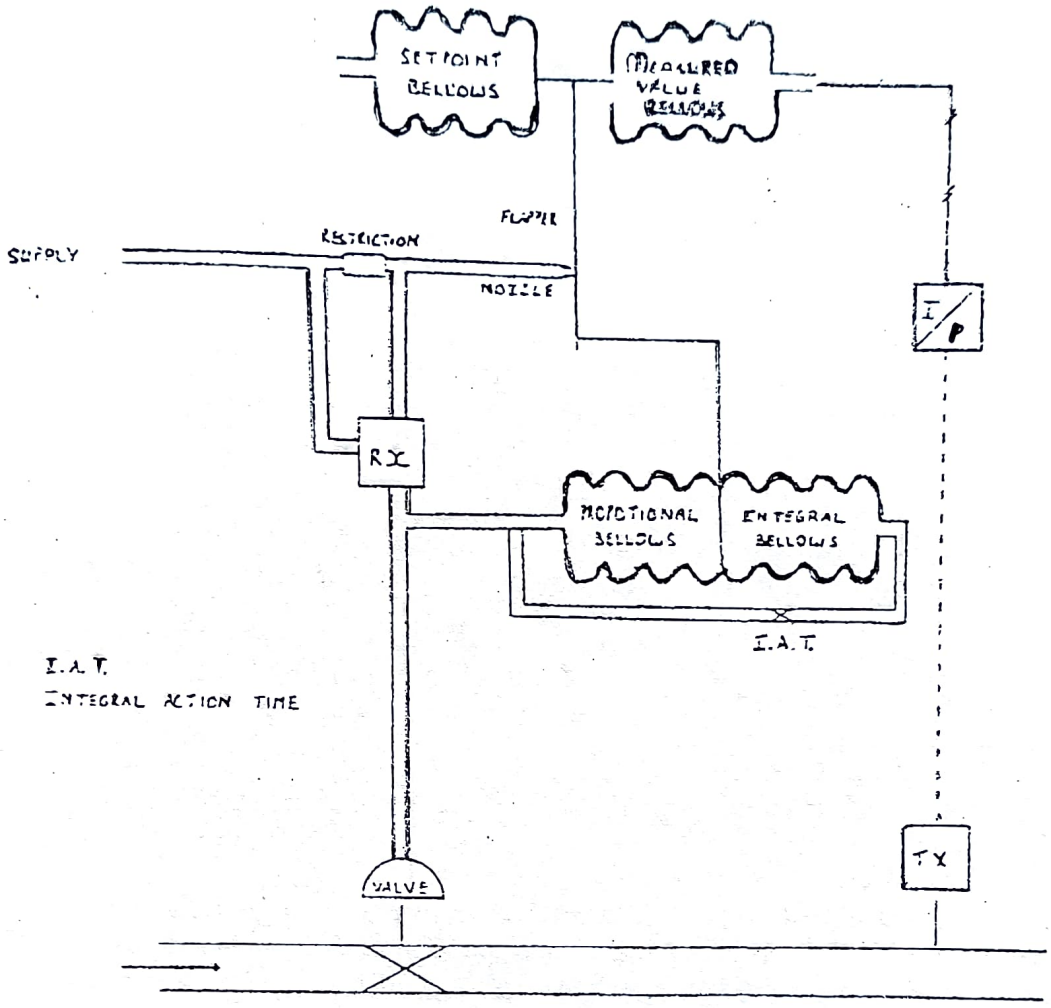


CLOSED LOOP RESPONSE

INTEGRAL CONTROLLER ACTION (Reset)

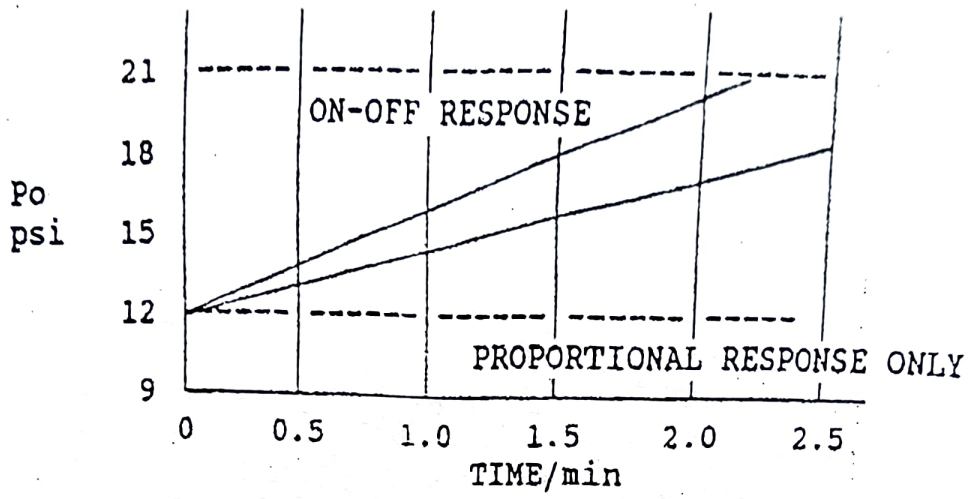
Offset, as we have seen, is inherent in proportional control, and another type must be introduced to give a fully compensatory effect. Integral controller action can achieve this. It is defined as an action in which the controller output signal changes at a *rate* proportional to the deviation.

Let us consider the Pneumatic Force Balance Controller with proportional and integral actions.



Proportional plus Integral controller

Assume equilibrium process conditions with a controller output pressure of 60 kPa and valve I opened approximately to its mid position. If product flow is increased, product temperature will decrease and the flapper will move away from the nozzle. Initial response is a decrease in output pressure P_o to the control valve motor and to bellows E. The pressure feedback to bellows E limits the output pressure change by repositioning the flapper through the gain adjustment mechanism. To be more specific, the controller provides proportional response. Due to the change in output pressure, a pressure differential now exists across valve I. This causes air to flow from bellows H through valve I and has the same result as resetting the zero spring. This moves the flapper away from the nozzle and produces and additional decrease in output pressure. The effect is to increase controller gain. This procedure of course takes time, which allows added steam flow to increase product temperature and reposition the flapper through bellows B. The pattern of events is hard to follow due to continuous changes and can be seen much better by examining the controller output to a fixed input change. Important points to note are (1) initial proportional response, (2) gain increase as a function of flow to or from bellows H, and (3) resetting effect of bellows H. It should be evident at this point why this type of control response was called automatic reset and has developed to reset control.



Proportional plus reset controller output responses to an input step change for different values of T_r and K represents proportional band

INTEGRAL ACTION TIME (IAT)

IAT, T_R or reset time is the time required for reset action to repeat the change caused by proportional action (min/repeat).

This time effect is directly related to the setting of valve I, as previously noted. For example, assume valve I was changed from its midpoint to a more closed position. Due to increased valve resistance, a different reset time would result.

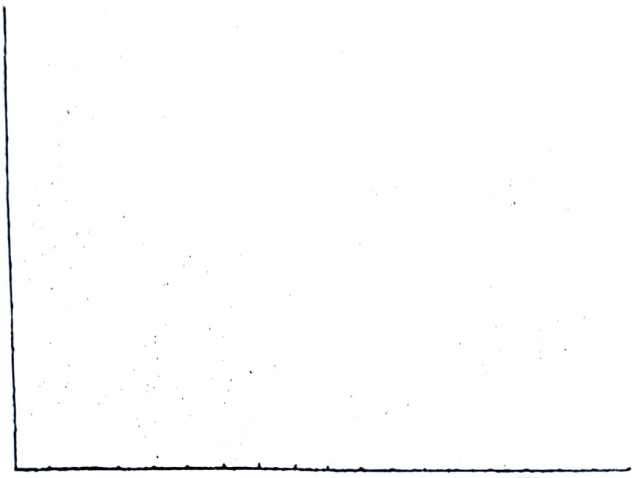
DERIVATIVE CONTROLLER ACTION (Rate)

It might be thought that a combination of proportional and integral controls gives all that is necessary for normal processes. This is true to a large extent, but plants with large transfer lag or distance velocity lag may be difficult to control.

To avoid continuous hunting, the proportional band may have to be made excessively wide and the integral time large. When load changes occur, big deviations take place and recovery time is long. To counteract this, derivative control is used. It is defined as a type of controller action in which the controller output signal is proportional to the rate of change of the deviation.

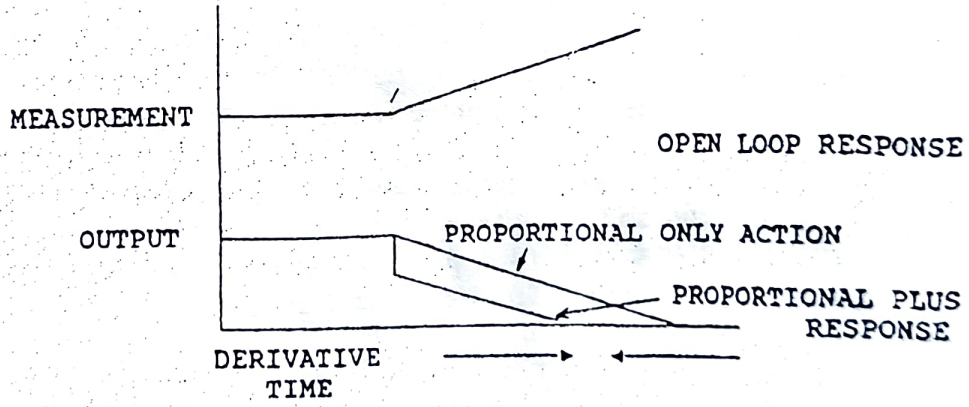
Whereas proportional and, to a lesser extent, integral control can be used alone, derivative control must be used in association with one or both of the others.

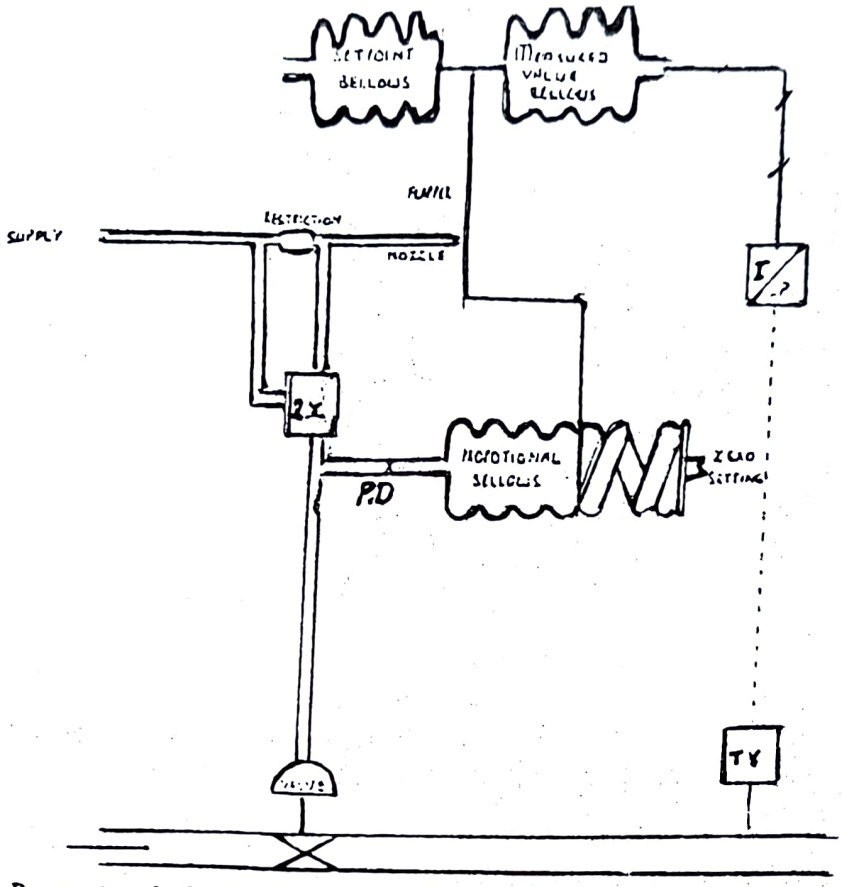
Although reset action eliminates offset, it does nothing to help problems of period and initial overpeak. In fact period and initial overpeak are increased when reset action is used.



Time responses for load changes using P and PI controllers

Derivative action occurs whenever the measurement signal changes. On a measurement change, rate action differentiates the change and maintains a level as long as the measurement continues to change at the given rate. Under steady-state conditions, the derivative acts as a 1 to 1 repeater. It has no influence on a controller's output. By reacting to a rate of input change, derivative allows the controller to inject more corrective action than is initially necessary in order to overcome system laziness.





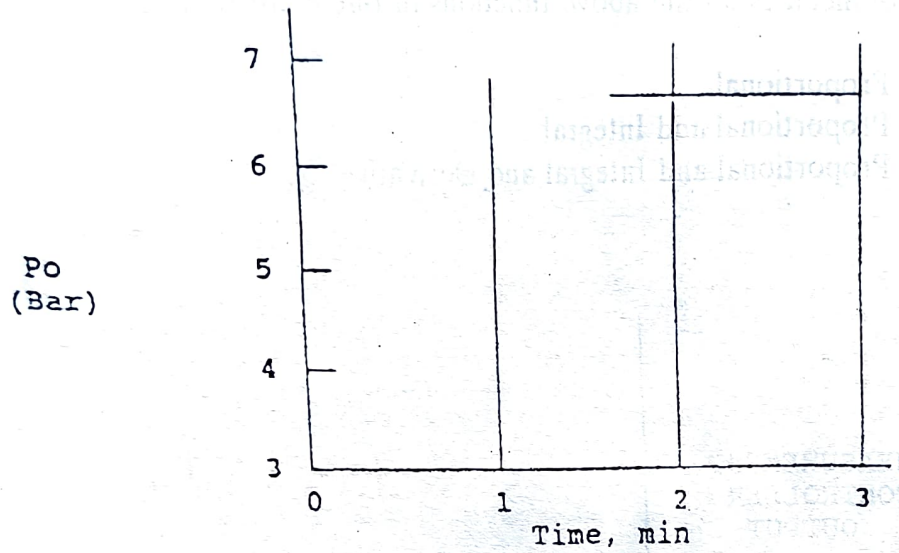
Proportional plus derivative controller

The basic controller circuit is identical to the proportional controller with one exception.

Valve D has been added in the feedback line to bellows E. If valve D is closed at some fixed pressure level proportional feedback is eliminated and the controller acts as a high-gain on-off controller. With valve D open, proportional control results. Thus, by varying the opening of valve D, controller gain can be varied from a high-gain on-off response to reduced gain of proportional action.

One problem in analyzing rate or derivative control is examining initial response. Assume valve D to be somewhere near its mid position with the controller in equilibrium and an output pressure of 300 kPa. As before, an increased product flow rate causes a decrease in product temperature and moves the flapper away from the nozzle. Theinitial response is decrease in output signal P_o to the control valve motor. With asimple proportional controller, the pressure in bellows E also decreases and limits change in P_o by repositioning the flapper through the gain adjustment mechanism. With valve D partially closed, however, a pressure decrease in bellows E is delayed. This in turn delays proportional feedback action and allows a greater change in output pressure. This is what provides high initial gain. If the original change is sufficiently large or in the form of a step input, it is possible to drive output pressure P_o to saturation conditions, namely zero or 630 kPa bar depending on direction of the input change. For this reason it is common practice to analyze rate response using a ramp-type input. Initial response in output pressure will still occur very quickly and be greater than what would be obtained using only proportional control. The change in P_o, however, causes a pressure drop across valve D and air now flows from bellows E. This initiates feedback action and a return to proportional control. The result is that change in controller output

pressure is a function of rate of change of input or error signal and positioning of valve D.

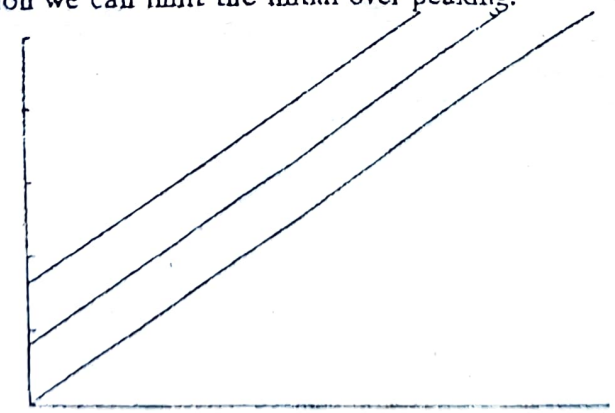


Proportional plus rate controller output response to a ramp input with different values for Td and K represents proportional band

DERIVATIVE ACTION TIME (DAT)

DAT, Td or rate time is the time required for the proportional action to maintain the value and size of the derivative action when the rate adjust valve is fully closed the rate time will be infinity giving on-off control. As the valve is opened the rate time will decrease and bleeding into the proportional bellows will occur to a faster rate.

With derivative action we can limit the initial over peaking.

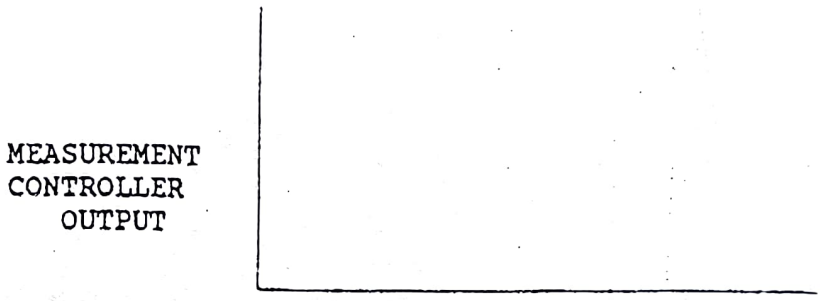


Time responses for load changes using P and Pd controllers

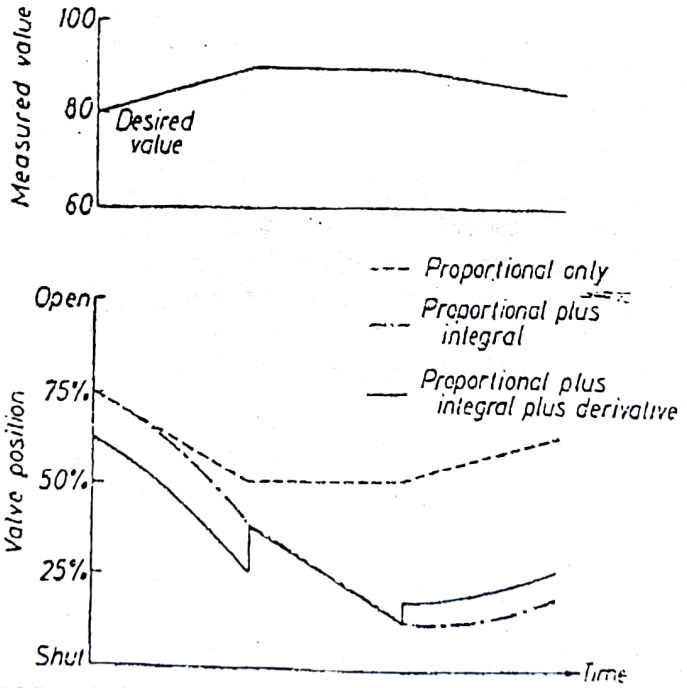
THREE TERM CONTROL (PID)

This control includes all the above functions in one control action.

----- Proportional
_·_ Proportional and Integral
_____ Proportional and Integral and Derivative



Response of PID Control modes

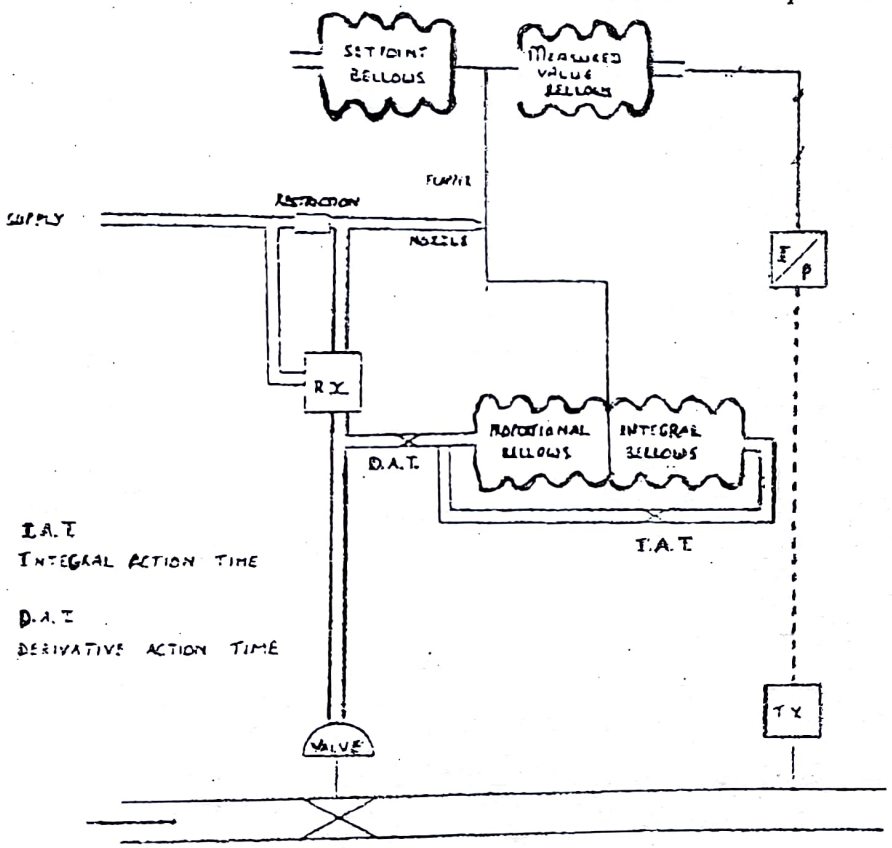


It should be obvious to the reader by this time that many "processes" will require both reset and rate control. Reset is needed to eliminate offset and rate action to decrease period and initial overpeaking.

The optimum controller is a high-gain or on-off controller. From considerations of simplicity, maintenance, initial cost, and offset, there can be no argument. A problem, of course, exists with stability. As "process" time constants decrease and gains increase, due to reduced mass and increased throughputs, controller gain must be reduced to obtain stability. The results is proportional control and resultant offset.

Reset "puts back" high gain at low frequency to eliminate offset. Associated phase lag of reset, however,. causes a reduction in controller gain and crossover frequency. This leads to an increase in initial overpeaking and period. To overcome these disadvantages and reduce initial overpeaking and period, rate control is added.

Rate "puts back" a portion of the high-frequency characteristics of on-off control. The end result is an effort to use best features of on-off control and still maintain a stable system. Another way of describing this overall picture is to say that the controller compensates for "process" conditions resulting from design limitations. In fact in electrical terminology, the controller is often referred to as a compensating network.



PROPORTIONAL INTEGRAL AND DERIVATIVE CONTROL

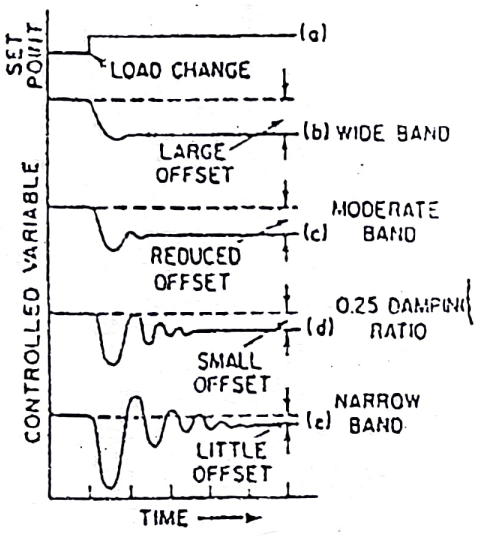
With valve D partially closed, however, a pressure decrease in bellows E is delayed. This in turn delays proportional feedback action and allows a greater change in output pressure. This is what provides high initial gain. If the original change is sufficiently large or in the form of a step input, it is possible to drive output pressure P_o to saturation conditions, namely zero or 45 kPa bar depending on direction of the input change.

Due to the change in output pressure, a pressure differential now exists across valve I. This causes air to flow from bellows H through valve I.

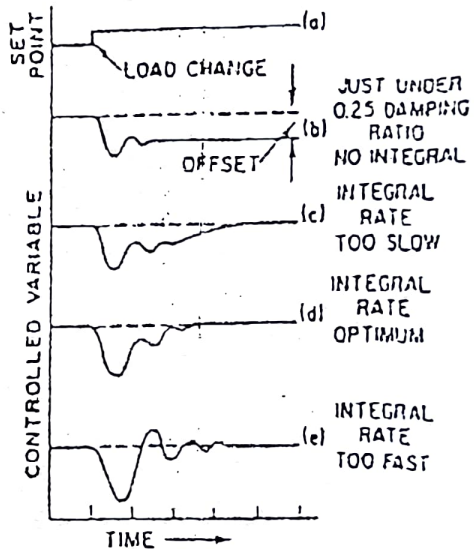
This moves the flapper closer to the nozzle and produces an additional increase in output pressure. The change in P_o, however, causes a pressure drop across valve D and air now flows from bellows E. This initiates feedback action and a return to

proportional control. The result is that change in controller output pressure is a function of rate of change of input or error signal and positioning of valve D and I.

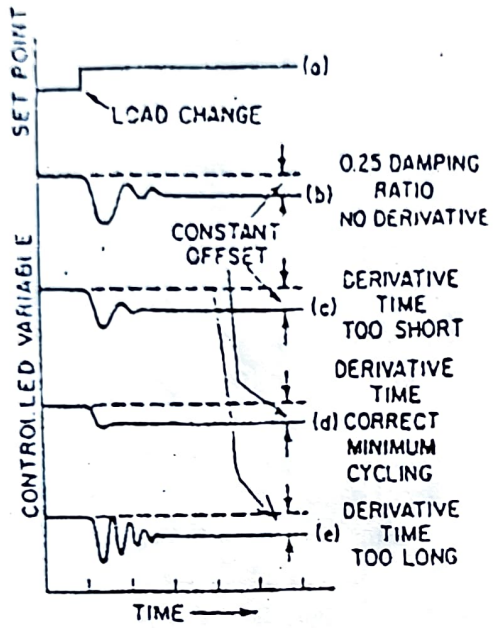
When proportional, integral and derivative controls are adjusted the following output curves are obtained.



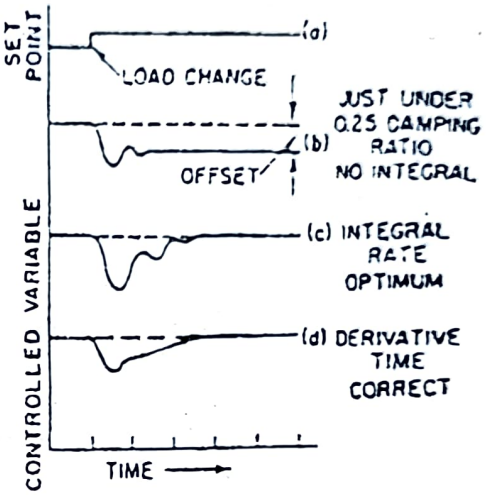
Effect of narrowing proportional band on recovery curves following set-point load change, showing how proportional-band adjustment is a compromise between instability and offset error.



Effect of increasing integral rate on recovery curves following set-point load change.



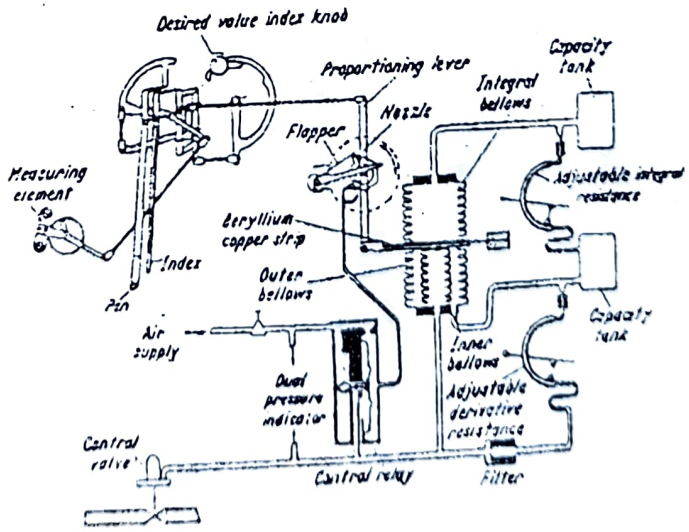
Effect of increasing derivative time on recovery curves following set-point load change.



Correct settings for a PID controller.

OTHER CONTROLLERS

POSITION BALANCE FOXBORO MODEL 40 THREE TERM CONTROLLER



Foxboro model 40 three-term controller (proportional with integral and derivative)

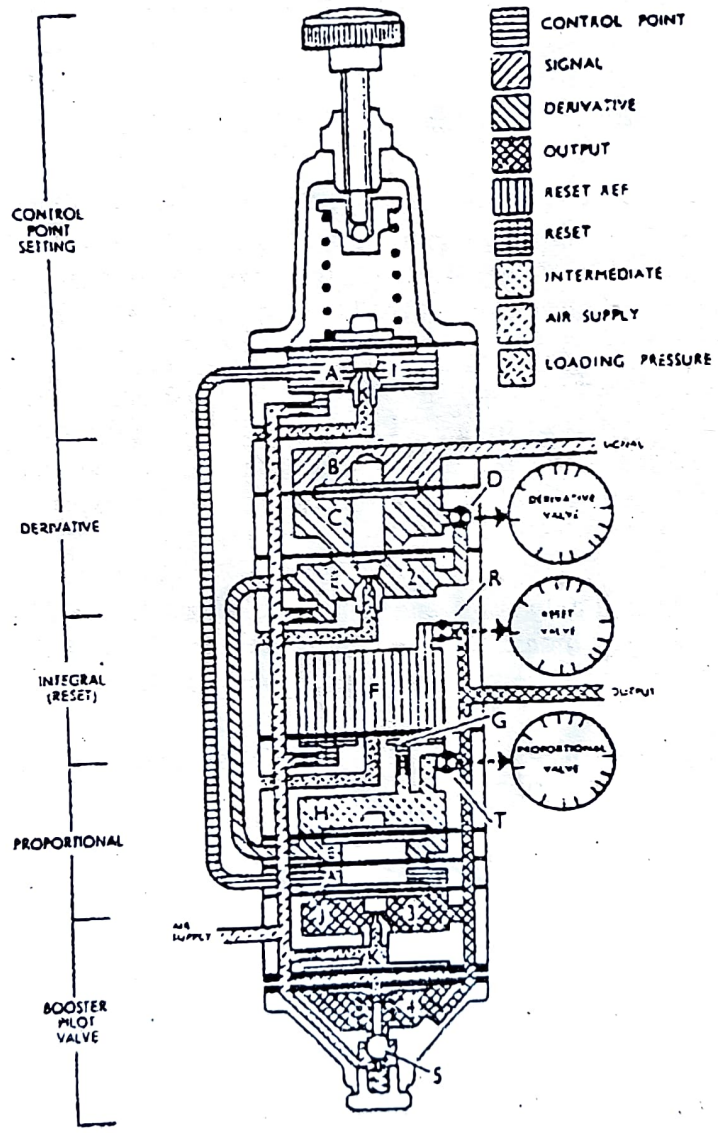
The adjustable resistances consist of a section of Bourdon tube, one end of which remains stationary and the other end may be moved to decrease the cross-sectional area

of the tube. By this method it is possible to change the flow through the restriction in the ratio of 500 to 1.

The purpose of the inner bellows is to increase the stability of the derivative action and render the system insensitive to high-frequency disturbances or "noise". If the derivative resistance is completely closed, the output pressure from the control relay P_1 will be applied to the inner bellows only. As the inner bellows has a cross-sectional area which is only one-fifth of that of the outer bellows, the force available to move the beryllium-copper strip per unit pressure change of P_1 will be only one-fifth the force available if P_1 were applied to the complete area of the outer bellows. The pressure in the inner bellows is proportional to the output, but owing to the reduced area the feed-back will be considerably reduced, so in effect a narrower proportional band is produced and the output pressure will be greater than that obtained for the same deviation if the output pressure was applied to the full area of the outer bellows. As air bleeds through the derivative resistance, the air pressure in the outer bellows will increase until it is also equal to the output pressure. The difference between the pressure in the inner and outer bellows will depend upon the setting of the derivative resistance and the rate of change of the deviation, so that the output will have a derivative component owing to the action of the dual bellows.

The function of the capacity tanks is to add volume to the bellows and assist in establishing the desired integral and derivative time constants.

STACK TYPE FORCE BALANCE CONTROLLER



Operation of a three-term force balance controller

It consists of a stack of pressure chambers, each separated from the next by a series of concentric flexible diaphragms. The transient position of the diaphragms, which is determined by the instantaneous summation of the pressures acting upon them, controls

the bleed of supply air to the atmosphere through specially designed bleed valves thus continuously balancing the forces involved. Proportional band, integral-action time, and derivative-action time adjustments are made by altering the setting of needle valves which restrict the otherwise free flow of air between the appropriate chambers. The desired value and the measured value are fed to the controller as air pressures, which are continuously compared, so that the output to the correcting unit varies in conformity with the deviation and the setting of the proportional action, integral action time, and derivative action time settings. When the process is stabilized at the desired value, the controller is in balance, and since there is then no flow through the needle valves, the resultant output signal is independent of the valve characteristics.

Proportional control unit - Assuming pressure H to be constant, any change in the pressure E, representing the signal pressure, repositions the central diaphragm unit. Owing to the action of the bleed valve 3, this produces an immediate change in the pressure J (proportional-action pressure) balancing out the change in the pressure E.

At the same time, depending upon the setting of the proportional valve, T, pressure H changes by a smaller amount than J. Pressures H and J effectively act in opposition, hence H changes still further in the same direction until the differential force between J and H balances the out-of-balance force between signal pressure E and desired value pressure A.

The ultimate differential between J and H necessary for balance is dependent upon the opening of the valve T, which therefore determines the width of the proportional band and is calibrated directly in percentage proportional band.

A simple switching plate allows the input to the chambers E and A to be exchanged, giving a reverse acting controller.

When the controller is supplied without a derivative unit, the signal pressure B is applied directly to the chamber E.

Integral action unit - Chambers J and F are connected through the integral valve R and any change in the pressure J, owing to signal variation, results in a smaller change in the pressures F and G.

Pressure G is maintained equal to pressure J by the 1:1 relay. Change in the pressure G causes a change in the intermediate pressure H and pressure J. The pressure J will continue to change in a manner which maintains the flow of air into or from the chamber F at a rate depending upon the setting of the valve R and the difference between the pressures E and A (i.e. the deviation) until the upper or lower limit of the output pressure is reached, or until the desired value is regained. The position of the valve R is calibrated in terms of integral-action time. The 1:1 relay ensures that pressures F and G are equal while allowing the proportional and integral terms to develop independently.

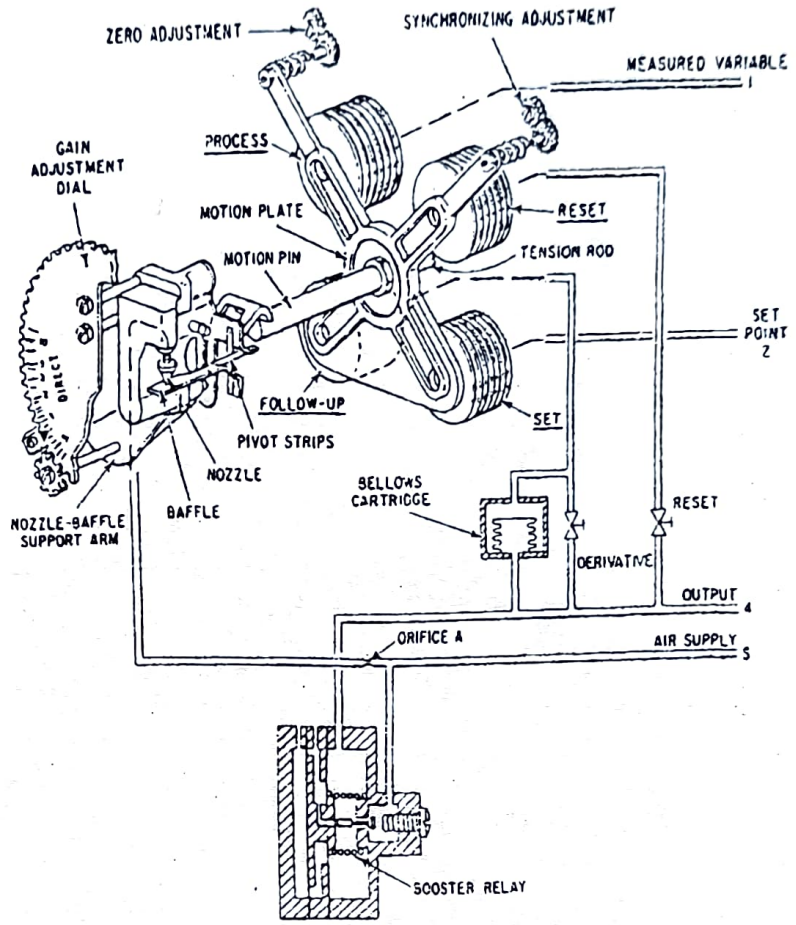
Derivative-Action unit - Provided the measured value signal is constant, the pressures B, C and E will be the same. When the measured value signal is changing, however, the pressure E will not be equal to the measured value signal owing to the difference of diaphragm areas exposed to the signal pressure B and the pressure E.

Suppose the signal is suddenly increased by 5 kPa and that the area exposed to pressure B is six times that exposed to pressure E, then in order to restore the balance of forces on the diaphragms the pressure E must increase by 30 kPa, thus greatly increasing the change in error signal fed to the proportional unit, and producing an increased change in the correcting signal. Change in the pressure E will produce a change in the pressure C. The pressure C will change at a slower rate than the pressure E, the actual difference in rates depending upon the setting of the valve D.

Owing to the fact that the area of the upper diaphragm exposed to the pressure C is six times the area of the lower diaphragm exposed to the pressure C, increase in pressure C will reduce the difference between the pressures E and B and tend to restore equilibrium. The rate at which these pressures approach equality will depend upon the rate at which pressure B is changing, and upon the rate at which air is flowing through the derivative valve D. Thus, the output signal to the correcting unit will contain a component which is dependent upon the rate-of-change of the measured value, i.e. it will have a derivative term and the magnitude of this term will depend upon the setting of D which is calibrated in terms of derivative-action time.

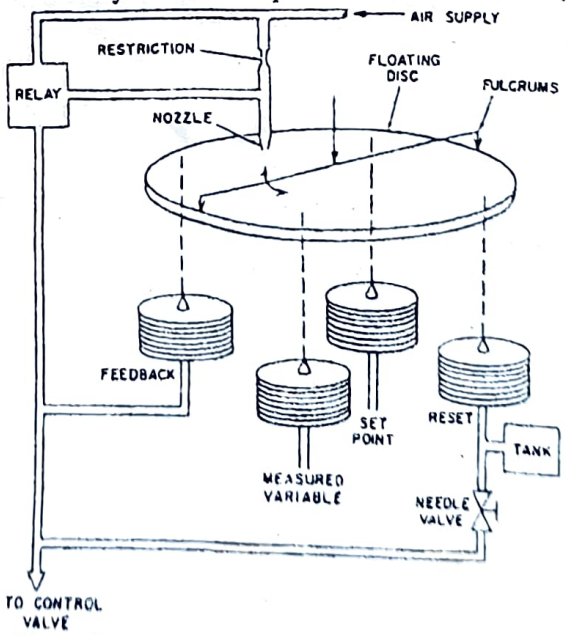
PNEUMATIC RECEIVER-CONTROLLERS

Some of these devices have gain continuously adjustable from 100 down to 0. A controller of this type, employing bellows as pressure elements, is shown in the figure below. Four bellows are arrange symmetrically about a motion plate supported by a central tension rod. The bellows which receives set point pressure is located opposite the bellows which receives transmitted pressure. Similarly, the follow-up bellows is located opposite the reset bellows and in quadrature with the set point bellows. Any difference between set and transmitted pressures imparts a motion to the central motion pin which moves the baffle. The output pressure change repositions the motion pin to restore balance. The relative effects of input and follow-up pressures are determined by rotatably positioning the gain adjustment dial around the motion pin. Integral and rate responses are adjusted by use of pneumatic restrictors.



Pneumatic receiver controller

This action can maybe better explained with a "wobble plate" action.



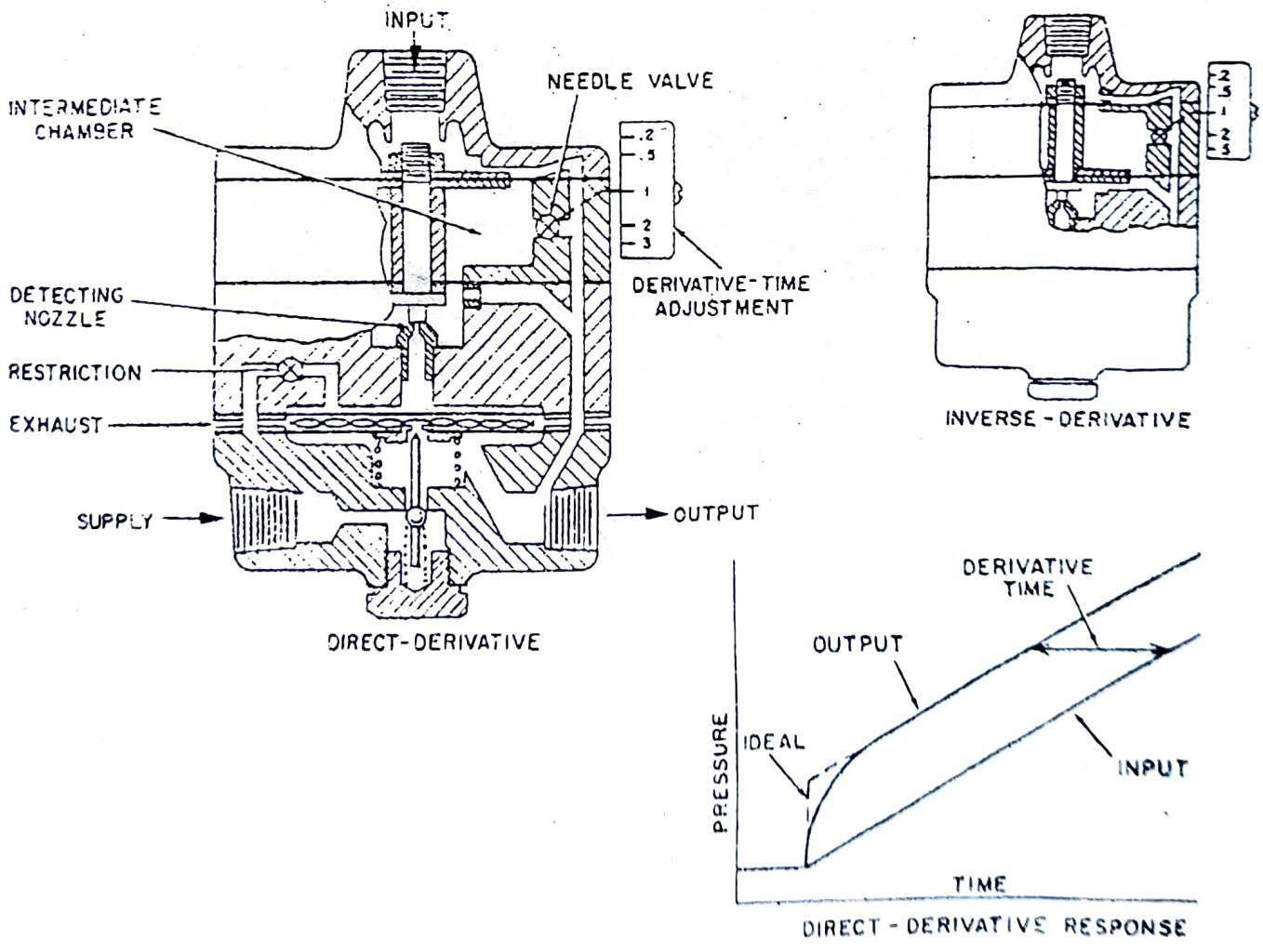
Typical moment-balance controller

The bellows all act from one side against a pivoted "wobble-plate". The wobble plate acts as the nozzle baffle. ROtating the pivot axis changes the proportional band. When the pivot axis coincides with the reset and feedback bellows, 0 percent proportional band results; when it coincides with the measured variable and set-point bellows, infinite band results. When the axis bisects the two bellows axis, 100 percent proportional band results.

FLUIDIC RELAYS

Derivative action is more effective if it is non-interacting and if it can be applied *ahead* of the proportional and reset action of the controller. Such a derivative unit can be built into the controller, or it can take the form of a separate relay.

This relay employs diaphragms, but the design can be executed with bellows just as well. The signal from some process transmitter, typically a temperature transmitter, would be connected to the input of the relay.



Direct and inverse derivative relays

Due to the difference between input- and output-diaphragm areas, a step change in input produces a largely amplified step change in output to maintain force balance.

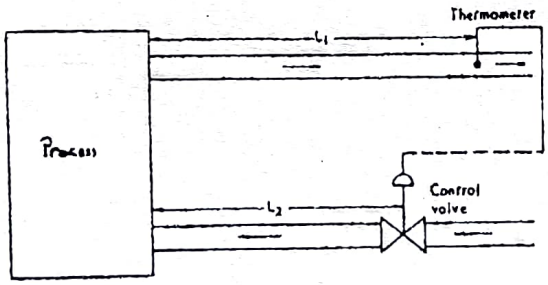
In a steady-state condition (with no change in input), the output pressure acts on both sides of the output diaphragm - and output pressure rebalances input pressure directly.

TIME LAGS

The controllability of a plant or process depends to a vital extent on the delay occurring between a change in conditions at one point in a plant and the manifestation of this change at another point. The important delays or time lags can be classified under two general headings: Distance Velocity Lags and Transfer Lags. Consider Distance Velocity Lags first.

Distance Velocity Lags

Distance Velocity lag is defined as "The time interval between an alternation in the value of a signal and its manifestation *unchanged* at a later part of the system, arising solely from the finite speed of propagation of the signal". The tendency is for the term to be restricted to cases where the medium is moving.



Distance Velocity lags in a control loop

The detecting element, a thermometer is situated at a distance L_1 from the plant in pipeline. A medium flows from the plant at a velocity Vm/second. The time for any signal, arising from a change in process conditions, to reach the thermometer is $\frac{L_1}{V}$ seconds.

If, for example, $L_1 = 20m$, and V is 5m/second, the distance velocity lag is 4 seconds.

A distance velocity lag can also occur when the correcting unit is at a large distance from the process. If the valve is away from the process to any great extent, a delay is

experienced between the adjustment of the valve and the new flow value arriving at the plant (L_2).

Transfer Lags

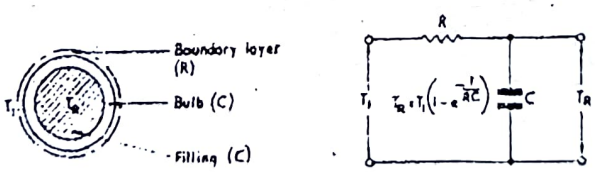
In the case of transfer lags, these must be considered at more than one point in the control loop. Detecting elements are responsible for some significant transfer lags and since they respond directly to changes in the values of the controlled conditions we may usefully consider them before the plant or process lags.

Detecting Element

Most serious detecting lags can occur in a temperature control system. A transfer of heat is necessary from the medium whose temperature is being controlled, through the walls of a packet or bulb to the sensitive element or filling. Any lag involved, therefore, is strictly speaking a transfer lag.

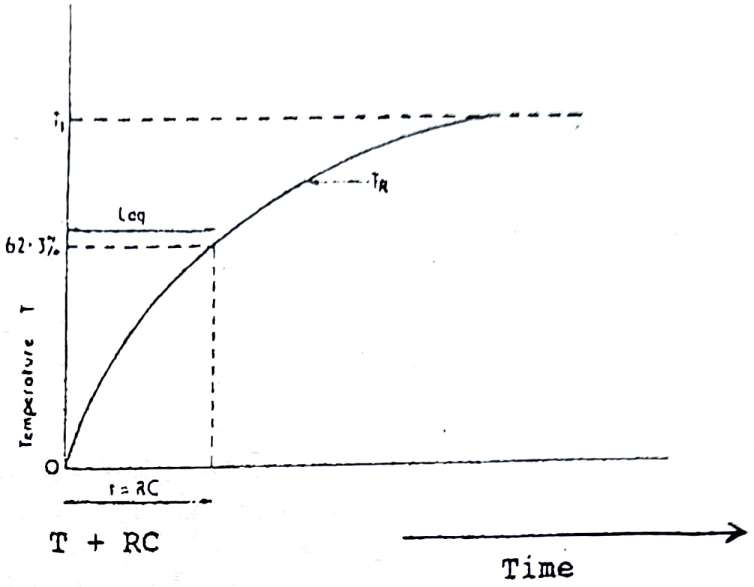
Tests are very often carried out on simple temperature indicating systems by plunging the element into a bath at a higher temperature and observing the progress of the indicated temperature with time. The time for the temperature to reach the 63.2 % mark is then taken as a measurement of the lag of the temperature element.

In the case of a mercury-in-steel bulb, for example, there is resistance - thermal resistance - formed by the boundary layer of fluid on the outer surface of the blub. Capacitance, in the form of thermal capacitance, exists in the bulb and liquid filling.



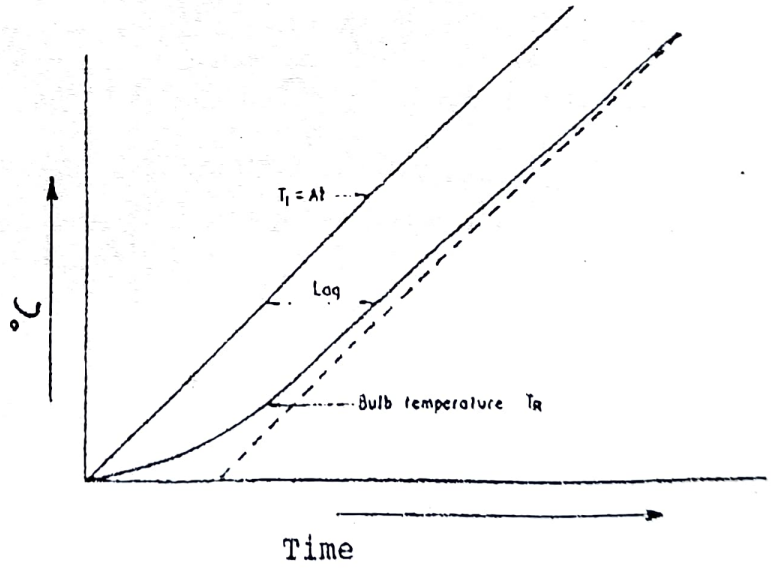
A mercury-in-steel bulb

If a sudden or step temperature change is applied to the bulb, following the electrical analogy we can plot the curve by plotting T_B against time.



Response to a step change

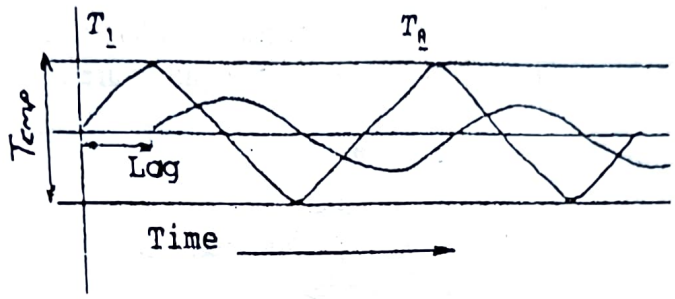
RC is known as the time constant of the bulb, and is actually the time for the temperature to reach 63,2% of its final value T_1.
Alternatively suppose that the bulb is exposed to a medium whose temperature is changing at a constant rate.



Response to a constant rate or ramp change

T_1 is a straight line. The bulb temperatures commences with a pronounced curve, and then tend to straighten out as the constant lag value is reached.

Then we must consider the behaviour of the element to a sinusoidal change in the temperature of the medium in which it is placed.

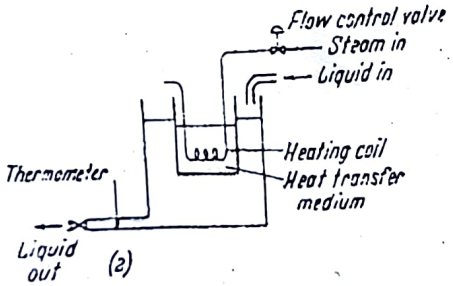


Response to a sinusoidal change

It can be seen that both frequency and the value of the time constant can influence the response to a significant extent.

Plant or Process

This the time taken for a process to rectify the deviation. Let us consider a process which must supply heated water.



Heating process plant

The water is being heated by passing it through a tank containing a heating coil through which steam is passed. The rate of rise of temperature of the water will depend upon the thermal capacity of the tank and it contents, and upon the rate of transfer of heat to the water when a drop in temperature is signalled to the valve, the steam flow will increase. Then the heating coil will transfer heat to the medium which in turn will heat up the water. This time elapse is known as the plant or process lag.

Transmission Lags

We have two main forms of transmission lines in the process control field: electrical and pneumatic. In the case of electrical lines the transmission time for a signal is negligible even for large distances. For pneumatic pipe lines, however, finite times are experienced

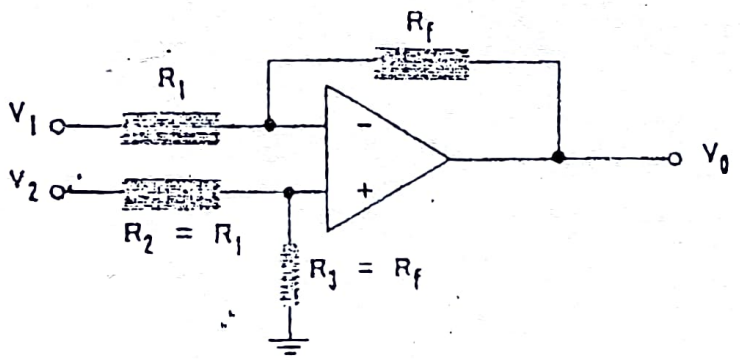
for the transmission of signals, and these, dependant on pipe internal diameter and length, can cause significant lags. The pneumatic transmission line can be considered analogous to an electrical line if the following broad assumptions are made:

a) Pressure disturbance values are small.
b) There is no steady flow of air in the line.
c) There is uniform distribution of the fluid resistance, capacitance and inductance parameters throughout the line.
d) The load is lumped.
e) The small disturbances that do occur do not vary the mean pressure level in the line.

ELECTRONIC CONTROL

DISCRIMINATION

The most commonly used circuitry or discrimination between reference and error levels will be the operational amplifier.



The differential amplifier or comparator

If $R_1 = R_2 = R_3 = R_f$ then $V_0 = V_2 - V_1$.

This circuit can be used to monitor the difference between two voltage levels. The comparator is a circuit which compares an input voltage with a reference voltage. The output of the comparator indicates whether the input signal is above or below the reference voltage. If the input voltage of an inverse comparator is above the reference voltage, the output approaches the negative supply potential and vice versa.

Other forms of discrimination is:

- Wheatstone bridge
- Linear variable deferential transformer

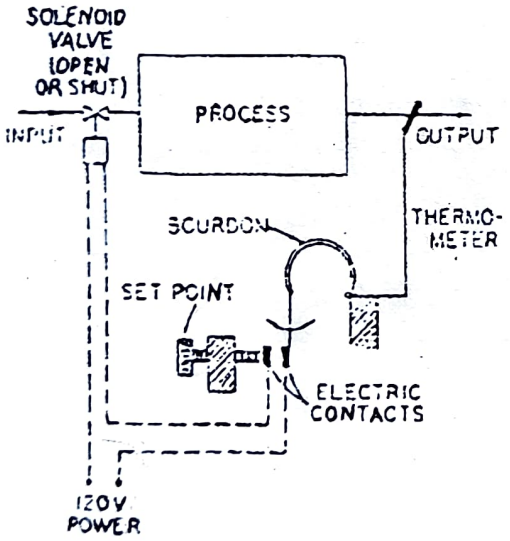
- Linear variable capacitive transducer

MODES OF CONTROL

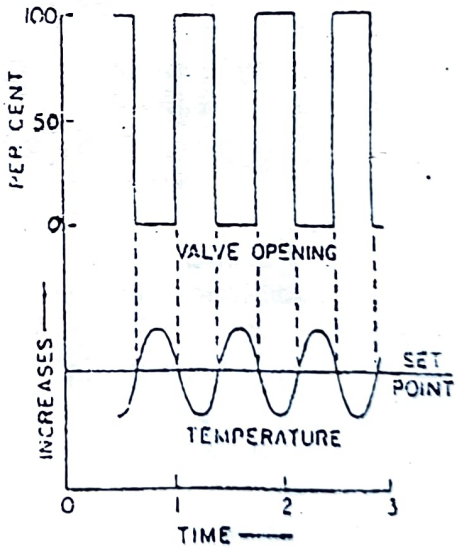
We use the same modes of control as for pneumatically operated control systems.

ON-OFF CONTROL

In this mode of control, the final control element is moved relatively quickly from one of two fixed positions to the other at a single value of the controlled variable. In as much as these two positions of the final control element are usually open and closed, this mode is often termed *on-off* control.



Typical tow-position electrically operated system

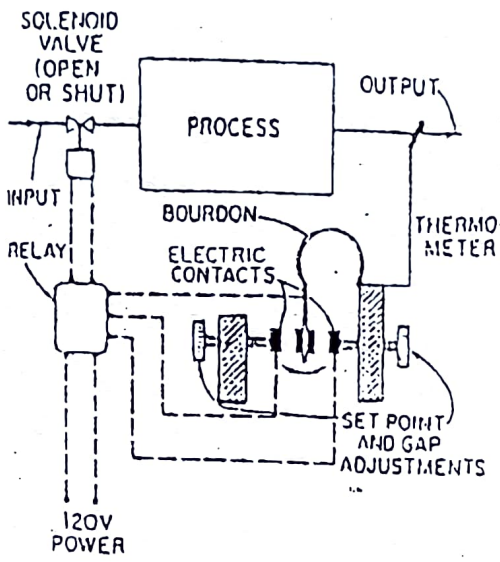


Relation between valve position and controlled variable in two-position action

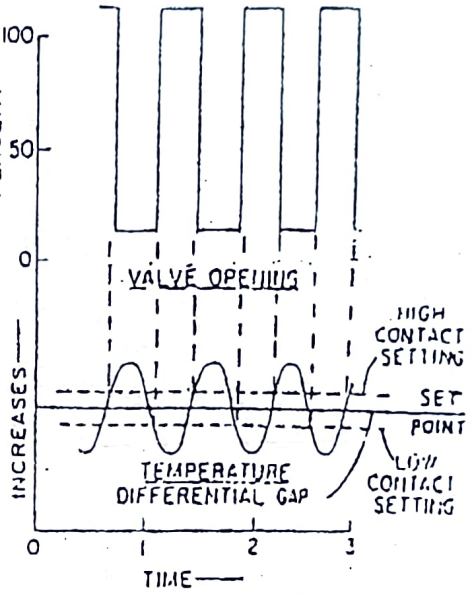
When the temperature is at or above the set-point value, the contact is closed and the valve closes; when the temperature is below the set point, the contact is opened and the valve opens. It is seen that two-position control cannot make an exact correction; its correction must be either greater or less than exact. Therefore, no stable balanced condition of input and output energy is ever achieved, and the controlled variable must continue to cycle up and down.

TWO-POSITION DIFFERENTIAL-GAP CONTROL ACTION

This is a common variation of the two-position action.



Typical two-position differential-gap electrically operated control system



Relation between valve position and controlled variable in two-position differential-gap control action

Here the final control element is moved relatively quickly from its first position to its second when the controlled variable reaches a set value from one direction, and can return to its first position only after the variable has passed in the opposite direction through a range of values (called the *differential gap)* to a second value.

A greater degree of control over the temperature cycling can be effected by designing the control law such that the switching point during heating occurs at a higher temperature compared with the switching point during cooling.

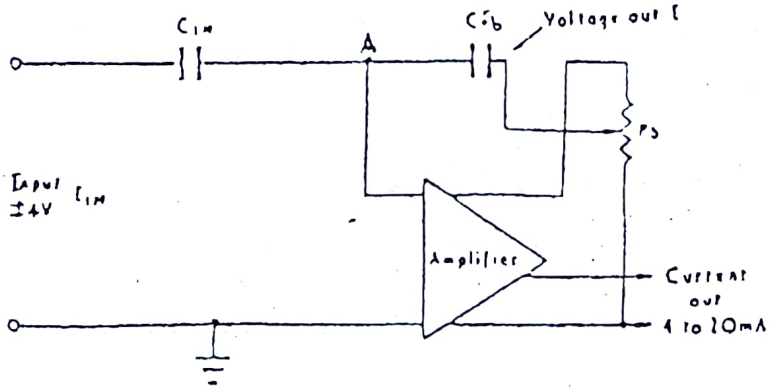
This type of control sometimes is referred to as "overlap" or "hysteresis". Increasing the overlap zone will result in a reduction in the frequency of the temperature cycling, but this will be accompanied by an increase in the temperature excursion.

MULTI POSITION CONTROL ACTION

In this action the final control element is moved to one of *three* or more fixed positions, each corresponding to a definite range of values of the controlled variable. The two-position differential gap electrically operated control system can produce a three-position action if it is used with a final control element that takes a third position when the variable value is within the differential gap. This is undertaken most readily with relays

and/or contactors, in the case of electric heaters; however, the simplicity of the hardware must be traded against reduced reliability and maintainability.

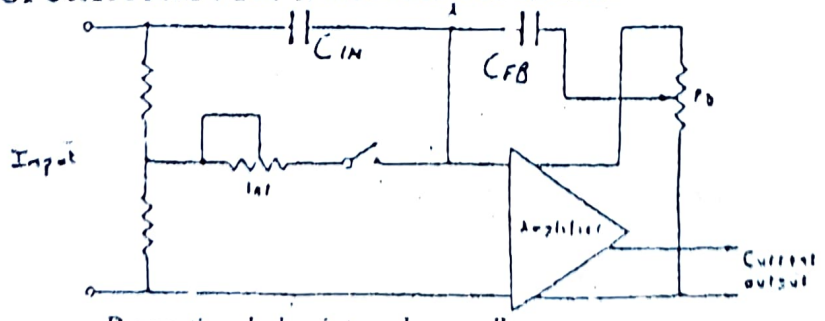
PROPORTIONAL CONTROL



Proportional controller

The capacitors are equal in value. There are two outputs from the amplifier. One comprises a current signal ranging from 4 to 20 mA for control purposes. The other is a voltage signal of ± 4 V for feedback purposes. With no input signal, Efb, the feedback voltage is zero and the potential at a relative to earth is zero. There will be an output signal, however, and this could conveniently be 12 mA the halfway value between 4 and 20 mA. Now if an error signal is impressed on the input capacitor C is charged and a potential appears at point A, forming in input signal to the amplifier. A feedback voltage is generated to Cfb to counteract this and the potential at A returns to zero. This is, of course, the same condition as we had for the original output signal of 12 mA. But for as long as there is a sustained input signal to the amplifier, there is an equally sustained feedback voltage generated and if this changes so does the output current in proportion. The proportional band can be adjusted to change the gain of the controller. This is done by adjusting the amount of feedback voltage to Cfb. Thus we have established a form of proportional control.

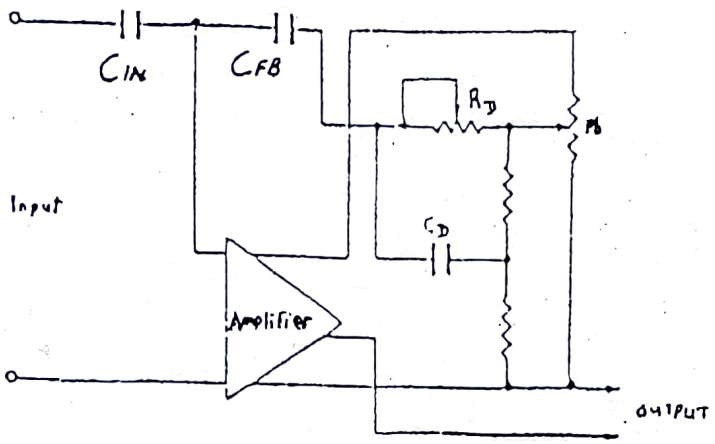
PROPORTIONAL PLUS INTEGRAL CONTROLLER



Proportional plus integral controller

For integral control, the input capacitor is effectively shunted by the integral potentiometer and other resistors in the input circuit. Initially it can be considered that the immediate proportional section occurs as before when an input signal appears. But the resistors provide a path for the discharge of C in, which would mean a potential appearing at A. But this would provide an input signal to the amplifier which would immediately supply a feedback signal to counteract it, and in effect recharge the capacitor. If the input signal is sustained the action is continued and will continue until the input signal is zero.

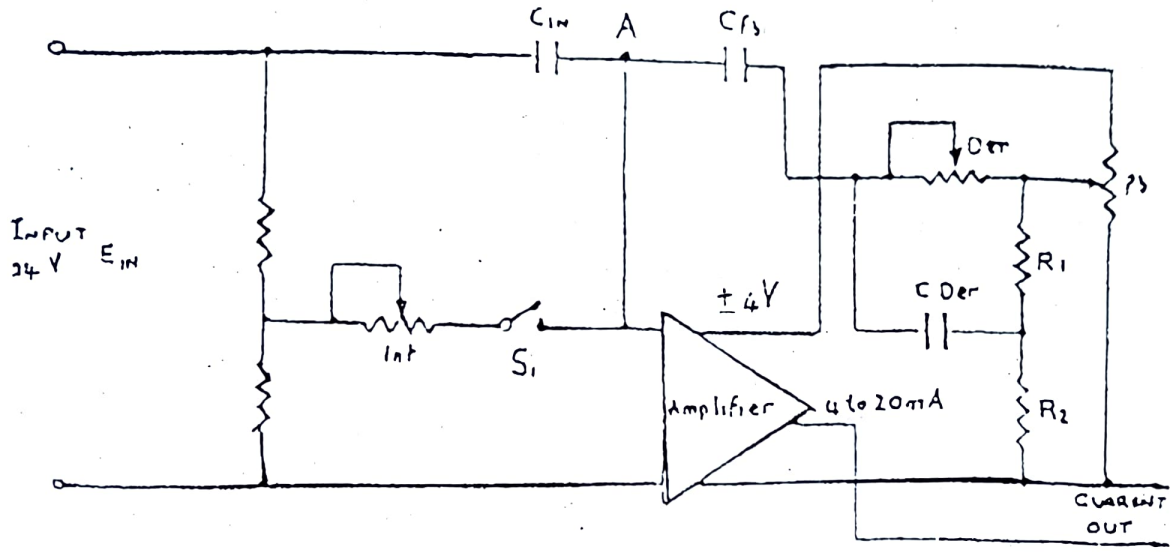
PROPORTIONAL PLUS DERIVATIVE CONTROLLER



Proportional plus derivative controller

Derivative control is included in the feedback path to capacitor Cfb. When an input signal appears there is a little variation to the normal proportional action. The derivative capacitor Cd, at the first instant of charge, transfers only a part of the voltage existing across the proportional band potentiometer FB because of the voltage divider arrangement R1 and R2. As a result to obtain the necessary feedback to Ctb the amplifier output is increased. But as the derivative capacitor becomes charged, the influence of the voltage divided is lessened and the voltage across PB is supplied to feedback capacitor Cfb. The time for this to happen is governed by the value of Cd and derivative resistor Rp. The voltage divider determines the amplitude of the derivative feedback.

If we add all these actions together we will get a three term PID controller.



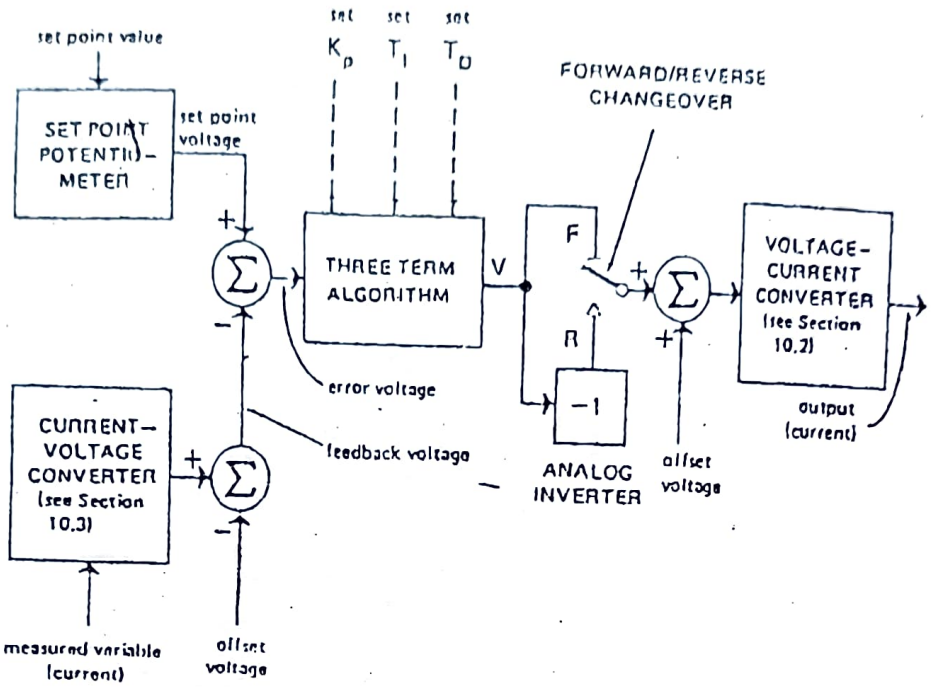
PID Feedback Controller

- There are two outputs from the amplifier, one a current signal, and the other a voltage signal for feedback purposes. These two outputs are proportional to one another but are electrically isolated.

- If a signal is present at the input of the amplifier a feedback voltage is generated to counteract this input so that the voltage at "A" returns to zero. By adjusting the resistor P_B we have a means of adjusting the proportional band.

- For the integral control, the input capacitor is effectively shunted by the integral potentiometer and other resistors. It can be considered that the proportional action occurs as before when the input signal appears but the resistors supply a path of discharge of "C in". Which would mean a potential appearing at "A". This would supply an input signal to the amplifier which would supply a feedback voltage to counteract it and in effect recharge the capacitor.

- The derivative is included into the feedback path to capacitor "Cfs". When an input signal appears there is little variation to the normal proportional action. The derivative capacitor only transfers a part of the voltage existing across. The proportional band potentiometer "P_B" because of the voltage divider arrangement $R_1 + R_2$. As a result to obtain the necessary feedback to Cfs the amplifier output increases thus causing the necessary derivative action.



PID controller

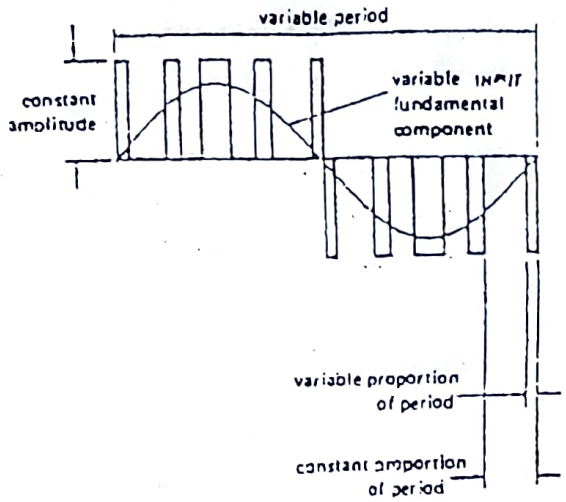
If the coefficients K_P, T_I and T_D can be adjusted independently of each other, each by means of its potentiometer (say), the control law is said to be "non-interacting": most modern electronic controllers are like this. In older electronic controllers, and in all pneumatic controllers, one control adjustment will change one coefficient in a major way but will also affect the value of at least one of the other coefficients to a minor extent: such controllers are said to be "interacting".

In may instances, not all of the terms would be used, in a particular loop. For example, derivative action can be eliminated by setting $T_D = 0$, whilst integral action can be removed by setting $T_I = \infty$. Huts, it is possible to configure P, PD, PI and PID control, and in some cases I control only (called "floating control"). Roughly speaking, the effect upon loop performance of the three terms can be summarised as follows:

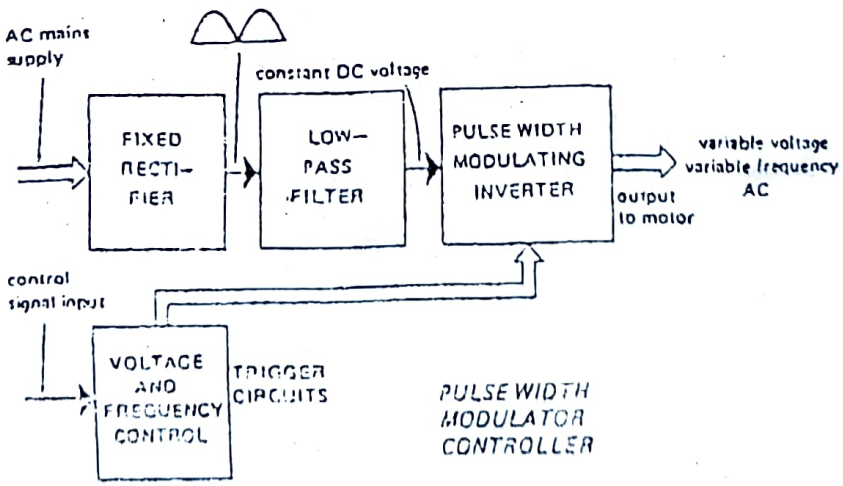
- Proportional action will reduce steady state error and increase the step response overshoot, as the proportional band K_P is reduced.
- Integral action will eliminate steady state error arising from most causes and, as the integral action time T_I is reduced, increase the step response overshoot; it is unlikely to be required whenever the plant process contains an inherent integration term, since the resulting loop would then contain two integrations.
- Derivative action will reduce the step response overshoot, as the derivative action time T_D is increased.
- It is difficult to generalise about the effect of the terms upon step response settling time.

PULSE DURATION CONTROLLER

The Pulse Duration Controller accepts a mA-input signal, compares it with the desired set point, and produces a series of on-off pulses to operate a control valve by means of a reversible electric motor for process control.



The duration of each pulse is proportional to the deviation of the measurement from the set point. The space and mark (on-off) time will be a constant value. When the deviation is small the space time will be longer than the mark time and for a large error the mark time will be longer than the space time.



There is one solid-state SCR contract for a condition or measurement less than set point. Contact closure energizes the reversible electric motor to drive the final operator in the necessary direction to obtain control.

The set point is adjusted by a dial located on the front of the controller. A local-remote set point selector switch is also located on the front of the controller as a standard feature, to facilitate adjustment of the set point from an external source.

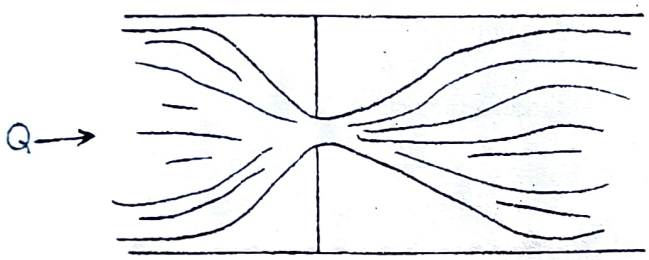
The time base of the output switching circuit is adjustable between 10 and 60 seconds. The dead band adjustment permits small measurement deviations without any corrective action being applied to the process.

FLOW RATE CALCULATIONS

We have theoretically deduced an equation or Bernoulli's theorem but in practise there are correcting factors to be included to the basic equation of $Q = A_2 E \sqrt{2gh}$.

CORRECTING FACTORS OF BERNOULLI'S THEOREM

Discharge Coefficient (C)



From the figure it can be seen that the momentum of the flowing liquid causes it to continue to contract after it has passed through the orifice.

The point at which the liquid reaches its minimum cross-sectional area is called the vena contracta. Unfortunately the position of the vena contracta moves closer to the orifice as the flow rate decreases, and further downstream as the flow increases.

Due to friction and velocity distribution, the practical flow figures do not line up with the theoretical ones.

Therefore to determine the measurement, the equation must include a correction fact or for the friction effects, and another for the contraction effect. It is customary to incorporate these two into two constant and denote it by the name discharge coefficient and letter C.

$$C = \frac{\text{true flow rate}}{\text{theoretical flow rate}}$$

EXPANSIBILITY FACTOR (ϵ)

The expansibility factor (ϵ) has a value of 1,00 for incompressible fluids. ϵ can be calculated or obtained from graphs.

Due to the temperature changes in a flow line this factor must be taken into consideration. This number is obtained by using the pressure relation, heat radiation (gamma) and the area relation.

$$Q = CA_2 E\epsilon \sqrt{2gh}$$

The density of the measured quantity will vary as well and the change will be exponentially, therefore it must be rooted.

$$Q = C.A_2 E\epsilon \sqrt{\frac{2gh}{\rho}}$$

REYNOLDS NUMBER

The term $\frac{vd\rho}{\mu}$ is known as the Reynolds number.

where
v = the mean velocity through the orifice
d = orifice diameter
ρ = the fluid density
μ = absolute viscosity

Definition

Rd is a dimensionless criterium which expresses the ratio of the inertia forces to viscosity forces of a fluid flowing in a closed system.

If the Reynolds number is above a certain value, called the limiting value, it is found that the flow coefficients remain substantially constant for all subsequent increases in the flow rate.

For lower Reynolds numbers an additional correction factor (Z) is included in the formulae. (Z becomes equal to 1 if Rd is large enough.) Z is the product of three multipliers and its value depends on Rd, m and the size and roughness of the pipe.

This factor will correct all changes in pipe size due to flow and is obtained by using the Reynolds number correction factor (Z_R) where a Reynolds number (R_d) is used and the pipe size correction factor (Z_s).

$$\therefore \quad Z = Z_R \times Z_s$$

From an engineering viewpoint, these variables effecting a fluid's flow rate cannot be evaluated for all possible pipeline conditions.

Therefore Reynolds number is used as an index figure correlating the combined effects of fluid velocity, pipe size and fluid viscosity to predict the type of flow profile.

(Essentially a ratio of inertia forces to viscous forces)

A low Reynolds number (up to about 1000 to 2000), indicates that the viscosity and velocity of the fluid, combined with the pipe size, produce a smooth, laminar flow.

Thus the equation becomes:

$$Q = Z.\epsilon C.A_2.E. \sqrt{\frac{2gh}{\rho}}$$

Constant values (0.01252)

It is always easier to measure the diameter of the pipe and we know that:

$$A = \frac{\pi d^2}{4}$$

where π and 5 are constant values. Gravitational value g and 2 under the root are constant values as well. We have to transfer some of the unit values as well. This bring us to the following equation for fluids and gasses:

$$Q = 0{,}01252.Z.\epsilon.C.d^2.E\sqrt{\frac{h}{\rho}}$$

ϵ = Epsilon

ρ = RHO

where
- Q = flow rate (m^3/hour)
- Z = Reynolds pipe size factor
- ϵ = expansion factor
- C = discharge coefficient
- d = orifice diameter section (mm)
- E = velocity approach factor
- h = differential pressure (mm H_2O or mm WG)
- ρ = density (kg/m^3)

This equation is used to calculate the quantity rate of flow (m^3/hr).

$$Q = 0{,}01252\ CEZ\epsilon d^2\sqrt{h/\rho} \qquad (m^3/hr)$$

$$Rd = \frac{3{,}54\ Q\rho}{\mu d} \quad \text{or} \quad \frac{3{,}54Q\rho}{\mu D\sqrt{m}}$$

$$m = \frac{d^2}{D^2} \qquad h = \left(\frac{Q}{0{,}01252CEZ\epsilon d^2}\right)^2\rho$$

$$E = \frac{1}{\sqrt{1-m^2}}$$

$$mE = \frac{N}{CZ\epsilon}$$

$$CmE = \frac{N}{Z\epsilon}$$

$$CmE = N$$

$$N = \frac{Q\sqrt{\rho}}{0,01252D^2\sqrt{h}}$$

$$Z = Z_R Z_D$$

$$\frac{\Delta\ kPa}{kPa(abs)\ x\ 27,2} = \frac{h}{\rho a}$$

Mass rate of flow (steam)

To calculate the mass rate of flow the equation becomes:

$$W = 0,01252\ Z.\epsilon C.d^2.E\ \sqrt{h\ x\ \rho}\ (kg/hr)$$

Formulas

Mass rate of flow (W)

$$W = 0,01252\ CEZ\epsilon d^2\ \sqrt{h.\rho}\quad (kg/hr)$$

$$Rd = \frac{3.54W}{\mu d} \quad or \quad \frac{3.54W}{\mu D\sqrt{m}}$$

$$*Diameter\ ratio = \frac{d}{D} \qquad Area\ ratio = \frac{d^2}{D^2} = m$$

$$\frac{\Delta kPa}{kPa(abs) \times 27,2} = \frac{h}{Pa} \qquad * \qquad h = \left(\frac{W}{\frac{0,01252 C E Z \epsilon d^2}{\rho}} \right)^2$$

$\frac{h}{\rho}$ = *pressure difference ratio*

$\left(\frac{inches\ H_2O}{lbf\ per\ in^2} \right)$

$$N = \frac{W}{0,01252\ D^2 \sqrt{h\rho}}$$

For both:

Nett pressure loss = line pressure x percentage (graph G)

Conversions

*1" Wg = 25 mm Wg = 249 Pa
1 kPa = 102 mm Wg = 1000 Pa = 7,5 mm Hg
1 lb/ft^3 = 16,0183 kg/m^3
Standard atmospheric pressure = 101,325 kPa
Gravitational acceleration = 9,81 m/s^2
1 lbf per in^2 = 6,88 kPa
1 bar = 100 kPa - 100 000 Pa
1" = 25,4 mm

$$d = \sqrt{\frac{Q}{0,01252\ C E Z \epsilon \sqrt{h/\rho}}} \qquad d = \sqrt{\frac{W}{0,01252\ C E Z \epsilon \sqrt{h x \rho}}}$$

$$\frac{d}{D} = \left(\frac{(mE)^2}{1+(mE)^2} \right)^{1/4}$$

When we do orifice calculation we must see that we use the following SI units:

	h	in mm H_2O (differential pressure)
	ρ	in kg/m^3 (density)
or	ρ	in lb/ft^3 x 16,0183 = kg/m^3
	Q	in m^3/hr (flow rate) - water and gasses
	W	in kg/hr (mass flow rate) - steam
	d	in mm (orifice diameter)

D in mm (pipe internal diameter)
P pressure in absolute (kPa)

$y = =$ specific heat of a gas at constant pressure (Cp) / specific heat of a gas at constant volume (Cv)

Notes

1. When Rd number becomes to large for table, Z_R becomes 1.
2. When the diameter becomes to large for table Z_D becomes 1.
3. For incompressible fluids ϵ becomes 1.
4. Calculations for inside orifice diameter

$$d = \sqrt{\frac{Q}{0{,}01252 Z.\epsilon.E\sqrt{h/\rho}}} \; x \text{ temperature multiplying factor}$$

(graph H bellow)

- calculate the N value

$$N = \frac{Q}{0{,}1252.D^2} \; x \; \sqrt{\frac{\rho}{h}} \simeq C.m.E$$

- Use graph D to read of M value

The following example will explain the equation in detail:

Calculate the rate of coke oven gas flow at a temperature of 35°C when the density of the gas is 0,1084 lb/ft³. A differential pressure of 1330 Pa is obtained with the pipe inside diameter of 250 mm and the inside orifice plate diameter of 150 mm when D and D/2 tappings are used. The pipe line pressure is 15" WG with the specific heat ratio of 1,6. What will the percentage of pressure loss be?

Solution

$$Q = 0,.1252\ Z\epsilon C d^2 E \sqrt{\frac{h}{\rho}}$$

$$= 0.01252\ 1,004\ 0,0973\ 0,608\ (150)^2\ 1,07157 \sqrt{\frac{MP}{1,7366}}$$

$$= 1408,78\ m^3/hr$$

Discharge coefficient C

The equation for C:

$$C = \frac{\text{true flow rate}}{\text{theoretical flow rate}}$$

If these values are not known we must use graph A where we need the area relation (m). The equation for M:

$$m = \frac{\text{inside orifice diameter}^2\ (d)}{\text{inside pipe diameter}^2\ (D)}$$

$$= \frac{150^2}{250^2}$$

$$= 0,36$$

From graph A:

$$C = 0,608$$

Velocity approach factor (E)

The equation for E:

$$E = \frac{1}{\sqrt{1-m^2}}$$

$$= \frac{1}{\sqrt{1-0{,}36^2}}$$

$$= 1{,}07187$$

Expansion factor ϵ

Consider graph B. To determine ϵ we must now

a) the relation between the pressure difference and absolute pressure ($^h/_p$)
b) the specific heat relation (Y) and
c) the area relation value (m)
d) $\frac{h}{P_{abs}} = \frac{\rho}{P_{ATM}+P_{gauge}} \times \frac{27{,}2}{1}$

where P_{gauge} = 15" WG

and 1"WG = 249 Pa

$\therefore P_{gauge}$ = 15 x 249

= 3735 Pa

a) $\frac{H}{\rho} = \frac{1}{101{,}32 + 3{,}735}$

= 0,2589

b) V = 1,6

c) m = 0,36

From graph B;

ϵ = 0,9973

a) $D = 250$ mm

but $1" = 25{,}4$ mm

$$\therefore \quad D = \frac{250}{25{,}4}$$

$$= 9{,}8"$$

b) $m = 0{,}36$

From graph F:

$Z_D = 1.001$

thus $Z = Z_R \times Z_D$

$= 1{,}003 \times 1{,}001$

$= 1{,}004$

Calculated values for rate of flow:

Z	=	1,004
ϵ	=	0,9973
C	=	0,608
d	=	150 mm
E	=	1,07187
h	=	102 mm WG
ρ	=	1,7366 kg/m^3

Percentage of pressure loss

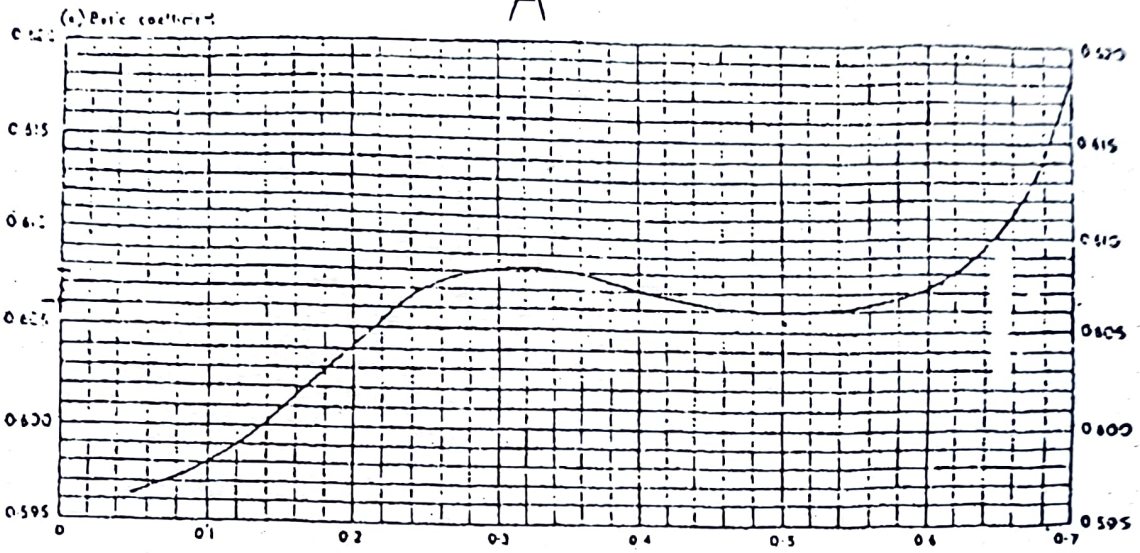
Consider graph G to determine loss we need

a) type of primary device; and
b) area relation (m)

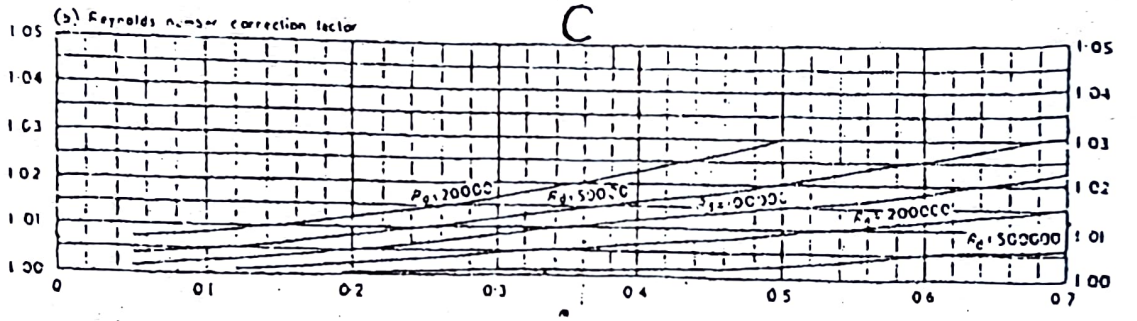
a) D and D/2 orifice plate
b) $m = 0{,}36$

See pages 189, 190 at the back of the book

A



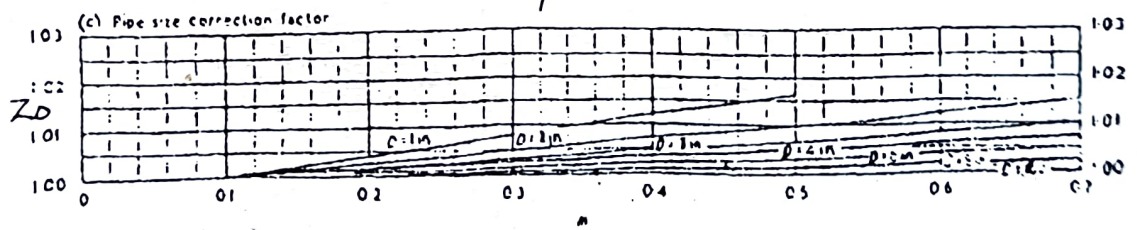
C



Z_R

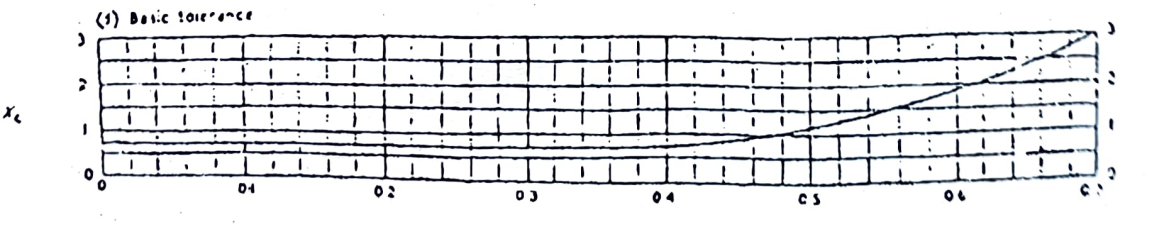
m

F



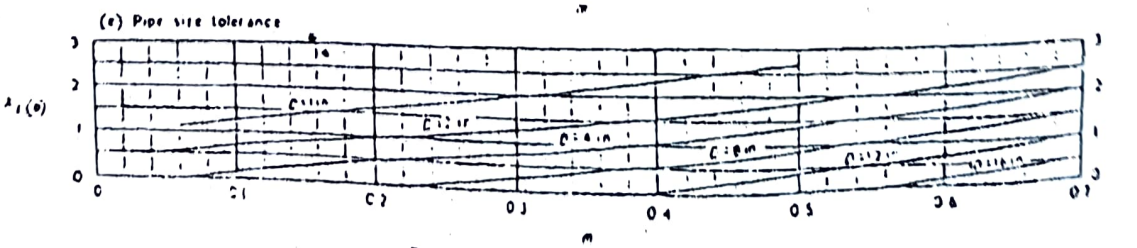
Z_D

m



X_c

m



$X_t(\theta)$

m

Data for orifice plate with D and D/2 tappings

B

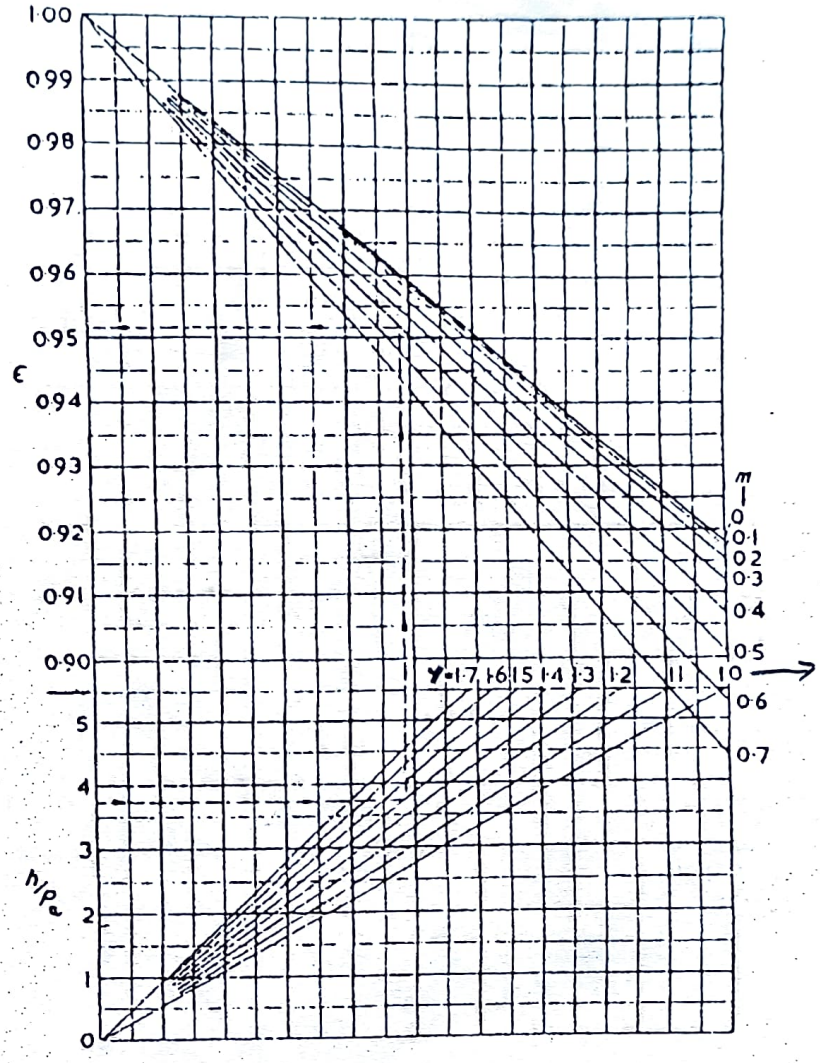
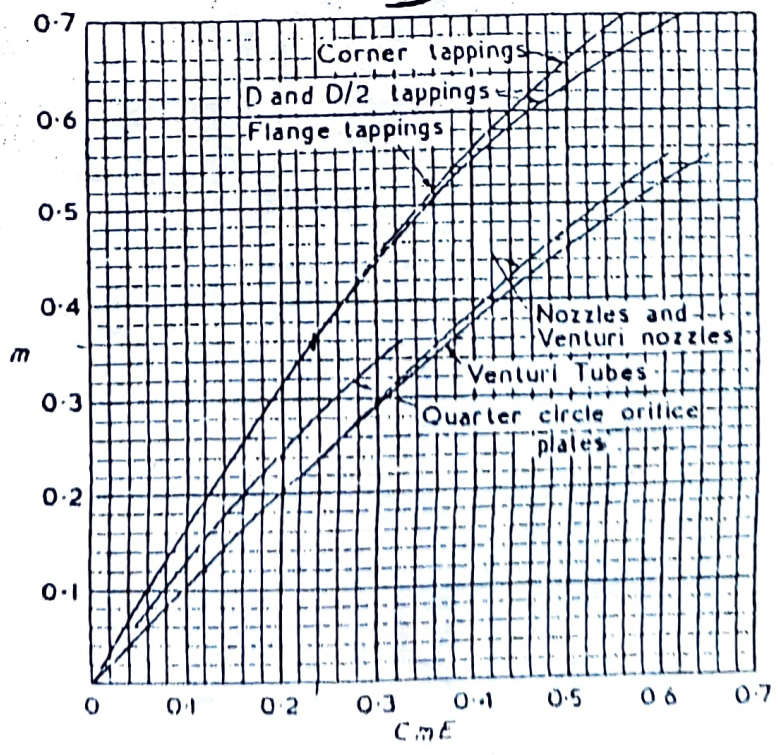


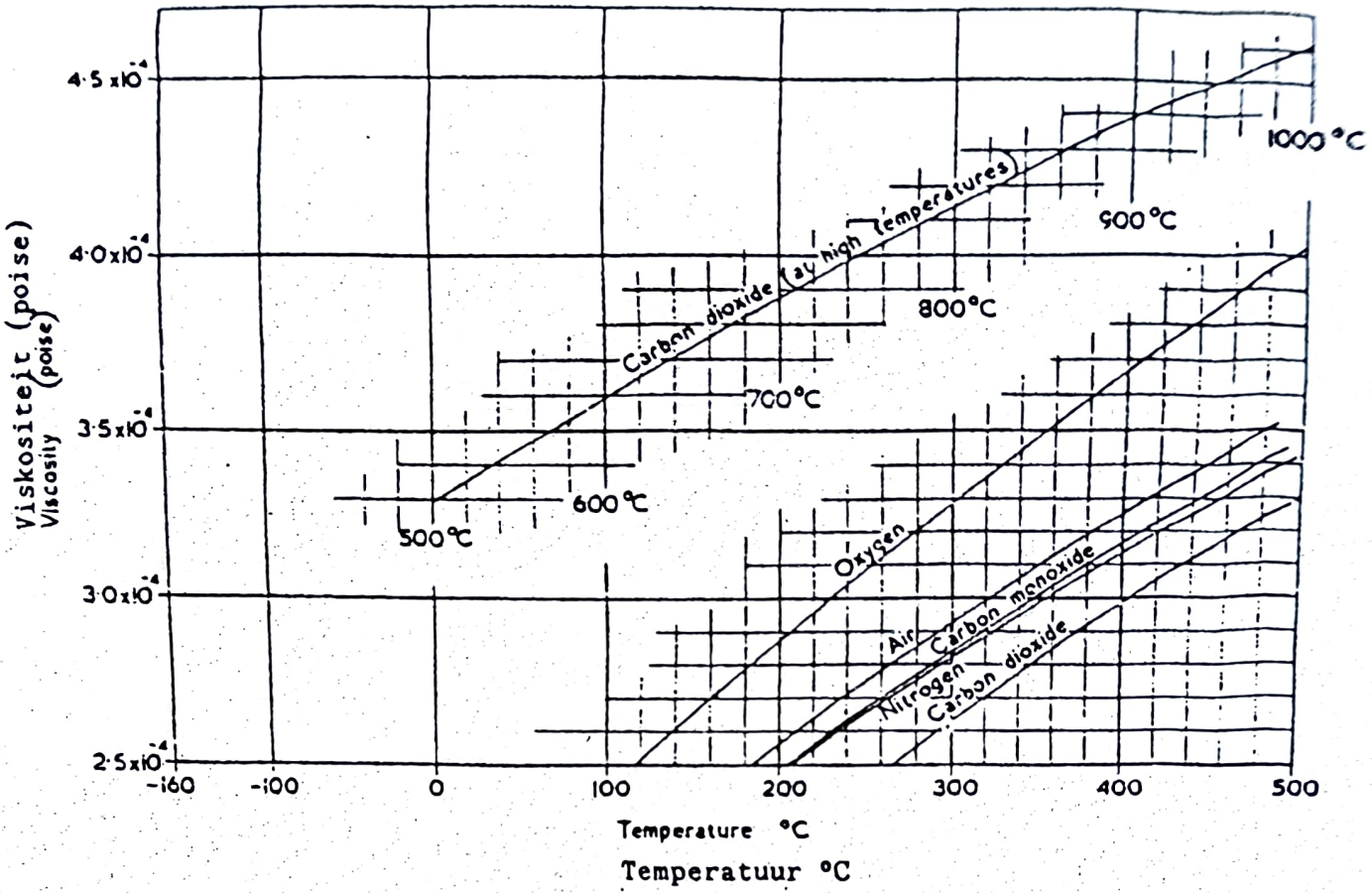
Fig. 39. Expansibility factor for orifice plate with D and D/2 tappings or with flange tappings

Uitsettings faktor vir meetskryf plaat vir D en D/2 en flens tappe.

D



E



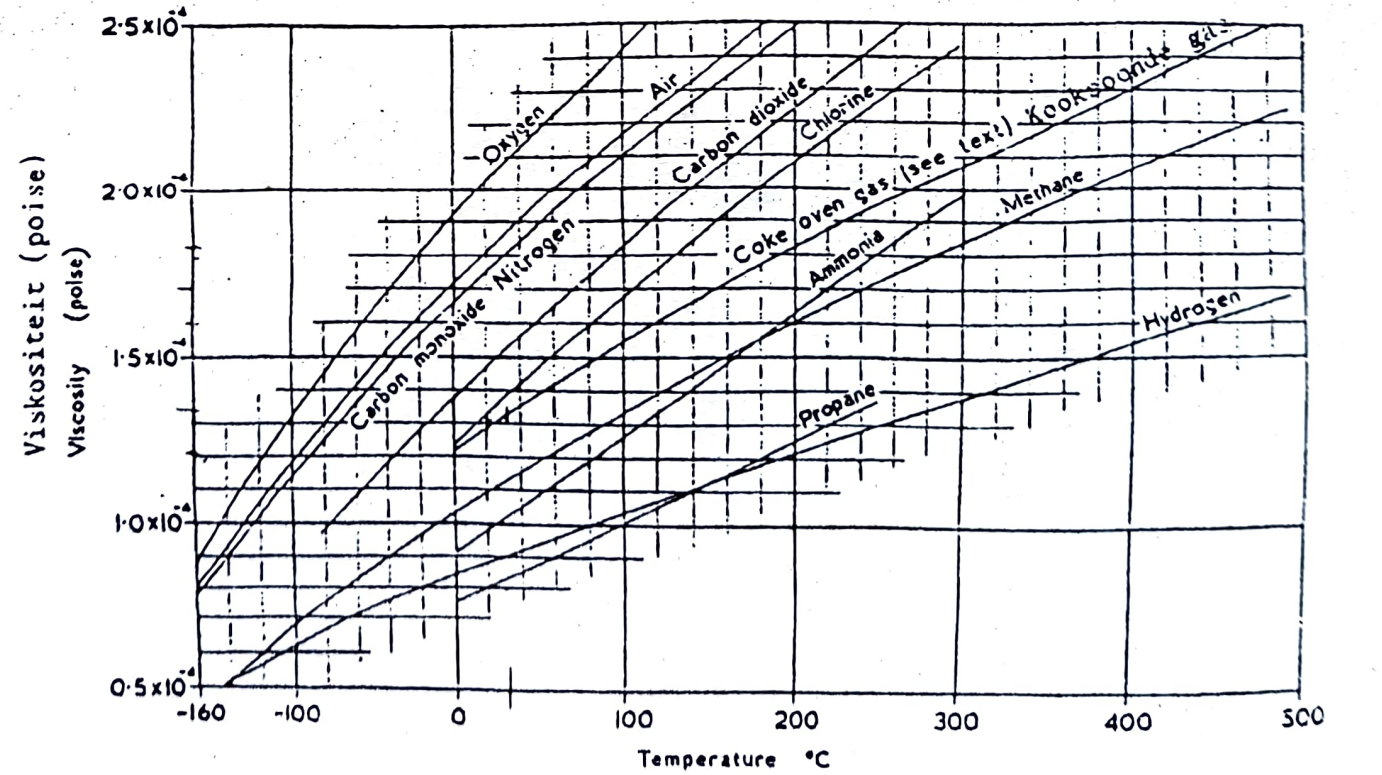
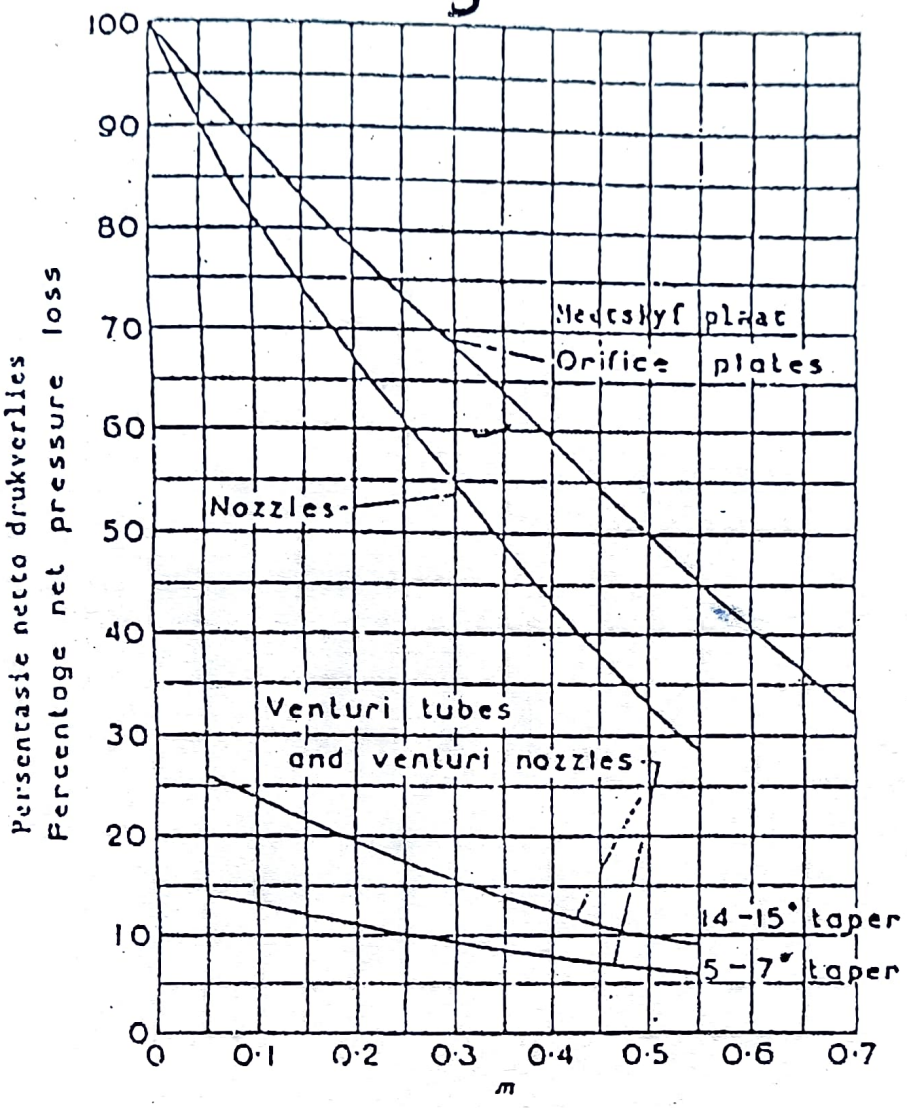


Fig. 18. Viscosity of selected gases at 1 atm


Persentasie netto drukverlies
Percentage net pressure loss
100
90
80
70
60
50
40
30
20
10
0
0
0·1
0·2
0·3
0·4
0·5
0·6
0·7
m
Meetskyf plaat
Orifice plates
Nozzles
Venturi tubes
and venturi nozzles
14 -15° taper
5 - 7° taper

Reynolds pipe size factor Z

The equation for Z:

Z = Reynolds number correction factor (Z_R) X pipe size correction factor (Z_D)

Reynolds number correction factor (Z_R)

Consider graph C. To determine Z_R we need

a) The Reynolds number (Rd)

b) the area relation (m)

a) $Rd = \frac{3.54\ Q\rho}{d}$

$= \frac{3.54 - 1406,9 - 1,7366}{1,5 \times 10^{-4}.150}$

$= 384400$

where Q = initial flow rate (m^3/hr)
ρ = density (kg/m^3)
μ = viscosity (poise)
d = inside orifice diameter (mm)

To calculate initial flow rate the equation for Q:

$Q = 0.01252.N.D^2\sqrt{h/\rho}$

$= 0,01252.0,2346(250)^2 \sqrt{\frac{102}{1,7366(mm)}}$

$= 1406,9\ m^3/hr$

where N = C.m.E constant
D = inside pipe diameter
h = Δ Pa (WG)
ρ = density (kg/m^3)

Calculations on Rd number

* N C.m.E

0,608.0,36-1,07187

0,2346

(Note: N constant can be read of graph D for an estimated value)

* D = 250 mm

* h = 1000 Pa

but 1 kPa = 102 mm WG

∴ h = 1 x 102

= 102 mm WG

* ρ = 0,1084 lb/ft^3

but 1 lb/ft^3 = 16,02 kg/m^3

∴ ρ = 0,1084 x 16.02

= 1,7366 kg/m^3

* Viscosity (μ)

COnsider graph E to determine μ we need

a) the temperature and
b) the type of material

a) T = 80°C
b) Coke oven gas

From graph E:

μ = 1,5 x 10^{-4} (poise)

b) m = 0,36

From graph C:Z_R = 1,003

Pipe size correction factor (Z_D)

Consider graph F to determine Z_D we need

a) inside pipe diameter (D); and
b) the area relation (m)